RWTH Aachen University
Software Engineering Group

Image Segmentation and Quantification of Synaptic Markers in the Mushroom Body Calyx of *Drosophila melanogaster*

Master Thesis

presented by
Aditya Rawat

1st Examiner: Prof. Dr. Abigail Morrison

2nd Examiner: Prof. Dr.-Ing. Johannes Stegmaier

Advisor: Prof. Dr. Abigail Morrison

The present work was submitted to the Chair of Software Engineering

Aachen, 28.06.2026

Zentrales Prüfungsamt/Central Examination Office

Eidesstattliche Versicherung
Statutory Declaration in Lieu of an Oath

Rawat, Aditya — 450954

Name, Vorname/Last Name, First Name — Matrikelnummer (freiwillige Angabe) / Matriculation No. (optional)

Ich versichere hiermit an Eides Statt, dass ich die vorliegende Arbeit/Bachelorarbeit/ Masterarbeit* mit dem Titel

I hereby declare in lieu of an oath that I have completed the present paper/Bachelor thesis/Master thesis* entitled

Image Segmentation and Quantification of Synaptic Markers in Mushroom body calyx of Drosophila melanogaster

selbstständig und ohne unzulässige fremde Hilfe (insbes. akademisches Ghostwriting) erbracht habe. Ich habe keine anderen als die angegebenen Quellen und Hilfsmittel benutzt. Für den Fall, dass die Arbeit zusätzlich auf einem Datenträger eingereicht wird, erkläre ich, dass die schriftliche und die elektronische Form vollständig übereinstimmen. Die Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

independently and without illegitimate assistance from third parties (such as academic ghostwriters). I have used no other than the specified sources and aids. In case that the thesis is additionally submitted in an electronic format, I declare that the written and electronic versions are fully identical. The thesis has not been submitted to any examination body in this, or similar, form.

Aachen, 20.06.2026 — Unterschrift/Signature

Ort, Datum/City, Date

*Nichtzutreffendes bitte streichen

*Please delete as appropriate

Belehrung:
Official Notification:

§ 156 StGB: Falsche Versicherung an Eides Statt
Wer vor einer zur Abnahme einer Versicherung an Eides Statt zuständigen Behörde eine solche Versicherung falsch abgibt oder unter Berufung auf eine solche Versicherung falsch aussagt, wird mit Freiheitsstrafe bis zu drei Jahren oder mit Geldstrafe bestraft.

Para. 156 StGB (German Criminal Code): False Statutory Declarations
Whoever before a public authority competent to administer statutory declarations falsely makes such a declaration or falsely testifies while referring to such a declaration shall be liable to imprisonment not exceeding three years or a fine.

§ 161 StGB: Fahrlässiger Falscheid; fahrlässige falsche Versicherung an Eides Statt
(1) Wenn eine der in den §§ 154 bis 156 bezeichneten Handlungen aus Fahrlässigkeit begangen worden ist, so tritt Freiheitsstrafe bis zu einem Jahr oder Geldstrafe ein.
(2) Straflosigkeit tritt ein, wenn der Täter die falsche Angabe rechtzeitig berichtigt. Die Vorschriften des § 158 Abs. 2 und 3 gelten entsprechend.

Para. 161 StGB (German Criminal Code): False Statutory Declarations Due to Negligence
(1) If a person commits one of the offences listed in sections 154 through 156 negligently the penalty shall be imprisonment not exceeding one year or a fine.
(2) The offender shall be exempt from liability if he or she corrects their false testimony in time. The provisions of section 158 (2) and (3) shall apply accordingly.

Die vorstehende Belehrung habe ich zur Kenntnis genommen:
I have read and understood the above official notification:

Aachen, 20.06.2026 — Unterschrift/Signature

Ort, Datum/City, Date

Abstract

Memory consolidation is associated with structural reorganization in neuronal circuits, making reliable quantification of synaptic markers an important prerequisite for studying learning-induced plasticity. In the mushroom body calyx of *Drosophila melanogaster*, this analysis is commonly performed on 3D microscopy data and still requires substantial manual effort for region selection, segmentation cleanup, and quantitative measurement. As a result, current workflows are difficult to scale and prone to user-dependent variation.

This thesis investigates an automated segmentation and quantification pipeline for synaptic structures in volumetric microscopy images of the mushroom body calyx. The proposed workflow combines microscopy-adapted foundation-model segmentation, based on MicroSAM, with pre-processing and post-processing steps tailored to noisy biological image stacks and visually similar surrounding tissue. The goal is to obtain robust object masks that support downstream measurements such as object count, volume, and marker-specific statistics with less manual interaction than conventional threshold-based procedures.

To assess the suitability of the approach, the pipeline is compared with alternative models and pre-processing variants on annotated data. The evaluation considers segmentation quality, robustness across image conditions, and practical usability for biological analysis. By reducing manual intervention while preserving quantitative interpretability, the presented workflow aims to provide a reproducible basis for larger-scale studies of synaptic reorganization and to support future experiments on memory-related circuit changes in *Drosophila melanogaster*.

Contents

Abstract iii

List of Figures viii

List of Tables ix

1 Introduction 1

1.1 Motivation and Biological Background 1

1.2 Problem Statement 2

1.3 Limitations of Existing Segmentation Approaches 2

1.4 Goal of This Thesis 3

1.5 Thesis Outline 3

2 Related Work 5

2.1 Biological Context of Bouton Quantification 5

2.2 Fluorescence Microscopy and 3D Image Characteristics 8

2.3 Image Pre-processing for Microscopy Segmentation 10

2.4 Segmentation Tasks in Biomedical Microscopy 13

2.5 Deep Learning for Biomedical Segmentation 15

2.5.1 nnU-Net 17

2.6 Instance Segmentation Models for Microscopy 18

2.6.1 StarDist 18

2.6.2 Cellpose 20

2.7 Foundation Models and Microscopy Adaptation 21

2.7.1 Transformer Encoders and Self-Attention 22

2.7.2 Swin UNETR 23

2.7.3 Segment Anything Model 25

3 Methods **29**

3.1 Datasets . . . 29

3.1.1 Microscopy Acquisition . . . 29

3.1.2 Dataset Composition . . . 30

3.1.3 Dataset Selection and Annotation Rationale . . . 31

3.1.4 Annotation Procedure . . . 38

3.2 Dataset Creation and Pre-processing . . . 40

3.3 Image Pre-processing and Post-processing . . . 43

3.3.1 Slice Extraction and Normalization . . . 43

3.3.2 Post-processing . . . 43

3.4 MicroSAM . . . 44

3.4.1 Model Variant . . . 44

3.4.2 Training Data Preparation . . . 45

3.4.3 Fine tuning Configuration . . . 45

3.4.4 Inference . . . 46

3.4.5 MicroSAM Base Comparison Variant . . . 47

3.5 Baseline and Comparison Models . . . 47

3.5.1 CellposeSAM . . . 47

3.5.2 Swin UNETR . . . 48

3.5.3 nnU-Net . . . 49

4 Experiments and Results **51**

4.1 Evaluation Framework . . . 51

4.1.1 Metric Selection . . . 52

4.1.2 Fair Comparison Framework . . . 55

4.2 Ablation Study . . . 56

4.2.1 Inference Pre-processing Selection for All-Pre-processing Models . . . 56

4.2.2 CellposeSAM Training and Inference Selection . . . 60

4.2.3 Preliminary MicroSAM Large Inference-Setting Check . . . 67

4.3 Per-Model Family Analysis . . . 68

4.4 Image-Specific Analysis . . . 70

4.4.1 Qualitative Image-Level Error Analysis . . . 71

4.5 Final Model Selection and Deployment Recommendation . . . 78

5 Conclusion and Outlook **81**

5.1 Conclusion 81

5.2 Limitations 82

5.3 Outlook 84

Bibliography **87**

A Prototype Software Workflow **93**

List of Figures

1.1 Representative Airyscan image and ground-truth volume 2

2.1 BRP-shortcherry bouton signal and Kenyon-cell channel 6
2.2 DA1 projections and mushroom body calyx 7
2.3 Calycal microglomeruli schematic 7
2.4 Airyscan and non-pre-processed LSM slice comparison 10
2.5 Microscopy image for segmentation concepts 14
2.6 Semantic and instance segmentation 14
2.7 Simplified U-Net architecture 17
2.8 StarDist radial-distance representation 19
2.9 Cellpose flow representation 20
2.10 Transformer encoder overview 22
2.11 Single-head and multi-head attention 23
2.12 Vision Transformer principle 24
2.13 Swin UNETR architecture 25
2.14 Segment Anything Model architecture 26

3.1 Close but separate bouton instances and corresponding mask 32
3.2 Intensity differences between bouton instances and corresponding mask . . . 33
3.3 Large complex bouton and borderline annotated instance 34
3.4 Distant puncta subclusters assigned to one test bouton 34
3.5 Extended training annotation for low intensity subclusters 35
3.6 Acquisition related high intensity artifact without annotation mask 36
3.7 Calyx region and marker based exclusion cases 36
3.8 Bouton like axonal signal and valid annotated slice 37
3.9 Two stage annotation workflow 39
3.10 MicroSAM training data flow 46

4.1 CellposeSAM raw Brain2 failure mode for XY-trained inference 60
4.2 CellposeSAM raw Brain2 failure modes for XYZ-trained inference 60
4.3 Whole-region CellposeSAM 2D and XYZ 2D comparison 62
4.4 Slice-level CellposeSAM 2D and XYZ 2D comparison 63
4.5 Brain1 one-sided 3D CellposeSAM comparison 65
4.6 Brain1 PSF-deconvolved 3D CellposeSAM comparison 66
4.7 Two-slice analysis of the disconnected Brain1 bouton prediction 67
4.8 MicroSAM Large three-dimensional prediction examples 72
4.9 MicroSAM Large two-dimensional slice examples 73
4.10 Additional MicroSAM Large two-dimensional examples 73
4.11 MicroSAM Base three-dimensional prediction examples 74
4.12 MicroSAM Base two-dimensional examples 75
4.13 Imaris brightness bias and volume mismatch 76
4.14 Imaris prediction for close boutons 77
4.15 Imaris slice-level error examples 77

List of Tables

3.1 Dataset distribution by input representation 30

4.1 Evaluation metrics used for instance-level bouton segmentation. 54
4.2 IoU-threshold sensitivity for all-pre-processing models 55
4.3 Pre-processing configurations used for fair comparison 55
4.4 All-pre-processing ablation for MicroSAM, SwinUNETR, and nnU-Net . . . 58
4.5 All-pre-processing ablation for CellposeSAM variants 58
4.6 CellposeSAM training and inference ablation across pre-processing settings. Best values are shown in bold and second-best non-tied values are underlined for the main comparison metrics. 61
4.7 Overall model comparison 69
4.8 Best-configuration image-specific metrics 70

Chapter 1

Introduction

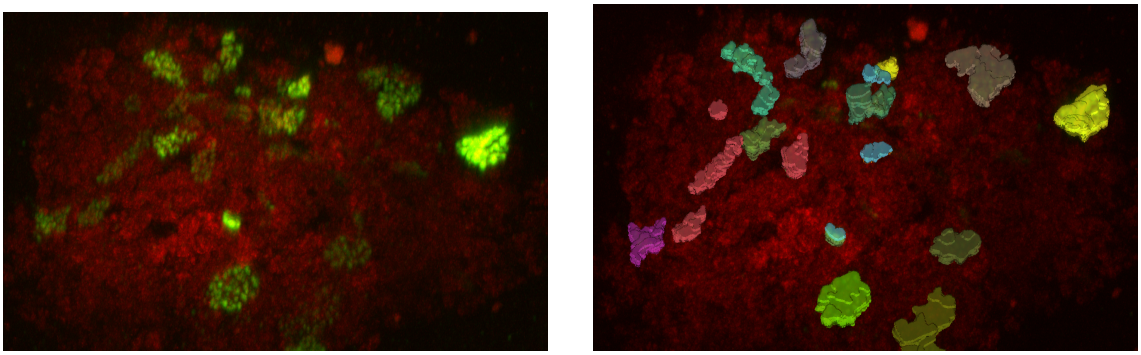
1.1 Motivation and Biological Background

Understanding the cellular and circuit mechanisms underlying learning and memory consolidation remains an active area of neuroscience research [1, 2]. The fruit fly, *Drosophila melanogaster*, serves as a highly tractable model organism for dissecting these processes [2]. Despite its small size, the *Drosophila* brain exhibits sophisticated behaviors and possesses anatomically distinct regions dedicated to sensory integration and memory, most notably the mushroom body [2, 3, 4]. Because many molecular and cellular pathways that govern neural development and synaptic function are evolutionarily conserved, studies in *Drosophila melanogaster* can provide insight into fundamental principles of nervous system organization and plasticity [2]. Moreover, *Drosophila* has proven valuable for modeling human neurological and neurodegenerative disease mechanisms, linking discoveries in the fly to processes relevant to human brain function and dysfunction [5, 6].

Recent advancements have revealed that memory is not merely a functional shift in neuronal firing, but can also involve structural plasticity and the reorganization of brain circuits [1, 7]. In the mushroom body calyx, learning and memory consolidation are accompanied by distinct structural reorganization [7]; the biological terms used in this paragraph, including the mushroom body calyx, projection neurons, Kenyon cells, microglomeruli, active zones, and boutons, are introduced in Section 2.1. Specifically, appetitive long-term memory consolidation has been associated with input-specific structural reorganization of the mushroom body calyx circuit, including an increase in microglomeruli number [7]. Baltruschat *et al.* identified odor-responsive microglomeruli by labeling DA1 projection-neuron boutons and synaptic active zones together with postsynaptic Kenyon-cell structures, then quantified structural properties such as microglomeruli number, projection-neuron bouton distribution, collateral axon length, and the volume occupied by DA1 boutons in the calyx [7]. These measurements illustrate the importance of scalable image analysis for this biological question, since researchers must localize the relevant region of interest and quantify the number and volume of synaptic-marker structures in 3D microscopy data. Mapping this physical rewiring provides valuable insight into how experience shapes neural circuits, but the required image analysis remains labor-intensive when regions and synaptic structures have to be selected, segmented, and cleaned manually.

1.2 Problem Statement

While the biological motivation for mapping these structural changes is clear, a central bottleneck lies in image analysis. The data considered in this thesis consists of 3D fluorescence microscopy stacks of the mushroom body calyx, where synaptic-marker structures have to be identified within a volumetric brain region. Figure 1.1 illustrates this setting using representative three-dimensional Airyscan image and ground-truth views. In the current collaborator workflow, researchers use Imaris to interactively define regions of interest and generate surface-based masks, a workflow consistent with Imaris' support for surface creation, masking, and manual correction in 3D microscopy data [8, 9]. However, such semi-automatic workflows can still require manual parameter choices, point-and-click interaction, and subsequent correction of imperfect masks. This limits scalability and makes the analysis sensitive to user-dependent decisions, especially when visually similar surrounding structures, such as axons, have to be excluded from the final segmentation.



(a) Original Airyscan image (b) Ground-truth annotations

Figure 1.1: Representative three-dimensional Airyscan visualization of the original image volume and the corresponding ground-truth bouton annotations in the mushroom body calyx. The example illustrates the volumetric nature of the target structures and the need to identify individual bouton candidates within a dense image region.

1.3 Limitations of Existing Segmentation Approaches

Image segmentation is the computational task of assigning pixels or voxels to meaningful regions or objects in an image. Existing segmentation approaches address parts of the bouton quantification problem, but each carries limitations when applied to 3D bouton analysis in this specific imaging domain. Classical pipelines based on thresholding, morphological filtering, and watershed-based separation are interpretable and straightforward to apply [10, 11, 12], but are sensitive to intensity variation, background signal from surrounding axonal processes, and manually chosen parameters. Projection-neuron boutons in the mushroom body calyx present a particular challenge because the relevant fluorescence signal does not form smooth, uniformly filled objects. Instead, it arises from a collection of discrete BRP-shortcherry-positive presynaptic active-zone puncta [7, 13], producing non-convex, lobulated shapes with internal intensity variation. Classical methods therefore tend to fragment individual bouton instances, produce false positives from visually similar background structures, or misclassify partial signal as independent objects. The Imaris-based workflow used for data collection follows the semi-automatic seed detection, region

growing, surface reconstruction, and manually adjusted thresholding strategy described by Kremer *et al.* [13]; however, it remains operator-dependent, sensitive to user-defined parameters, and difficult to scale to larger collections of volumetric image stacks.

Supervised deep-learning methods reduce some of these limitations by learning image features directly from annotated data, but require representative ground-truth annotations that are costly to produce for specialized 3D microscopy data and may not transfer reliably across imaging conditions or biological preparations [14, 15]. More recent foundation-model approaches, most notably the Segment Anything Model (SAM), offer interactive and promptable segmentation that can reduce the need to train a task-specific model from scratch [16]. Microscopy adaptations such as MicroSAM further improve applicability to biological image data [17]. Nevertheless, general applicability is not the same as domain-specific optimization. Applied without adaptation, these models have not been exposed to the irregular active-zone puncta morphology of the target structures, the anisotropic resolution of the confocal z-stacks used in this study, or the specific fluorescence characteristics of the imaging preparation. A further constraint is that no publicly available annotated dataset of this structure exists, meaning that supervised adaptation requires generating ground-truth annotations from scratch, a process that is both time-intensive and domain-specific.

1.4 Goal of This Thesis

To address these limitations, this thesis develops and evaluates a domain-adapted 3D segmentation pipeline for the automated quantification of projection-neuron boutons in the mushroom body calyx. Four instance segmentation models, MicroSAM, Cellpose 3D, Swin UNETR, and nnU-Net, are fine-tuned on a custom annotated dataset of confocal fluorescence volumes acquired using the biological preparation and extraction workflow established in prior structural plasticity studies [7, 13]. A dedicated pre-processing and post-processing pipeline is developed to handle the specific signal characteristics of BRP-shortcherry-labeled bouton structures. All four models are systematically benchmarked against each other and against the Imaris-based workflow currently employed by the collaborating biology group, using object-level detection metrics as well as bouton count and volume estimation accuracy as evaluation criteria. The goal is to determine whether domain-adapted deep-learning segmentation can provide a reliable and scalable alternative to the existing semi-manual quantification workflow.

1.5 Thesis Outline

Chapter 2 reviews related work on deep learning-based segmentation in 2D and 3D microscopy as well as foundation-model approaches such as SAM and its microscopy adaptations. Chapter 3 describes the proposed method, including the dataset, model configuration, and the quantification procedure. Chapter 4 presents the experimental evaluation, results, and deployment recommendations. Chapter 5 summarizes the main findings, discusses limitations, and outlines directions for future work.

Chapter 2

Related Work

This chapter reviews the biological and technical work needed to position the segmentation problem addressed in this thesis. The goal is not to provide a broad neurobiology survey, but to explain why projection-neuron bouton segmentation matters, why the corresponding microscopy data are difficult, and which computer vision methods are most relevant for the proposed pipeline and benchmark. General architectural concepts such as convolutional networks and transformers are therefore kept brief; the focus lies on biomedical and microscopy segmentation methods that are directly connected to bouton counting and volume estimation.

2.1 Biological Context of Bouton Quantification

Drosophila melanogaster has served as one of the most productive model organisms in biological research for more than a century. Its short generation time of roughly ten days, large offspring numbers, low maintenance cost, and extensive genetic toolkit make it suitable for experiments that would be difficult to perform at comparable scale in vertebrate systems [2, 18]. Its relevance is not limited to insect biology. Approximately 75% of human disease-associated genes have a functional homologue in the fly, and many cellular pathways involved in neural development, synaptic function, and disease mechanisms are conserved between flies and mammals [5, 19, 20]. Consequently, *Drosophila* has become an important *in vivo* model for human neurological and neurodegenerative disease research, as well as for studying the circuit mechanisms underlying learning and memory [2, 5, 6]. A key enabler of circuit-level imaging in this system is the GAL4/UAS binary expression system [21], which allows specific neuronal populations to be targeted and visualized with fluorescent markers. In the experimental context of this thesis, such genetic tools are used to visualize presynaptic regions in the mushroom body (MB) neuropil, providing the biological basis for the bouton segmentation task.

Within the *Drosophila* brain, the mushroom body (MB) is a paired brain structure and region of dense neural tissue that has been studied extensively as a major center for associative learning and olfactory memory [2, 3, 22]. The mushroom body is populated by Kenyon cells (KCs), which are the intrinsic neurons of the mushroom body and serve as a key neuronal substrate for memory formation and retrieval [22]. Sensory input reaches the mushroom body via olfactory projection neurons (PNs), which are specialized relay neurons that transmit odor information from the antennal lobe, the primary olfactory

processing center of the fly brain, to the mushroom body [22, 23, 24]. The calyx forms the main input region of the mushroom body, where incoming PN axonal terminals make synaptic contact with KC dendritic claws [23, 25]. This arrangement makes the calyx a biologically relevant region for studying how sensory input is represented and how synaptic structures are modified during learning and memory consolidation.

Within the calyx, PN axonal terminals and KC dendritic claws form discrete synaptic units known as microglomeruli [7, 13, 23, 25]. Each microglomerulus can be understood as a presynaptic PN bouton associated with multiple postsynaptic KC dendritic claws, forming an anatomical unit of olfactory information transfer into the mushroom body [23, 24, 25]. Figure 2.1 shows this marker context in an image crop: the BRP-short.mCherry, or BRP-shortcherry, bouton signal is displayed in green, while the Kenyon-cell-associated channel is displayed in red. Prior studies have used such structures to investigate structural plasticity in the mushroom body calyx [7, 13]. Kremer *et al.* analyzed long-term structural changes at mushroom body input synapses, while Baltruschat *et al.* linked appetitive long-term memory consolidation to circuit reorganization in the calyx [7, 13]. Their analyses used quantitative readouts such as microglomeruli number, bouton distribution, active-zone number, terminal area, and volume measures of DA1-positive microglomeruli or bouton containing calycal regions [7, 13]. These readouts motivate the present segmentation task because the image analysis pipeline must not only identify signal rich regions, but must separate individual bouton instances with sufficient consistency to support counting and volume estimation.

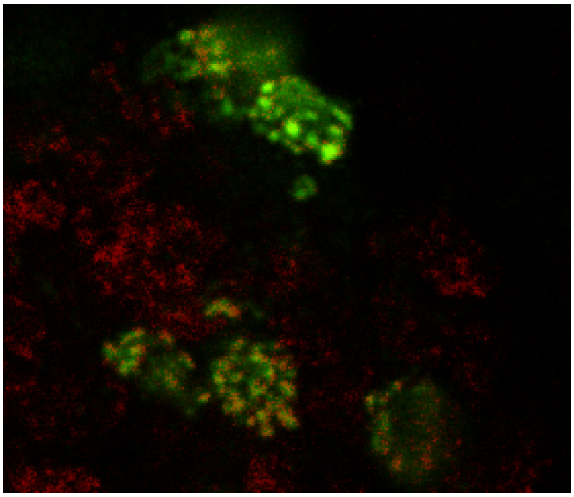


Figure 2.1: Representative two-channel crop showing presynaptic bouton signal together with Kenyon-cell context. In this pseudocolored display, BRP-short.mCherry, also referred to as BRP-shortcherry, is a shortened Bruchpilot construct fused to mCherry; it is shown in green and labels presynaptic active-zone puncta within projection-neuron boutons. The Kenyon-cell-associated channel is shown in red and provides the surrounding postsynaptic/anatomical context; the red channel was reduced by 30% for display balance.

The calyx is not a simple, sharply delineated object in the image. It is a region of dense neural tissue in which KC dendrites and PN inputs form distributed and selectively overlapping connectivity patterns [24]. Identifying and delineating individual microglomeruli within this region is non-trivial, as the structures lack sharp boundaries in the fluorescence signal and their detection requires careful parameter tuning even in semi-automated work-

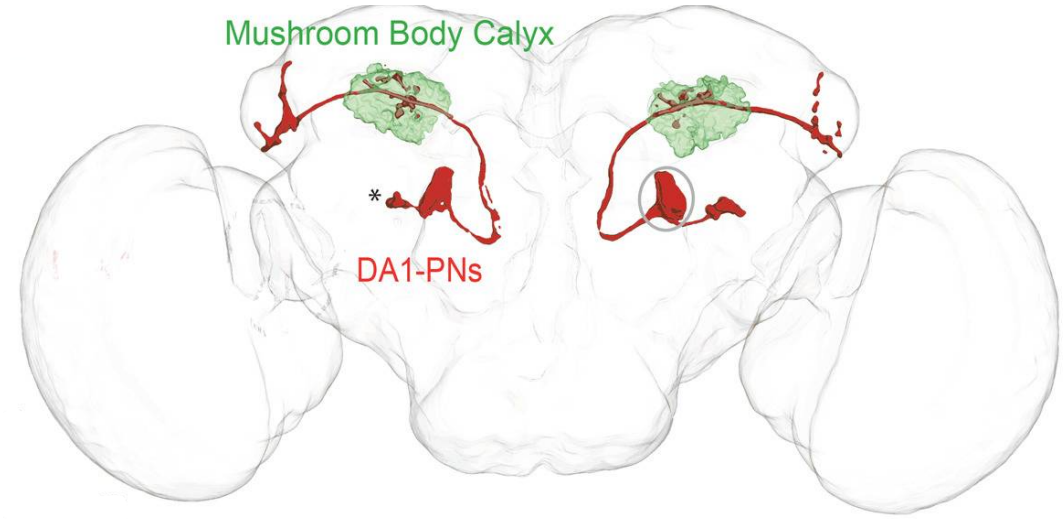


Figure 2.2: Anatomical position of DA1 projection-neuron projections and the mushroom body calyx in the *Drosophila* brain. The calycal regions are shown in green and DA1 projection-neuron structures in red, illustrating the brain region in which bouton quantification is performed. Adapted from Figure 1B of Baltruschat *et al.* [7].

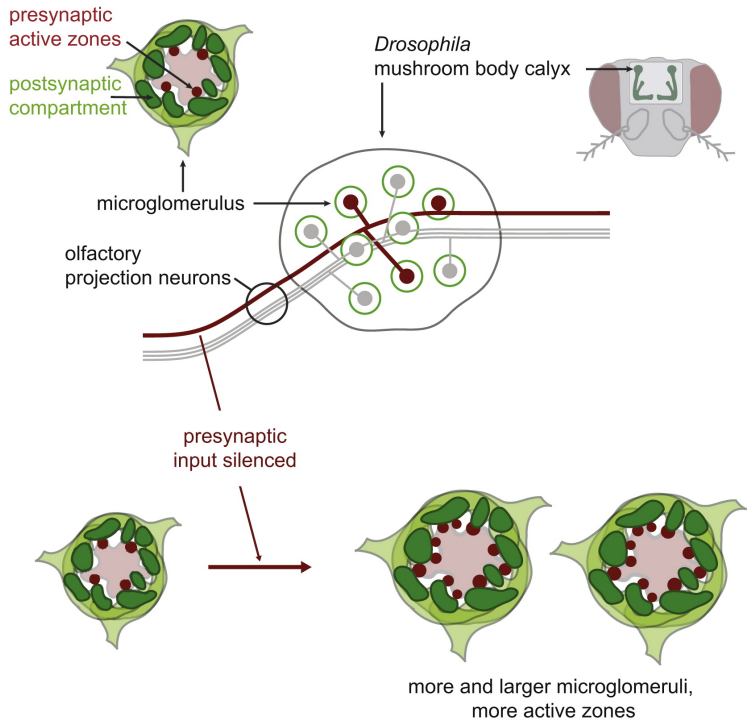


Figure 2.3: Schematic representation of calycal microglomeruli. Presynaptic projection-neuron boutons containing active zones are associated with postsynaptic Kenyon-cell dendritic claws, and structural plasticity can alter microglomerular size and active-zone number. Adapted from the graphical abstract of Kremer *et al.* [13].

flows; in the workflow described by Kremer *et al.*, automated detection recovered more than 30% of manually identified microglomeruli with a low false-positive rate [13]. In this experimental system, presynaptic active zones are visualized using BRP-short.mCherry, also referred to as BRP-shortcherry, a genetically targeted truncated fragment of the active-zone scaffold protein Bruchpilot fused to the red fluorescent protein mCherry; this thesis uses BRP consistently when referring to the Bruchpilot marker and the corresponding fluorescent signal [7, 13]. Because Bruchpilot localizes to presynaptic active zones, the resulting BRP/mCherry-positive signal appears as clusters of presynaptic active-zone sites visualized in the 568 nm red fluorescence channel and is used here as a proxy for tracking presynaptic structural changes [7, 13]. Although the microglomerulus as a whole comprises both pre- and postsynaptic components, the present thesis focuses specifically on the presynaptic PN axonal bouton, which is directly visible in this channel as a cluster of BRP-short.mCherry-positive active-zone puncta, where *puncta* refers here and throughout the thesis to small, locally bright spots in the fluorescence signal that arise from individual or closely spaced presynaptic active-zone sites. The number and volume of these bouton structures serve as the primary quantitative readouts of synaptic plasticity [7, 13]. The practical details of how the calyx region and relevant channel are selected for the dataset are described in the dataset section of Chapter 3; here, the important point is that the biological target already imposes non-trivial object-definition constraints on the segmentation problem.

At the level of the segmentation target, a PN bouton does not correspond to a homogeneous filled sphere. The BRP-positive active-zone signal appears as clustered puncta, so a bouton-related object can contain several bright local maxima, weaker connecting signal, and internal intensity variation. A single bouton can contain several puncta, while visually similar puncta can also belong to neighboring boutons, axonal processes, or non-target background signal. Prior studies also use postsynaptic KC claw markers to relate presynaptic bouton signal to microglomerular organization [7, 13]. For image segmentation, this means that puncta are informative markers of synaptic structure, but they are not identical to the final object instances to be counted. A model must therefore learn when multiple puncta belong to a single bouton, when close puncta should be separated into different bouton instances, and when visually similar signal should be rejected as background or as a neighboring non-target structure. In practice, this ambiguity appears as close but separate puncta clusters, strong intensity differences between puncta, distributed sub-clusters within a large bouton, and weak or discontinuous signal between parts of the same biological bouton.

This biological context leads directly to the computer vision formulation used in the rest of the chapter. The goal is not merely to detect fluorescence, but to obtain instance level bouton objects whose number and volume can be used as quantitative biological readouts. The following sections therefore move from biological motivation and microscopy image characteristics to segmentation tasks, classical workflows, biomedical deep learning models, and volumetric evaluation criteria.

2.2 Fluorescence Microscopy and 3D Image Characteristics

Fluorescence microscopy is a light microscopy technique in which structures of interest are labeled with fluorescent markers and visualized by detecting the light they emit after excitation by a laser at a suitable wavelength [26]. Unlike brightfield microscopy, where

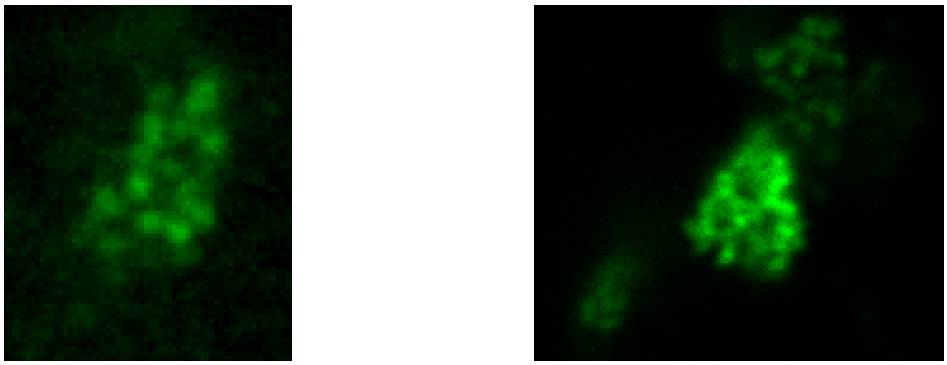
tissue components are observed through general transmitted-light contrast, fluorescence microscopy uses the spectral selectivity of labels to make targeted molecular or cellular structures visible against low background [26]. This selectivity is especially useful in neural tissue, where many cell types and subcellular compartments coexist in a densely packed volume [27]. In prior calyx studies, different neural structures were marked either by genetic fluorescent reporters or by antibody staining and imaged in separate fluorescence channels, allowing projection-neuron boutons to be inspected together with surrounding anatomical context [7, 13]. Confocal laser scanning microscopy (CLSM) is well suited to this type of data because a focused laser beam is scanned across the sample and fluorescence passing through a pinhole aperture is recorded, rejecting much of the out of focus light and enabling optical sectioning in thick fluorescent samples [26, 27, 28]. To form a three-dimensional image, the focal plane is stepped through the sample in the axial direction and a two-dimensional fluorescence image is recorded at each depth position. The resulting sequence of optical sections, commonly referred to as a z-stack, samples the volumetric distribution of fluorescent signal and forms the primary data format used in this thesis [27].

The imaging data considered in this thesis were acquired using two confocal modalities available on the ZEISS LSM 880 platform: standard CLSM and Airyscan. In standard CLSM, optical lateral resolution is governed primarily by wavelength and objective numerical aperture, while the lateral pixel size is set by the selected scan field, zoom factor, and number of pixels used for sampling [27]. Standard CLSM therefore provides volumetric fluorescence data through optical sectioning, but the finite point spread function still introduces blur at the scale of small subcellular structures, and no pixel reassignment or linear deconvolution step is inherent to the standard acquisition [29]. In the resulting z-stacks, each image element is a voxel with a defined physical size in x, y, and z [27]. This voxel geometry is important for segmentation because lateral and axial resolution are not equivalent: structures that are sharply separated in the image plane may appear blurred, elongated, or ambiguous across neighboring optical sections. In the data considered here, the relevant presynaptic bouton channel appears as clustered BRP-shortcherry-positive puncta, while additional channels provide anatomical context such as Kenyon-cell associated signal [7, 13]. The segmentation task is therefore not simply to detect bright pixels in one optical section, but to determine which punctate signals form a single bouton instance across neighboring slices while accounting for the different appearance of structures in lateral and axial directions.

Airyscan is a specialized detector module for the ZEISS LSM 880 that uses a 32-element GaAsP detector array, arranged in a compound-eye configuration, instead of the conventional single-pinhole, single-detector detection path [29, 30]. In super-resolution mode, the detector array samples the pinhole plane such that each element acts as a small sub-Airy detector, and the resulting channel signals are processed using pixel reassignment and linear deconvolution applied automatically during image reconstruction by the acquisition software [29, 31]. This reconstruction uses the additional spatial information captured by the detector array to improve signal-to-noise ratio and yields an approximately 1.7-fold resolution improvement in lateral and axial directions compared with standard CLSM [29, 30]. Because finer lateral sampling requires more pixels to cover a comparable field of view, Airyscan acquisitions produce larger image files than standard CLSM acquisitions, with practical consequences for memory use and computation in downstream image analysis.

The resulting acquisition-specific appearance difference is shown in Figure 2.4. In the Airyscan example, the bouton signal appears as a combination of several compact blob-like substructures, whereas the non-pre-processed LSM slice shows a more diffuse signal

with less clearly defined boundaries between neighboring blob-like regions.



(a) Airyscan slice, 1.5-fold brightness (b) Non-pre-processed LSM slice

Figure 2.4: Representative two-dimensional bouton-region slices from the Airyscan and non-pre-processed standard confocal LSM data. The Airyscan panel is displayed with a 1.5-fold brightness increase for visibility. In this example, the Airyscan bouton signal appears as several compact blob-like substructures, whereas the LSM slice shows more diffuse signal with no clear boundary between neighboring blob-like regions.

Beyond the modality-specific differences between standard CLSM and Airyscan, the resulting confocal z-stacks share two practical characteristics that are central for bouton segmentation. First, the data are anisotropic in resolution and sampling, so the same bouton can appear more continuous in the lateral plane than along the z-axis [26, 27]. Second, fluorescence intensity can vary across a stack: signal attenuation with depth, non-uniform illumination across the scan field, and background fluorescence from neighboring structures can change the contrast of bouton related signal and introduce visually similar non-target signal [26, 27]. These characteristics mean that a fixed global intensity threshold cannot reliably separate bouton related signal from background across the full imaging volume, and that segmentation approaches must account for both three-dimensional structure and spatial variation in signal intensity.

2.3 Image Pre-processing for Microscopy Segmentation

Raw fluorescence microscopy images are not neutral inputs for segmentation. They can contain acquisition-related artifacts and variability, including background fluorescence, noise, optical blur, intensity differences across the field of view or imaging depth, and differences in pixel or voxel spacing between acquisitions [26, 27, 32]. If these effects are passed directly to a segmentation model or classical computer vision workflow, the same structure type may appear at different intensities or with different boundary sharpness across images acquired under different conditions, reducing the consistency and reliability of downstream analysis. Pre-processing therefore denotes image processing operations applied before semantic or instance segmentation, with the aim of reducing artifacts, compensating for blur, or bringing sampling and image dimensions into a common format across datasets [26, 33, 34]. These operations are not segmentation methods in themselves, but they shape the image representation from which downstream algorithms identify object boundaries, local maxima, and intensity contrasts.

Intensity normalization is a basic pre-processing step that maps raw image values from their original numeric range to a standardized output range, typically the continuous interval from zero to one. Raw digital microscopy images are often stored as integer arrays whose possible values depend on the bit depth and acquisition settings of the imaging device; for example, an 8-bit image can contain values between 0 and 255, whereas a 16-bit image can contain values between 0 and 65535. Directly using these raw values as input to image processing algorithms or deep learning models can lead to inconsistent behavior across acquisitions, because absolute fluorescence intensities depend on acquisition parameters and do not necessarily have the same meaning between images. Normalization reduces this dependence by rescaling intensities to a fixed range, making image representations more comparable across bit depths and acquisition conditions [35]. A common robust variant is percentile based normalization, in which a low percentile, such as the first percentile, and a high percentile, such as the 99th percentile, are mapped to zero and one, respectively, instead of using the absolute minimum and maximum. Values outside this percentile range are typically clipped. This reduces the influence of extreme outliers or saturated pixels while preserving the relative intensity structure of the image [35]. More broadly, pre-processing and intensity standardization are important considerations in deep learning based medical image analysis [36].

Background subtraction is a complementary pre-processing step for images with slowly varying illumination, out of focus haze, or broad autofluorescent signal. Rolling ball background subtraction estimates a smooth local background by treating the intensity image as a surface and conceptually rolling a spherical structuring element underneath that surface; the resulting background estimate is then subtracted from the original image [37]. The ball radius determines the spatial scale that is interpreted as background, so it should be larger than the expected foreground object scale when the goal is to preserve compact target structures while removing broad background variation [37].

Beyond intensity normalization and background correction, two restoration-oriented pre-processing operations are commonly applied to address these characteristics in fluorescence microscopy pipelines. The first is image sharpening, in which the image is processed to amplify structures within a specific size range while suppressing slowly varying background and high-frequency noise, producing an output in which target structures appear more prominent relative to their surroundings [35]. The second is deconvolution. In image formation, convolution describes how the true object signal is spread by the response of the imaging system, commonly represented by a kernel or point spread function; deconvolution refers to the inverse restoration problem of estimating a sharper underlying signal from the observed, blurred image when this response is known or approximated [26, 35]. Both approaches are particularly relevant to pipelines that target subcellular structures such as presynaptic active-zone puncta, where the signal of interest occupies only a small number of voxels and is easily obscured by blur or background contributions.

Difference of Gaussians (DoG) filtering is a scale-selective image sharpening operation that subtracts two Gaussian-smoothed versions of the same image computed at different scales, thereby suppressing slowly varying background while retaining structures whose spatial extent lies between the two smoothing scales [35, 38, 39]. The direct output of this subtraction is a band-pass detail image rather than the final sharpened image. If I denotes the original image and G_{σ_1} and G_{σ_2} are Gaussian smoothing kernels with $\sigma_2 > \sigma_1$, the DoG detail component can be written as $D = G_{\sigma_1} * I - G_{\sigma_2} * I$. This component contains positive values at bright structures that are stronger at the smaller scale than in the broader background estimate, negative values around opposite intensity transitions, and

values close to zero in approximately homogeneous regions. To obtain the final sharpened image, the detail component is added back to the original image as $I_{\text{sharp}} = I + \alpha D$, where α controls the sharpening strength, and the result is typically normalized to a consistent intensity range before downstream processing [35]. The connection between Gaussian scale space filtering and blob enhancement is rooted in classical work on Gaussian derivative image operators [38, 39]. DoG filters have been widely applied as pre-processing and detection steps in fluorescence microscopy pipelines that target small spatially localized structures such as synaptic puncta, where the signal of interest is embedded in spatially diffuse background fluorescence [40]. The scale parameters of the filter must be chosen to match the expected size of the target structures: overly small scales amplify noise, whereas overly large scales suppress the target signal together with the background.

Convolution is a mathematical operation that describes how a signal is transformed when passed through a system: the output is the result of the input signal being spread and mixed according to the system response function, known as the kernel [35]. Deconvolution refers to the inverse problem of recovering the original input signal from the convolved output, given knowledge or an estimate of the kernel [35]. In fluorescence microscopy, the optical point spread function acts as the convolution kernel, and deconvolution is applied to recover a sharper estimate of the underlying fluorescence distribution from the blurred observed image [26].

Richardson–Lucy deconvolution is an iterative method for solving this inverse problem under a Poisson noise model, which is commonly used for photon-limited fluorescence imaging [41, 42]. Starting from an initial estimate of the underlying image, the algorithm updates that estimate by comparing the observed image with a forward-convolved version of the current estimate, thereby seeking an image that better explains the observed data. The method originates from independent work on Bayesian iterative image restoration and on rectifying observed distributions [41, 42]. In fluorescence microscopy, Richardson–Lucy deconvolution can improve apparent contrast and the separation of nearby structures when an appropriate point-spread function or approximation is available. However, because the method is iterative, noise can be amplified and ringing artifacts can appear around bright structures if the number of iterations is not controlled. In practice, the iteration count therefore has to balance improved sharpness against noise amplification [26].

Image resampling and interpolation are pre-processing operations applied when datasets differ in pixel spacing, voxel spacing, or field-of-view size. A common motivation for resampling in multi-modal imaging pipelines is to make datasets acquired at different resolutions comparable before processing. Resampling defines a new output grid with fewer pixels or voxels, and interpolation determines which intensity value each location on this grid should receive from the original image. Such operations are widely used in medical and biological image processing for resizing, registration, and standardizing inputs across acquisition conditions [43].

Bilinear interpolation provides a smoother two-dimensional alternative for image-plane resampling. For each output pixel, it uses the four nearest pixels in the original image plane and computes a distance-weighted average, so closer pixels contribute more strongly than farther pixels [35]. For volumetric data, the corresponding three-dimensional extension is trilinear interpolation, in which a new voxel value is estimated from the eight neighboring voxels in the original volume. If only the lateral image grid is changed while the axial sampling remains fixed, a three-dimensional stack can also be resampled by applying bilinear interpolation independently to each two-dimensional slice and then stacking the

resampled slices back into a volume. When downscaling fluorescence volumes, any interpolation strategy changes the effective voxel size and can partially smooth small structures near the resolution limit, which has implications for their apparent size and detectability in downstream segmentation.

2.4 Segmentation Tasks in Biomedical Microscopy

Image segmentation is a central task in biomedical microscopy because many biological questions cannot be answered directly from raw intensity images. Microscopy data first provide spatial intensity measurements, but downstream analysis often requires a more structured representation: foreground regions, object boundaries, labeled compartments, or individual object identities. Segmentation provides this conversion from image data to analyzable biological structures by assigning labels to pixels in 2D images or voxels in 3D volumes [44]. This enables quantitative measurements such as area, volume, morphology, spatial distribution, and object count. In practice, biomedical microscopy segmentation is challenging because target structures may have weak contrast, blurred boundaries, heterogeneous intensity, touching or overlapping objects, imaging noise, and acquisition-dependent artifacts [26, 27, 45]. Segmentation therefore acts as the interface between visual microscopy data and computational quantification [36, 45].

Segmentation tasks can be broadly divided into two formulations that differ in what the output labels represent. Figure 2.5 shows an example microscopy image, while Figure 2.6 illustrates the difference between semantic and instance segmentation. In semantic segmentation, each pixel or voxel is assigned to a predefined class such as foreground or background, but no distinction is made between individual objects of the same class. The output is a class map in which all objects belonging to the same category share the same label, making it possible to determine where a structure type is present but not how many instances of it exist. Instance segmentation addresses this limitation by assigning a unique identity to each individual object, so that separate instances of the same class receive distinct labels even when they are adjacent or visually similar. The output is therefore an object map in which each detected structure can be addressed and measured independently. In the context of this thesis, where the biological readouts of interest are bouton count and bouton volume, semantic segmentation is insufficient because it cannot determine how many boutons are present or what volume each individual bouton occupies. The task therefore requires instance segmentation, in which each bouton is identified as a separate object with its own boundary, enabling both counting and per-object volume estimation from the same output.

Segmentation in fluorescence microscopy differs substantially from segmentation in natural images. Natural images often contain color, texture, shadows, and object-level context that provide strong cues for separating objects, and large-scale datasets such as ImageNet and COCO have supported the development of general purpose computer vision models [47, 48]. Fluorescence microscopy images differ fundamentally in their signal characteristics: structures are visible only through marker-specific emission, boundaries are often defined by the spatial extent of molecular labeling rather than by sharp visual edges, and intensity can vary across preparations, acquisition settings, and depth within a volume [26, 27]. These properties make segmentation in microscopy a domain-specific image analysis problem [45]. In 3D microscopy, anisotropic resolution further changes object appearance between lateral and axial directions, as discussed in Section 2.2. In addition, reference annotations require

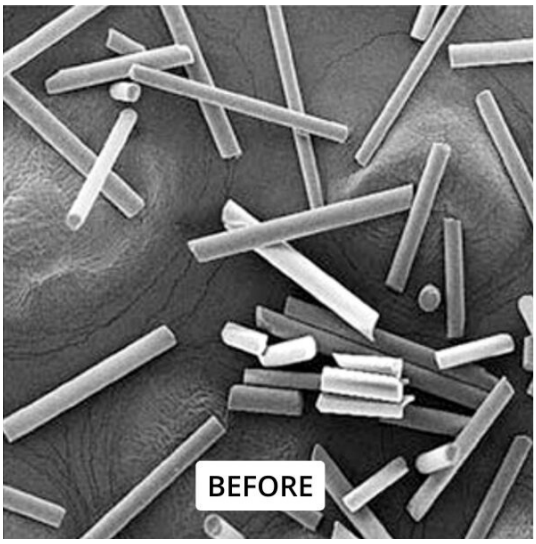


Figure 2.5: Example microscopy image used to illustrate segmentation concepts [46].

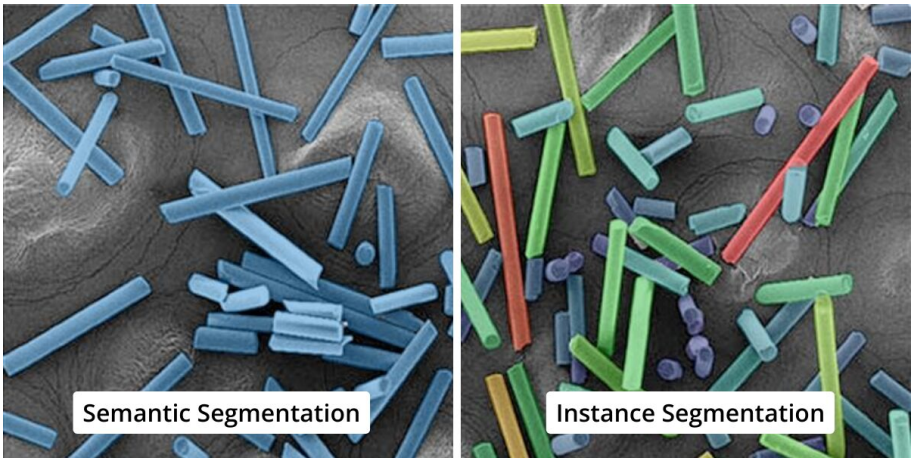


Figure 2.6: Illustration of semantic and instance segmentation in microscopy data [46].

domain expertise because annotators must decide which signals correspond to the target structure and which represent background, neighboring structures, or imaging artifacts [36, 45]. Together, these factors mean that direct transfer of models trained on natural images or generic biomedical datasets cannot be assumed and motivate domain-specific adaptation for specialized fluorescence microscopy tasks [17].

Classical segmentation pipelines for biomedical microscopy typically combine thresholding, morphological filtering, connected component analysis, and watershed based separation [10, 11, 12]. Thresholding partitions an image into foreground and background by comparing pixel or voxel intensities against a fixed or locally adapted cutoff value [10, 35]. Otsu's method selects such a threshold automatically by minimizing intra-class intensity variance, avoiding the need to manually specify a global threshold value [10]. Morphological operations such as erosion, dilation, opening, and closing are then used to refine binary masks by removing small noise objects, filling holes, and smoothing object boundaries [11]. Connected component analysis assigns unique labels to spatially contiguous foreground regions, providing a simple mechanism for extracting individual objects from a binary mask. Watershed based separation treats the image as a topographic surface and simulates flood-

ing from seed points, creating boundaries at intensity ridges between adjacent regions and enabling the separation of touching objects that share a common foreground mask [12]. These classical methods are attractive because they are interpretable and do not require annotated training data. Their computational cost is also typically low compared with deep learning models, since they rely on simple local, morphological, or region based image operations that can be implemented with standard image processing libraries. This makes them accessible in biological research settings where annotated datasets are scarce or unavailable.

However, classical segmentation methods depend strongly on image contrast, background uniformity, and manually chosen parameters, which limits their reliability when applied to fluorescence microscopy data with the characteristics described in Section 2.2 [45]. Thresholding methods can fail when intensity varies across the imaging volume or field of view, because a single global threshold cannot simultaneously capture weak target signal and suppress bright background or non-target fluorescence [26, 27, 35]. Morphological filtering can remove biologically relevant signal fragments when object boundaries are irregular or punctate rather than smooth and well defined [35, 45]. Connected component analysis merges structures that share foreground voxels, which is problematic when background fluorescence or weak connecting signal bridges the gap between adjacent target structures [26, 27, 35]. Watershed based separation can oversplit irregular structures when internal intensity variation creates spurious ridges within a single object [12, 35]. Furthermore, threshold values and the scale parameters of morphological operations often have to be re-tuned for new imaging conditions, preparations, or acquisition modalities, making such pipelines difficult to apply consistently across datasets [36, 45]. These limitations arise from a mismatch between the assumptions of classical methods and the signal properties of fluorescence microscopy data described in the preceding sections.

Interactive tools such as Imaris partially address these limitations by combining semi-automatic detection with manual parameter adjustment and correction, allowing expert users to compensate for cases where automated classical methods produce unreliable results [8, 9]. Kremer *et al.* established a workflow for quantifying presynaptic active-zone signal in the mushroom body calyx using Imaris based seed detection, surface reconstruction, and manually adjusted thresholds, and this workflow has been applied in subsequent structural plasticity studies of the same preparation [7, 13]. Such expert guided workflows are valuable for biological analysis where the number of samples is small and manual correction is feasible. However, they introduce user-dependent variability, require repeated parameter decisions for each image volume, and do not scale to larger datasets [45]. These constraints motivate data driven segmentation approaches that learn domain specific features directly from annotated examples and can be applied consistently and automatically across large collections of volumetric image data, which are described in the following sections.

2.5 Deep Learning for Biomedical Segmentation

Deep learning shifted biomedical image segmentation from manually designed image processing rules toward trainable feature representations learned from annotated examples. Instead of relying on fixed thresholds, handcrafted filters, or explicit object shape assumptions, neural networks can learn image features that are useful for predicting dense pixel or voxel labels directly from data [49]. For biomedical microscopy, the key advantage is that such models can learn appearance patterns specific to a modality or marker, including

weak boundaries, heterogeneous intensities, and contextual cues that are difficult to encode manually [36]. This makes deep learning especially suitable for segmentation tasks where classical pipelines require repeated parameter tuning or fail under changes in staining, acquisition settings, or object morphology [36, 45].

Convolutional neural networks (CNNs) are widely used deep learning architectures for visual data because their convolutional structure is well suited to images, where informative patterns often recur across spatial locations. By exploiting this spatial regularity, deep convolutional networks have driven major progress in image and video processing and have become a standard architectural starting point for many visual analysis tasks [49]. CNNs apply learned filters across the spatial dimensions of an input to detect local patterns such as edges, intensity gradients, and texture structures. Stacking convolutional layers allows CNNs to build increasingly abstract representations: earlier layers typically capture low-level local features, whereas deeper layers capture higher-level semantic information [49]. To process large inputs and enlarge the effective receptive field, CNNs often reduce spatial resolution through downsampling operations such as max pooling or strided convolution [49]. Max pooling retains the strongest activation within a local neighborhood, whereas strided convolution applies filters with a step size greater than one, producing a smaller feature map. Downsampling reduces memory requirements and computational cost while allowing deeper layers to integrate information over larger regions of the input. A key development in CNN design was the introduction of residual connections, in which the input to a block of layers is added directly to its output; this improves gradient propagation and enables the training of much deeper networks [50]. Together, convolutional feature extraction, hierarchical abstraction, downsampling, and residual connections form core building blocks on which many modern biomedical segmentation models are built.

Early CNN architectures were commonly designed for image level classification, with fully connected layers mapping features extracted by convolutional layers to a fixed-size output such as a single class label for the whole image [49, 51]. This design is suitable for classification but not for segmentation, where spatial structure must be preserved and a label has to be estimated for each pixel or voxel. CNNs therefore had to be adapted from image level prediction to dense prediction [51]. Fully convolutional networks (FCNs) were an important step because they replaced the fully connected classification layers of earlier CNNs with convolutional prediction and upsampling layers, allowing the network to produce spatial label maps and enabling end-to-end pixel-wise segmentation of inputs of arbitrary size [51]. A foundational architecture for biomedical image segmentation is U-Net, which was designed for tasks requiring both semantic context and precise spatial localization, such as cell segmentation in microscopy images [14]. Its architecture consists of a contracting encoder path and an expanding decoder path connected by skip connections, giving the network its characteristic U-shaped structure. The encoder progressively reduces spatial resolution through repeated convolution and downsampling operations, extracting increasingly abstract feature representations at each scale, while the decoder restores spatial resolution through upsampling. At each scale, skip connections concatenate decoder features with the corresponding high-resolution encoder features, reintroducing spatial detail that would otherwise be lost during downsampling and enabling more accurate localization of object boundaries [14]. Figure 2.7 summarizes this encoder–decoder structure. For volumetric microscopy and medical imaging, 3D U-Net extended this principle from pixel-wise to voxel-wise prediction by using three-dimensional convolutions and sparse volumetric annotation, enabling models to incorporate spatial context across adjacent slices [15]. U-Net and its variants have since become a central architectural paradigm for biomedical image

segmentation and form the basis of several models evaluated in this thesis [36, 52].

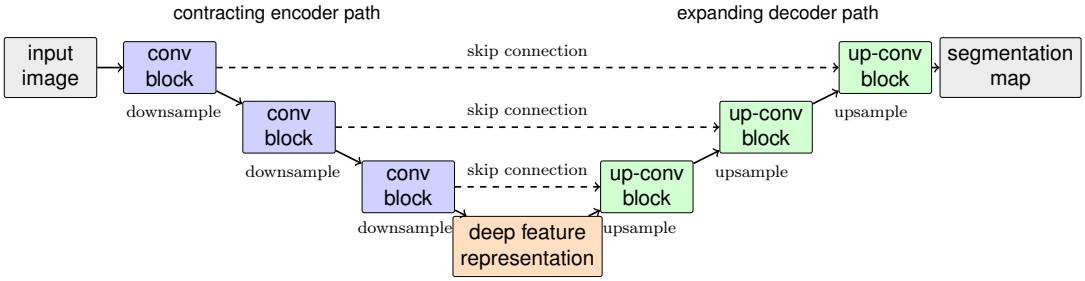


Figure 2.7: Simplified U-Net architecture. The encoder extracts increasingly abstract feature representations while reducing spatial resolution, the decoder restores spatial resolution, and skip connections transfer high-resolution encoder features to support precise localization. Own schematic based on Ronneberger *et al.* [14].

These architectural developments, including convolutional feature extraction, dense prediction through fully convolutional design, encoder–decoder structure with skip connections, and volumetric extension through 3D convolutions, collectively define the design space from which modern biomedical segmentation models are drawn. The models evaluated in this thesis each build on or extend this foundation in different ways: some adapt the learning objective to produce instance level output directly, some replace the convolutional encoder with a transformer based alternative to capture longer-range spatial dependencies, and some leverage large scale pretraining to reduce the annotation burden associated with supervised training on specialized biological data. These approaches provide the basis for the model specific adaptations and evaluations described in the following sections.

2.5.1 nnU-Net

nnU-Net, introduced by Isensee *et al.* [52], extends the U-Net framework described above by addressing a persistent practical limitation of supervised deep learning segmentation: the need for extensive manual configuration of pre-processing, network architecture, and training decisions for each new dataset. Rather than proposing a new architectural paradigm, nnU-Net contributes a self-configuring pipeline that analyzes the properties of a given dataset through dataset fingerprinting, in which the median voxel spacing, image dimensions, intensity distribution, and class frequencies are extracted to characterize the segmentation task [52]. This fingerprint drives three categories of configuration decisions: fixed parameters that remain constant across all datasets, such as the U-Net architecture template and the combination of Dice and cross-entropy losses; rule-based parameters derived directly from the fingerprint, including patch size, network depth, normalization strategy, and batch size; and empirical parameters that are learned from the training data itself [52]. Based on this analysis, nnU-Net automatically configures and trains up to three U-Net variants, consisting of a 2D U-Net, a 3D full-resolution U-Net, and a cascaded 3D U-Net in which a low-resolution network first segments the full volume and a second network subsequently refines the predictions at full resolution.

Each configured model is trained using five-fold cross-validation for 1000 epochs per fold, together with a comprehensive data augmentation pipeline that includes random rotations, scaling, elastic deformation, gamma correction, and mirroring [52]. This training regime makes the full nnU-Net workflow computationally demanding, because several folds and,

depending on the dataset, several candidate configurations may need to be trained before the final model selection step [52]. After training, nnU-Net automatically evaluates possible ensembles of the trained configurations and selects the combination with the best validation performance, so that model selection and ensemble construction require little manual intervention [52]. The network produces semantic voxel-wise foreground predictions rather than instance level outputs directly; recovering individual object instances from these predictions therefore requires a post-processing step whose appropriate form depends on the spatial arrangement of the target structures in the specific application domain.

In the context of this thesis, nnU-Net is relevant as one of the supervised segmentation baselines fine-tuned and evaluated on the custom annotated dataset of projection neuron boutons. As established in Section 2.4, semantic foreground predictions do not directly assign instance identities; an instance recovery step is therefore required before object counts and per-object volumes can be computed. If predicted bouton regions are separated in the foreground mask, connected component analysis can convert semantic predictions into instance labels without an additional learned instance separation mechanism. This assumption is reasonable for the annotated bouton objects considered here, which are treated as separate non-overlapping instances even though their fluorescence signal can be irregular and close to neighboring structures, as described in Section 2.1. nnU-Net therefore serves as a supervised semantic segmentation baseline that tests how far self-configuring voxel-wise segmentation, followed by connected component recovery, can support bouton counting and volume estimation on the target dataset.

2.6 Instance Segmentation Models for Microscopy

The deep learning architectures described in Section 2.5, including U-Net and 3D U-Net, are predominantly semantic segmentation frameworks whose original formulations produce pixel-wise class predictions without distinguishing between individual objects of the same class [14, 15]. A foreground and background label map is useful for identifying where target signal is present, but it does not assign separate identities to objects of the same class. Instance level output must therefore be recovered by an additional grouping step, such as connected component analysis for spatially separated foreground regions or watershed based separation for touching objects [12]. The limitation of this semantic mask plus grouping strategy is that object identity has to be reconstructed from spatial connectivity, distance transform peaks, or intensity valleys, which can be unreliable when neighboring objects touch or when boundaries are weak [53].

These limitations motivate approaches that predict object specific information in addition to foreground probability, so that instance recovery is guided by learned instance-relevant representations rather than reconstructed from a semantic mask alone [53, 54]. The following subsections describe two models that are relevant to the segmentation pipeline evaluated in this thesis and that each address these limitations through a distinct combination of learning objective, output representation, and instance recovery strategy [53, 54, 55].

2.6.1 StarDist

StarDist was introduced by Schmidt *et al.* as an instance segmentation method for fluorescence microscopy that addresses the limitations of semantic mask based instance recovery

described above [53]. Its central contribution is the shift from predicting only where foreground signal is present to predicting an instance aware representation from which individual objects can be recovered more directly [53]. This changes the role of post-processing: instead of reconstructing object identities from connectivity, distance transform peaks, or intensity valleys alone, post-processing selects among learned object candidates.

Mechanically, StarDist uses a U-Net based backbone to extract image features and branches into two output heads: one predicting an object probability map and one predicting radial distance maps, with one distance channel for each predefined angular direction [53]. These radial distances are supervised from annotated instance masks during training: for each pixel inside a ground truth object, the target distances are computed as the actual distances from that pixel to the object boundary along the fixed radial directions, and the network learns to predict these distances from the image data. Figure 2.8 illustrates how these distances define a star-convex object candidate. At inference time, the predicted distances and object probabilities together form a large set of overlapping star-convex object candidates, from which final instance labels are recovered by ranking candidates according to their object probability and applying non-maximum suppression to retain high confidence candidates while suppressing overlapping alternatives [53]. The 3D extension generalizes the same idea from polygons to star-convex polyhedra, using radial distances along directions distributed over the unit sphere so that instance candidates can be predicted and filtered directly in volumetric microscopy data [55].

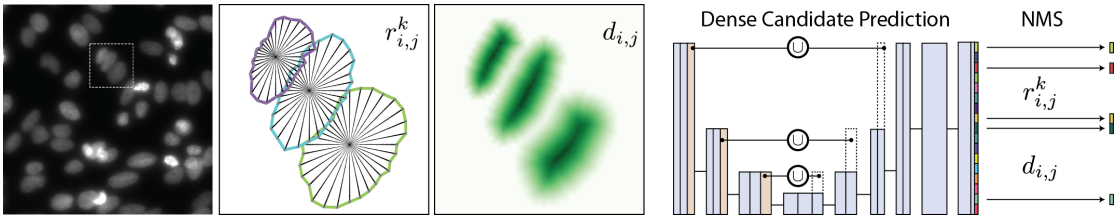


Figure 2.8: Illustration of the StarDist radial-distance representation. For a candidate pixel, distances are measured along predefined radial directions until the object boundary is reached; the resulting distances define a star-convex polygon that can be ranked by object probability and filtered by non-maximum suppression. Adapted from Schmidt *et al.* [53].

StarDist is included in this review as a foundational reference for learned geometric instance prediction in fluorescence microscopy, rather than as an evaluated model. It showed that modifying the learning objective so that the network predicts instance specific geometric information directly can be an effective strategy, providing a useful reference point for comparison with later instance segmentation methods such as Cellpose [53]. The star-convex representation is well suited to compact objects such as cell nuclei, but it also imposes a geometric constraint: the object boundary must be reachable from a candidate center along straight radial directions. This constraint becomes limiting when objects are elongated, branched, weakly connected, or internally fragmented [53]. In the context of this thesis, the constraint is relevant because the bouton target described in Section 2.1 is not a homogeneous filled body; BRP-shortcherry-labeled boutons appear as clustered active-zone puncta with weak connecting signal, internal intensity variation, and ambiguous boundaries against neighboring axonal fluorescence. This mismatch between the star-convexity assumption and the actual bouton morphology motivates comparison with approaches that encode object geometry less explicitly.

2.6.2 Cellpose

Cellpose was introduced by Stringer *et al.* [54] as an instance segmentation method that also learns an instance-specific output representation, but replaces StarDist's explicit star-convex boundary parameterization with a learned vector flow representation. In this sense, it follows the same general idea highlighted above for StarDist, namely making instance recovery part of the network output, while relaxing the specific requirement that each object boundary be describable by radial distances from a candidate center. Instead of predicting radial distances to a star-convex boundary, Cellpose predicts a cell-probability map and a gradient flow field whose vectors guide foreground pixels toward the center of their corresponding object. Instance masks are then recovered by following the predicted flows: foreground pixels are iteratively displaced along the vector field and grouped according to their convergence points. This removes the fixed star-convex boundary parameterization, although the learned flow field still encodes object shape through the training masks [54].

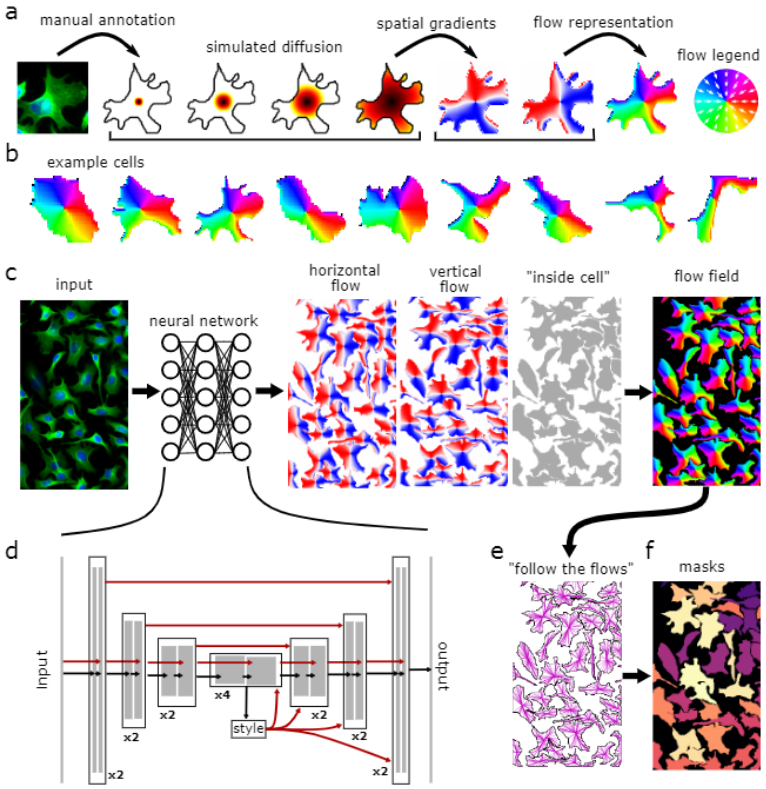


Figure 2.9: Illustration of the Cellpose flow representation. The model predicts a foreground probability map and spatial flows that move pixels or voxels toward the center of their corresponding object; pixels or voxels whose trajectories converge to the same region are grouped into one instance. Adapted from Stringer *et al.* [54].

Mechanically, Cellpose uses a U-Net based backbone, as described in Section 2.5, to produce three output channels: the horizontal and vertical components of the gradient flow field and a cell probability channel [54]. Figure 2.9 illustrates these predicted outputs and the resulting flow based instance recovery principle. Ground truth instance masks are not used directly as supervision targets; instead, they are converted into flow-field targets through a simulated diffusion process. For each annotated object, heat is allowed to diffuse from the object center across the object interior, and the spatial gradients of the resulting heat distribution define smooth horizontal and vertical flow vectors that point toward the center

from each interior pixel [54]. These diffusion derived flow fields, together with a binary foreground map derived from the same masks, form the training targets against which the network predictions are compared during loss computation. The network therefore learns to predict, from raw image features, the flow structure that diffusion would produce from the ground truth mask, so that instance flows can be estimated for unseen images at inference time without requiring manual annotations.

The three-dimensional processing mode of Cellpose has remained architecturally consistent since its introduction in the original publication: flows are computed independently on the three mutually orthogonal planes of the input volume, specifically the XY, XZ, and YZ planes, and the resulting two-dimensional flow predictions are averaged into a consensus three-dimensional flow field from which instances are recovered through gradient integration in the volumetric domain [54, 56, 57]. Since confocal z-stacks are anisotropic, with axial slice spacing typically larger than lateral pixel size as discussed in Section 2.2, the volume is resampled along the z-axis before processing to match the lateral pixel spacing; without this step, object proportions would appear distorted on the XZ and YZ planes and the flow predictions from the three orthogonal views would be inconsistent with each other. This orthogonal plane mechanism is inherited unchanged by CellposeSAM, which is described in Section 2.7.3. Cellpose 3 introduced a distinct contribution by addressing a persistent limitation of generalist segmentation models in microscopy, namely degraded image quality arising from noise, blur, or undersampling [57]. Rather than training a restoration network to minimize pixel reconstruction error independently of the segmentation task, Cellpose 3 chains a trainable image restoration network with a pre-trained Cellpose segmentation network and optimizes the restoration network to minimize the segmentation loss directly, so that the restored image is specifically adapted to support downstream segmentation rather than to general perceptual reconstruction [57].

2.7 Foundation Models and Microscopy Adaptation

The models described in the preceding sections are trained for specific segmentation tasks on task specific annotated datasets, with architecture, loss function, and training configuration adapted to the properties of the target domain [36]. Foundation models follow a different paradigm: they are models trained on broad data at scale and adapted to a wide range of downstream tasks, for example through fine-tuning, prompting, or other lightweight adaptation strategies [58].

In image segmentation, this paradigm is attractive because large-scale pretraining can provide transferable visual representations, while adaptation to a downstream task can reduce the amount of task specific annotation and training needed compared with developing a fully supervised model from scratch [16, 58]. Promptable foundation models for image segmentation, such as SAM, extend this further by enabling segmentation through user provided prompts, reducing the need for task specific retraining for each new application [16]. However, generality does not imply domain reliability: foundation models are not optimized for fluorescence microscopy, and their pretraining data typically differs substantially from confocal z-stacks in signal characteristics, resolution, object morphology, and labeling conventions, meaning that direct application without domain specific adaptation cannot be assumed to produce reliable quantitative segmentation in specialized biomedical settings.

2.7.1 Transformer Encoders and Self-Attention

The transformer architecture was introduced by Vaswani *et al.* [59] for sequence modeling in natural language processing, where long-range dependencies had traditionally been handled by recurrent architectures that process inputs sequentially and can struggle to retain information across long distances. The transformer replaces sequential recurrence with attention, which is a mechanism that models relationships between all elements of a sequence simultaneously, enabling parallel computation and direct long-range dependency modeling independent of the distance between sequence elements [59]. A complete transformer encoder block combines multi-head self-attention with a position-wise feed-forward network, using residual connections and layer normalization around both components to stabilize training and preserve gradient flow [59]. Figure 2.10 summarizes this encoder structure schematically.

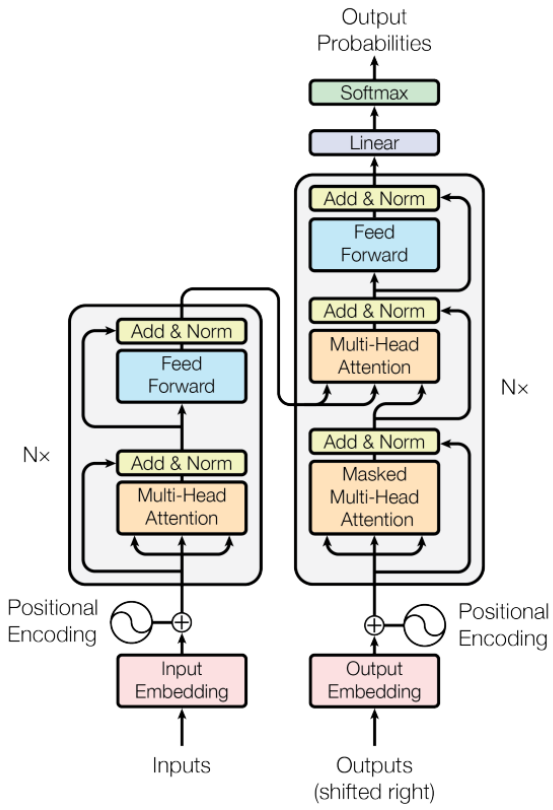


Figure 2.10: Schematic overview of a transformer encoder. Input tokens are combined with positional information and processed by repeated encoder blocks containing multi-head self-attention and feed-forward layers, enabling contextual feature representations to be learned from token relationships. Own schematic based on Vaswani *et al.* [59].

The core operation underlying the transformer is scaled dot-product attention, which computes a weighted combination of value vectors based on the compatibility between query and key vectors [59]. Queries, keys, and values are learned linear projections of the input: each query represents an element seeking contextual information, each key represents an element that can be matched against a query, and each value contains the information to be aggregated. For query matrix Q, key matrix K, and value matrix V, scaled dot-product

attention is defined as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{2.1}$$

where d_k is the dimensionality of the key vectors. The scaling factor $\sqrt{d_k}$ prevents the dot products from growing too large in magnitude, which helps stabilize the softmax weights [59]. The resulting attention weights express how strongly each query attends to each key, so that the output for each element is a weighted aggregation of the corresponding values. In self-attention, which is the form used in transformer encoders, queries, keys, and values are all derived from the same input sequence, allowing every element to attend to every other element and capture contextual relationships across the full input. To allow the model to jointly capture information from multiple representation subspaces, scaled dot-product attention is extended to multi-head attention, in which the attention operation is applied in parallel across multiple independently learned projection subspaces and the outputs are concatenated and projected back to the original dimension [59]. Figure 2.11 contrasts single-head and multi-head attention schematically.

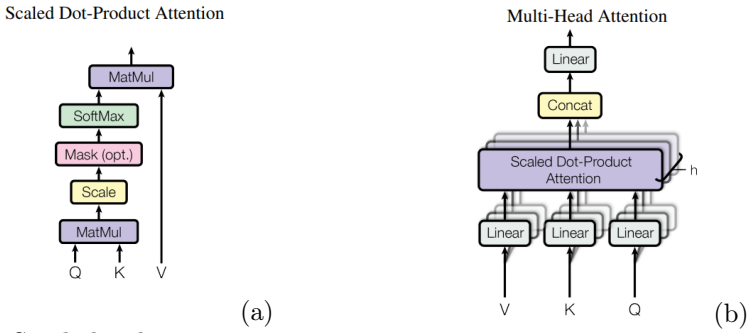


Figure 2.11: Schematic comparison of single-head and multi-head attention. Multi-head attention computes several attention operations in parallel on learned projection subspaces and combines their outputs, allowing different relationship patterns between tokens to be represented simultaneously. Own schematic based on Vaswani *et al.* [59].

Transformer based architectures have since been applied to vision models that treat image data as sequences. In the Vision Transformer, this is achieved by dividing the input into fixed-size spatial patches, projecting each patch into a token embedding, adding positional information, and processing the resulting patch token sequence with stacked transformer encoder blocks [60]. Figure 2.12 illustrates this patch-token representation. This formulation allows the model to capture dependencies between distant image regions more directly than convolutional architectures, whose receptive field grows only through stacked local operations [60]. These transformer and vision transformer principles form the architectural basis of the models described in the following subsections.

2.7.2 Swin UNETR

While the Vision Transformer demonstrates the viability of transformer based feature extraction for visual tasks, applying global self-attention directly to high-resolution images presents a practical challenge: the cost of computing attention between all patch tokens

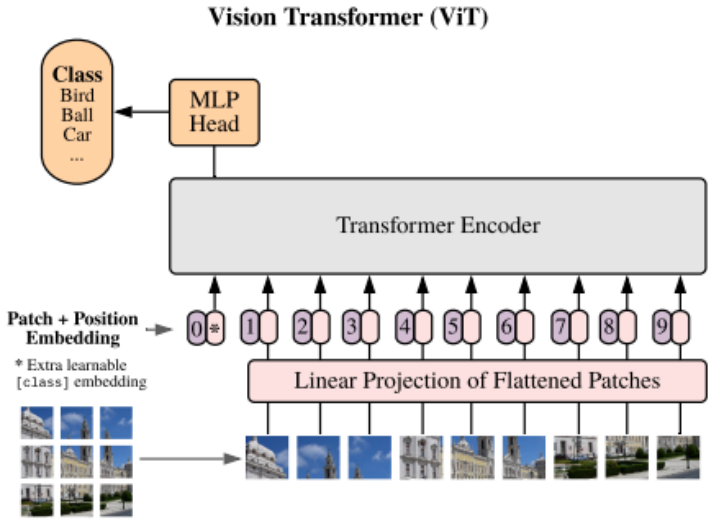


Figure 2.12: Schematic illustration of the Vision Transformer principle. The input image is divided into fixed-size patches, each patch is projected into a token embedding, positional information is added, and the resulting token sequence is processed by transformer encoder layers. Own schematic based on Dosovitskiy *et al.* [60].

scales quadratically with the number of tokens, making it expensive for dense prediction tasks such as segmentation [60, 61]. This challenge becomes even more pronounced when transformer encoders are adapted to volumetric biomedical data, where the number of spatial tokens can be large [62]. The original Swin Transformer addresses this image-level limitation by computing self-attention within non-overlapping local windows of fixed size rather than across the full image, which gives linear computational complexity with respect to image size for a fixed window size [61]. To enable information exchange across window boundaries, Swin Transformer blocks alternate between regular and shifted window partitions, allowing tokens to interact with neighboring regions across layers [61]. This hierarchical windowed design also produces multi-scale feature maps by merging neighboring patches between stages, allowing the encoder to capture both local structure and broader contextual information [61].

Swin UNETR, introduced by Hatamizadeh *et al.* [62], follows the U-Net design described in Section 2.5 but replaces the convolutional encoder with a Swin Transformer encoder, combining hierarchical windowed self-attention with the localization oriented encoder-decoder structure established as effective for dense biomedical prediction. The input volume is partitioned into non-overlapping three-dimensional patches, which are linearly projected to token embeddings and processed through four encoder stages, as illustrated in Figure 2.13 [62]. Within each stage, consecutive Swin Transformer blocks alternate between W-MSA, which computes attention within fixed local three-dimensional spatial windows, and SW-MSA, which shifts window positions between layers to introduce cross-window connectivity, with each attention operation followed by a position-wise feed-forward network with layer normalization and residual connections [62]. Patch merging operations between stages progressively reduce spatial resolution while increasing feature dimensionality, producing a hierarchy of representations at four scales that captures both fine-grained local structure and broader contextual information across the full volume. These multi-scale encoder features are passed to a residual convolutional decoder with skip connections that progressively restore spatial resolution to the full input dimensions, and the final output is a

voxel-wise semantic prediction map trained using a combined Dice and cross-entropy loss on annotated volumetric data [62].

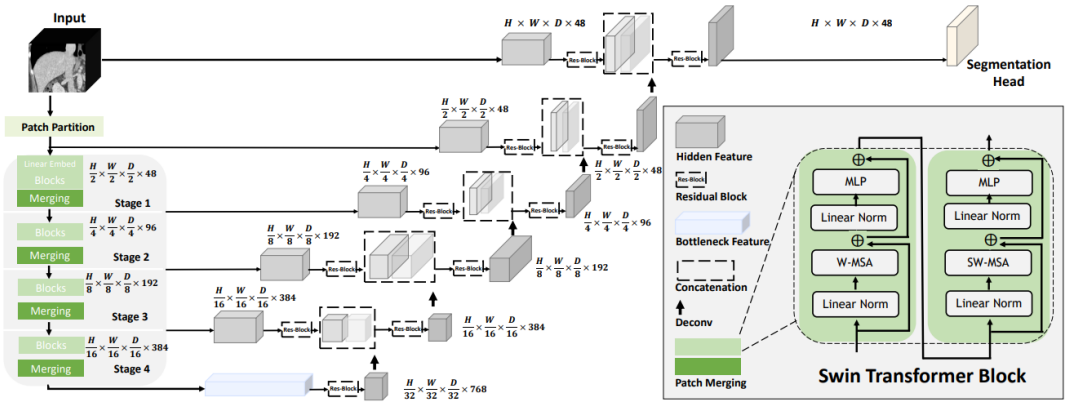


Figure 2.13: Swin UNETR architecture for volumetric semantic segmentation. The input volume is split into three-dimensional patch tokens and encoded by hierarchical Swin Transformer blocks with regular and shifted-window self-attention, while skip-connected decoder stages progressively restore spatial resolution to produce a voxel-wise segmentation output. Adapted from Hatamizadeh *et al.* [62].

2.7.3 Segment Anything Model

The Segment Anything Model (SAM) was introduced by Kirillov *et al.* [16] as a promptable foundation model for image segmentation, designed to address a broad range of segmentation tasks within a unified framework rather than being optimized for a single application domain. Its central contribution is a promptable segmentation paradigm: the model receives user-provided spatial prompts, such as foreground and background point clicks, bounding boxes, or existing masks, and predicts a segmentation mask for the indicated region without requiring task-specific retraining [16]. To support this general capability, SAM was trained on SA-1B, a dataset comprising more than one billion segmentation masks across eleven million images, assembled through a combination of manual annotation, semi-automatic annotation, and fully automatic mask generation [16]. This large-scale training provides SAM with broad visual representations that can transfer to diverse segmentation settings and reduce the annotation burden for new tasks, since users can obtain segmentation masks through interactive prompting rather than by providing dense pixel-level annotations for each new domain.

The SAM architecture consists of three components that operate in sequence: an image encoder, a prompt encoder, and a mask decoder, as shown in Figure 2.14 [16]. The image encoder is a Vision Transformer, as described in Section 2.7, that processes the full input image and produces a dense image embedding capturing the spatial feature representation at a fixed resolution; because this encoding step is computationally the most expensive component, it is performed once per image and the resulting embeddings can be reused across multiple prompt queries without re-encoding [16]. The prompt encoder maps user-provided prompts to embeddings that can be combined with the image representation: sparse prompts such as point clicks and bounding boxes are represented as positional embeddings combined with learned tokens indicating prompt type, while dense prompts

such as existing mask inputs are downscaled using convolutional layers and added elementwise to the image embeddings [16]. The mask decoder is a lightweight transformer based module that attends jointly to the image embeddings and prompt embeddings through cross-attention layers to produce one or more predicted segmentation masks together with associated confidence scores, enabling SAM to handle ambiguous prompts by predicting several plausible masks simultaneously rather than committing to a single prediction [16]. This separation of the heavy image encoding step from the lightweight prompt encoding and mask decoding makes SAM well suited to interactive segmentation workflows, where a fixed image can be segmented interactively in near real time by varying the prompts without re-encoding.

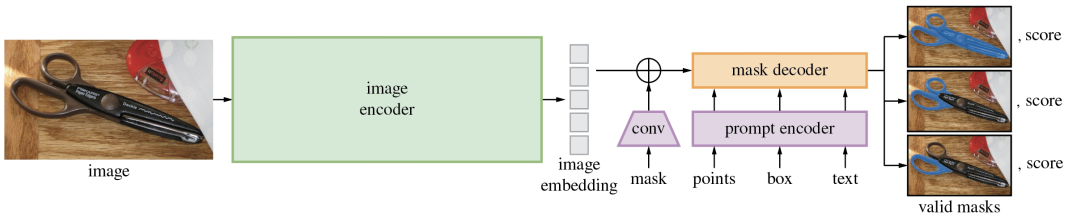


Figure 2.14: Schematic overview of the Segment Anything Model architecture. An image is encoded once into an image embedding, user prompts are encoded separately, and a lightweight mask decoder combines both representations to predict candidate masks with confidence scores. Adapted from the Segment Anything project diagram [16].

MicroSAM

MicroSAM, introduced by Archit *et al.* as Segment Anything for Microscopy [17], directly addresses the limitations of applying SAM to microscopy data by fine-tuning the SAM framework on a curated dataset of over 17,000 microscopy images spanning diverse imaging conditions, biological structures, and sample preparation methods. This domain specific fine-tuning produces two independently trained generalist model variants that share the same SAM based architecture but are trained on entirely separate domain-specific datasets: one for light microscopy, which covers fluorescence and brightfield imaging of cells and tissues, and one for electron microscopy, which covers subcellular structures imaged at nanometer scale resolution [17]. Both variants retain the SAM encoder-decoder architecture and the promptable segmentation paradigm described above, so that the model can still be guided through point, box, or mask prompts for interactive annotation, while the domain-specific fine-tuning substantially improves segmentation quality for microscopy structures compared with applying the original SAM weights directly.

In addition to the interactive prompting workflow inherited from SAM, MicroSAM introduces a dedicated automatic instance segmentation mode, referred to as AIS, which extends the model with an additional segmentation decoder trained jointly with the fine-tuned SAM model [17]. This decoder predicts three outputs at each pixel: a foreground probability indicating whether the pixel belongs to an object, and normalized distance maps encoding the distance to the nearest object centroid and to the nearest object boundary respectively [17]. These predictions are combined as input to a watershed algorithm that uses the distance and foreground information to delineate individual object instances without requiring any manual prompts. This approach is distinct from the automatic mask generation mode of the original SAM, which places a regular grid of point prompts and merges

resulting mask candidates through non-maximum suppression [16]. For three-dimensional volumetric data, MicroSAM extends this by segmenting each slice individually in two dimensions and merging the resulting instance labels across slices based on object overlap, producing volumetric instance segmentations without requiring a three-dimensional model architecture [17].

CellposeSAM Extension

CellposeSAM, introduced by Pachitariu and Stringer [63], combines the architectural contributions of SAM and Cellpose by replacing the U-Net based image encoder of Cellpose, described in Section 2.6.2, with the Vision Transformer image encoder of SAM, described in Section 2.7.3. The flow-based decoder, diffusion derived training targets, gradient integration, and instance recovery mechanism established in Section 2.6.2 remain entirely unchanged. The motivation for this encoder substitution is that the SAM ViT encoder was pretrained on a substantially larger and more diverse dataset than any microscopy-specific training corpus, and its learned representations therefore encode richer and more generalizable visual features than a U-Net encoder trained from scratch or fine-tuned on domain-specific data alone [63]. This combination retains the instance segmentation quality of the Cellpose flow approach while substantially improving generalization to new imaging domains where the target structures differ from standard training distributions.

Chapter 3

Methods

3.1 Datasets

This section describes the microscopy data, dataset composition, and annotation procedure used for model training and evaluation. For details of the underlying biological context of the segmentation target, refer to Section 2.1.

3.1.1 Microscopy Acquisition

All imaging data used in this thesis were acquired on a ZEISS LSM 880 laser scanning confocal microscope using an oil immersion objective with a numerical aperture of 1.4 and an immersion oil refractive index of 1.518. The microscopy image stacks for both the Airyscan and standard confocal LSM modalities were acquired by Pallavi Syam, a doctoral researcher in the Tavosanis laboratory at RWTH Aachen University, and were provided for the computational analysis conducted in this thesis. The biological preparation and bouton analysis context are grounded in related Tavosanis laboratory studies of the *Drosophila* mushroom body calyx [7, 13]. Imaging was performed on fixed whole mount *Drosophila melanogaster* brains in which projection neuron axonal boutons were labeled with BRP-short.mCherry, as described in Section 2.1. This made the boutons visible as discrete fluorescent puncta in the red channel using Alexa Fluor 568, with emission collected around 603 nm. A second Alexa Fluor 488-labeled channel, with emission collected around 519 nm, captured Kenyon cell dendritic structures for anatomical context. Two confocal acquisition modalities available on the LSM 880 platform were used across different brain samples: Airyscan super resolution mode, which uses the integrated 32-element GaAsP detector array [31], and standard confocal laser scanning microscopy, which uses the conventional single pinhole detection path, both described in Section 2.2.

Airyscan acquisitions were collected at a native lateral voxel size of 0.0425×0.0425 µm per pixel and an axial step size of 0.3 µm per optical section, yielding images of 1834×1834 pixels per slice. Pixel reassignment and deconvolution based processing were performed in the ZEISS ZEN software during Airyscan reconstruction, improving signal to noise ratio and lateral resolution. Standard confocal acquisitions were collected at a lateral voxel size of 0.0709×0.0709 µm per pixel and an axial step size of 0.3 µm, yielding images of 1100×1100 pixels per slice without automated reconstruction. Both modalities share the same axial spacing, producing a z-to-lateral anisotropy ratio of approximately 7.1 for

Airyscan and 4.2 for standard confocal, which is relevant to all downstream processing steps. To ensure consistent spatial sampling across modalities during training and inference, Airyscan volumes were downsampled from their native resolution to 1100 × 1100 pixels using bilinear interpolation, matching the lateral voxel size to 0.0709 µm per pixel.

3.1.2 Dataset Composition

The dataset used in this thesis is derived from 13 volumetric image stacks acquired from the available brain preparations. These stacks did not arrive as a final thesis specific ground truth dataset. Training and validation labels were annotated by the author to support model fitting and model selection, whereas the held out test ground truth was defined by Pallavi Syam according to biological criteria after reviewing and editing the test annotations. Masks generated by the author or model predictions for the test volumes were therefore treated only as support material, not as authoritative biological ground truth. At the biological volume level, these stacks are split into training, validation, and test partitions of nine, two, and two volumes respectively. Six volumes were acquired using the Airyscan modality and seven using standard confocal laser scanning microscopy. Both modalities are represented in each partition. For model input construction, each Airyscan volume contributes one reconstructed original input, whereas each standard confocal LSM volume contributes its original input plus two additional pre-processed inputs: one Difference-of-Gaussians (DoG) sharpened version and one Richardson–Lucy (RL) deconvolved version. These two pre-processing operations are introduced conceptually in Section 2.3 and their thesis specific implementations are specified in Section 3.2. The training partition therefore contains 4 Airyscan inputs and 5 LSM originals plus 2 additional inputs for each of the 5 LSM volumes, giving $4+5+2\times5=19$ training inputs. The resulting distribution of input representations is summarized in Table 3.1. The biological split was fixed using a manual seed to ensure reproducibility and fair comparison.

Table 3.1: Dataset distribution by partition and stored input representation. Airyscan volumes contribute one reconstructed original input, while each standard confocal LSM volume contributes original, DoG sharpened, and RL deconvolved inputs.

Partition	Airyscan original	LSM original	LSM DoG	LSM RL	Total inputs
Training	4	5	5	5	19
Validation	1	1	1	1	4
Test	1	1	1	1	4

The validation and test volumes were intentionally selected rather than sampled at random from the available image collection, because each held out partition contains only two biological volumes. The goal was to make these small partitions as representative as possible by covering the broadest feasible range of bouton morphologies, signal conditions, and annotation challenges. Each held out partition therefore contains one Airyscan volume and one standard confocal volume. After input representation construction, this yields one Airyscan input, one LSM original, and two additional pre-processed LSM inputs for validation, giving $1+1+2\times1=4$ validation inputs; the same one plus three distribution is used for testing. This intentional partitioning makes the validation and test metrics more informative about performance across the relevant variability of the dataset than would be expected from a homogeneous or randomly selected subset of the same small

size. The split was fixed at the volume level before two-dimensional slice export and was stored in partition-specific training, validation, and test directories. Consequently, data loader shuffling during model fitting changed only the order or crop sampling of examples within a fixed partition; it did not randomly decide which brain volumes or rare bouton morphologies entered a given run. This design was used to reduce run-to-run variability and to avoid cases in which one training run contains specific difficult bouton structures while another otherwise identical run does not.

To make the voxel geometry consistent across acquisition modalities, all Airyscan volumes were resampled slice by slice from their native 1834×1834-pixel lateral grid to 1100×1100 pixels using bilinear interpolation with anti-aliasing, and the resampled slices were then stacked back into three-dimensional volumes, as introduced in Section 2.3. This operation preserves the physical field of view: the native Airyscan lateral extent is $1834 \times 0.0425 = 77.945$ µm, so resampling this extent to 1100 pixels yields a lateral pixel size of $77.945/1100 \approx 0.0709$ µm, matching the standard confocal pixel size within rounding. Because both modalities share the same axial step size of 0.3 µm, the downsampled Airyscan volumes and the standard confocal volumes both have voxel dimensions of $0.0709 \times 0.0709 \times 0.3$ µm^3 and a z-to-lateral anisotropy ratio of approximately $0.3/0.0709 \approx 4.23$. This pre-processing step ensures that all stored inputs in the dataset use consistent spatial sampling and anisotropy. It also remains compatible with the optical resolution limit: by the Nyquist–Shannon criterion [27], the sampling interval of 0.0709 µm corresponds to a minimum sampled feature size of $2 \times 0.0709 = 0.142$ µm, which remains below the approximate standard confocal lateral resolution of $0.51 \times 0.603/1.4 \approx 0.219$ µm at the 603 nm BRP-shortcherry emission wavelength. The tradeoff of this resampling step is that Airyscan specific high resolution lateral detail is not fully retained in the stored model inputs: fine structures sampled at the native 0.0425 µm pixel spacing may be smoothed or merged during downsampling. The results for Airyscan data should therefore be interpreted as performance on resolution matched Airyscan inputs rather than on the full native resolution acquisitions.

3.1.3 Dataset Selection and Annotation Rationale

A substantial part of the methodological work in this thesis consisted of defining what should count as an annotatable bouton instance and constructing dataset partitions that expose models to the resulting ambiguity. The purpose of this subsection is therefore not to provide a visual catalog of examples, but to document the annotation principles that guided dataset selection. The examples were chosen because they represent recurring or biologically important ambiguity classes in the data: instance boundary ambiguity, intensity and morphology variation, and exclusion of visually similar non-target signal. Examples from the test partition are used only to characterize the held out evaluation data and to document the annotation convention; they were not used for model fitting or parameter selection.

Annotation Target and Instance Boundaries

As established in Section 2.1, projection neuron boutons are not homogeneous filled objects with clear boundaries but are visible as clusters of BRP-shortcherry-positive presynaptic active zone puncta. The central annotation problem is therefore not merely to detect

puncta, but to decide which puncta constitute one biological bouton instance. Figure 3.1 illustrates one side of this problem: spatial proximity alone is insufficient because neighboring boutons can appear as a single cluster. The corresponding mask assigns separate instance identities to the three boutons, while the fluorescent structures in the upper part of the field remain unannotated because their three-dimensional extent was smaller than the minimum volume accepted for valid bouton instances in this dataset. This example was retained in the training partition because it simultaneously exposes the model to closely adjacent target instances and to visible non-target fluorescence below the operational bouton size range.

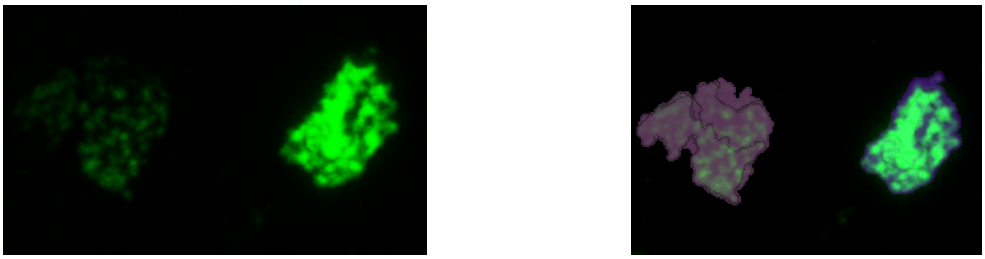
(a) Sample of raw image's 3D representation (b) Corresponding 3D instance mask

Figure 3.1: Representative three-dimensional crop from the training partition showing spatially proximate BRP-shortcherry-positive puncta clusters, while the annotation mask assigns separate instance identities to the corresponding boutons.

In addition to the proximity challenge introduced above, Figure 3.1 also shows fluorescent structures in the upper region that were excluded for an image specific reason: their reconstructed volume fell below the smallest valid bouton volume used in the annotation protocol. This makes the exclusion reproducible as a size based operational decision rather than as a purely subjective judgment about brightness or appearance. The example also introduces the broader principle that not all fluorescent signal present in an imaging volume corresponds to annotatable bouton instances. Further below, additional provisions are documented for other bouton like but non-target cases, including acquisition related high intensity artifacts, structures outside the calyx or without sufficient BRP-shortcherry marker evidence, and axon like BRP-shortcherry-positive signal.

Intensity variation across the fluorescence signal in a confocal z-stack, both between slices at different imaging depths and between structures present within the same slice, represents a fundamental challenge for instance segmentation models that must generalize reliably across the full volumetric extent of an image, as discussed in Section 2.2. Because the BRP-shortcherry-positive presynaptic active zone puncta signal can differ substantially in brightness between bouton instances even within the same optical section, a model trained predominantly on high intensity examples would risk suppressing weaker instances as indistinguishable from background fluorescence, while a detection sensitivity calibrated to recover dim instances would simultaneously risk treating background fluorescence elsewhere in the volume as spurious bouton signal. Figure 3.2 shows a representative training example in which two bouton instances are present in the same slice alongside one another, with the left instance exhibiting substantially weaker and more diffuse fluorescence than the clearly resolved high intensity puncta cluster of the right instance, directly instantiating this challenge in a single training image. This example is included in the training partition so that the model is exposed during training to the full range of per bouton fluorescence intensities present in the dataset and learns to detect valid instances regardless of their

absolute signal level.

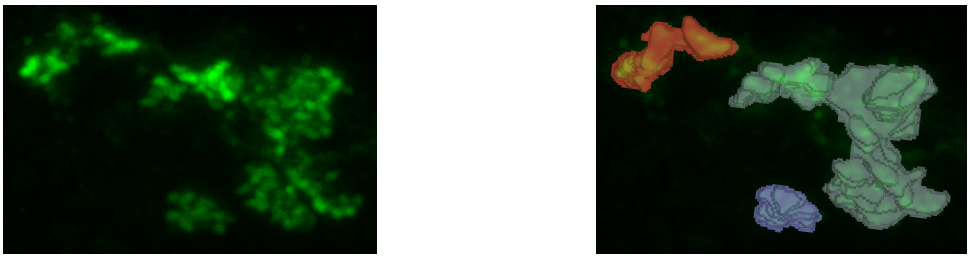


(a) Raw image (b) Corresponding instance mask

Figure 3.2: Representative training crop showing two annotated bouton instances with strongly different BRP-shortcherry fluorescence intensities. The corresponding mask shows that both the weaker, diffuse bouton and the brighter punctate bouton were included as target instances.

Large and morphologically complex boutons introduce a different failure mode because their puncta may appear as several spatially distributed subclusters rather than as one compact object. Figure 3.3 shows such a field, containing one markedly large projection neuron bouton instance alongside smaller compact instances. The large instance has an irregular, nonconvex puncta distribution that extends across a substantially greater spatial area than the typical compact cluster morphology described in Section 2.1. It occupies a manually measured annotation volume of approximately 19 µm^3, compared to the typical range of 8 to 9 µm^3 that characterizes the smaller instances visible in the same field. This example specific measurement is consistent with published ultrastructural measurements of *Drosophila* mushroom body calyx projection neuron boutons, for which Yang *et al.* report a volume range of 1.3–39.9 µm^3, a mean of 12.35 µm^3, and 80% of boutons between 10 and 30 µm^3 [64]. Because the puncta of a large bouton can appear as separated subclusters with gaps between them, the annotation regions for such instances were intentionally extended to encompass the full spatial extent of the bouton rather than being restricted to the individual puncta signal alone. This extension is motivated by the requirement for accurate bouton counting: if the subclusters of a single large bouton were annotated as separate instances, the resulting count would be inflated relative to the true number of boutons present, whereas a unified annotation that covers the full spatial extent of the large instance ensures that the downstream count remains correct and that the manual correction burden following automated detection is minimized.

The same example also illustrates an axon associated ambiguity that arises from the difference between two-dimensional training slices and the full three-dimensional biological interpretation. The red mask region in Figure 3.3(b) appears in the displayed two-dimensional slice like a compact punctate bouton candidate and is visually difficult to distinguish from nearby valid bouton instances. However, inspection of the neighboring slices showed that the upper part of this signal is connected to an axonal structure. Since axonal structures are not part of the target bouton definition, this region was not biologically interpreted as a true bouton. It was nevertheless retained as an annotated training instance because the model is trained slice by slice: if one visually similar compact structure is labeled as bouton while another is labeled as background for a reason only visible in three dimensions, the two-dimensional training signal becomes inconsistent. The annotation is therefore a conservative training label decision intended to avoid contradictory supervision, not biological confirmation that the region is a true bouton.



(a) Sample of raw image's 3D representation (b) Corresponding 3D instance mask

Figure 3.3: Representative sample showing a large, morphologically complex bouton instance and the corresponding instance mask.

Figure 3.4 shows a bouton instance from the test partition that was identified as a motivating case during the annotation process: the constituent puncta subclusters of this instance are separated by a spatial gap large enough to resemble two distinct objects under a strict connectivity criterion, yet they are biologically part of the same presynaptic terminal and are therefore assigned a single instance label whose mask extends across both subclusters and the intervening gap, consistent with the microglomerular structure described in Section 2.1. For visualization, the raw image panel is shown with a linear sixfold brightness increase so that the weak puncta remain visible in print, while the corresponding mask panel is left unadjusted. This case represents one of the most difficult instance definition problems in the dataset because no signal based criterion alone can distinguish between two genuinely separate boutons in close proximity and two subclusters of one bouton separated by a gap in their puncta distribution, and its resolution requires biological judgment about the correct instance boundary. This image belongs to the Airyscan test volume, which is one of only two test volumes and covers a wider range of morphological challenges than any single training volume, and it was therefore retained in the test partition rather than used for training despite the annotation challenge it presents.

(a) Airyscan test image crop (b) Corresponding instance mask

Figure 3.4: Airyscan test crop showing a bouton whose puncta appear as spatially separated subclusters. For visibility, the raw image panel is displayed at 6 times the original brightness, while the corresponding mask is unadjusted. The mask assigns both subclusters and the intervening low signal gap to one instance, preserving the biologically defined bouton identity.

The observation of this distant subcluster structure in the test partition motivated an intentional annotation decision when a structurally similar case was identified in a training volume, shown in Figure 3.5, in which the constituent puncta are distributed across both bright and dim spatial regions with low or absent signal between the brighter subclusters. Rather than restricting the annotation to the bright signal alone, which would risk splitting the instance into multiple separate detected objects and inflating the bouton count, the

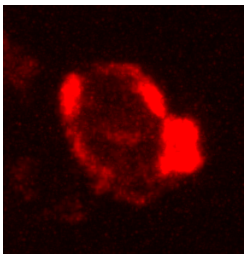
annotation mask was intentionally extended to cover the full spatial extent of the bouton including the dim intercluster regions, following a preference for false positive over false negative errors to preserve the single instance identity and maintain count accuracy. This extended annotation strategy additionally ensures that complex multiple subcluster instances remain spatially connected in the annotation and can be correctly merged during three-dimensional label reconstruction, and this image is included in the training partition so that the model learns to recognize spatially distributed subclusters as belonging to a unified instance rather than treating each bright region as an independent object.

(a) Low intensity training crop (b) Extended instance mask

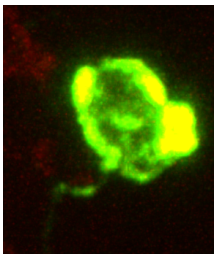
Figure 3.5: Training crop showing a structurally similar low intensity bouton in which the annotation mask is extended across dim intercluster regions so that spatially distributed puncta remain one connected instance.

The extended annotation strategy described above can therefore lead to apparent false positives or extra detected regions in later model predictions, because the model is encouraged to include dim intercluster areas around difficult boutons rather than restrict each instance to only the brightest puncta. This tradeoff was accepted intentionally: for the downstream counting task, preserving hard, spatially distributed boutons as complete single instances is more important than avoiding every additional candidate region. Extra detections can be removed during review or downstream filtering, whereas a missed or split complex bouton would directly compromise the biological count.

Confocal fluorescence microscopy can produce regions of anomalously high intensity that arise from acquisition related factors rather than from the presence of the biological target structure, as discussed in Section 2.2, and such regions can appear simultaneously across both imaging channels. Unlike the BRP-shortcherry-positive presynaptic active zone puncta that characterize annotatable bouton instances as described in Section 2.1, the spurious high intensity region shown in Figure 3.6 does not exhibit the discrete subcluster structure of a genuine bouton signal and was therefore excluded from annotation on the recommendation of the biological collaborator. The figure presents this region in both the BRP-shortcherry channel and the Kenyon cell channel without any annotation mask, so that the model is exposed during training to the appearance of anomalously bright non-target signal across both channels and learns to withhold predictions in such cases rather than treating high fluorescence intensity alone as sufficient evidence for a bouton instance. This example is included in the training partition to improve the robustness of the model to acquisition related intensity artifacts that may appear in previously unseen imaging volumes.



(a) BRP-shortcherry channel



(b) Kenyon cell channel

Figure 3.6: Representative training crop showing an anomalously bright acquisition related signal in both the BRP-shortcherry and Kenyon cell channels. No annotation mask is shown because the high intensity region was excluded from the target bouton annotations.

Figure 3.7 summarizes the calyx-region and marker-based exclusion cases used during training annotation.

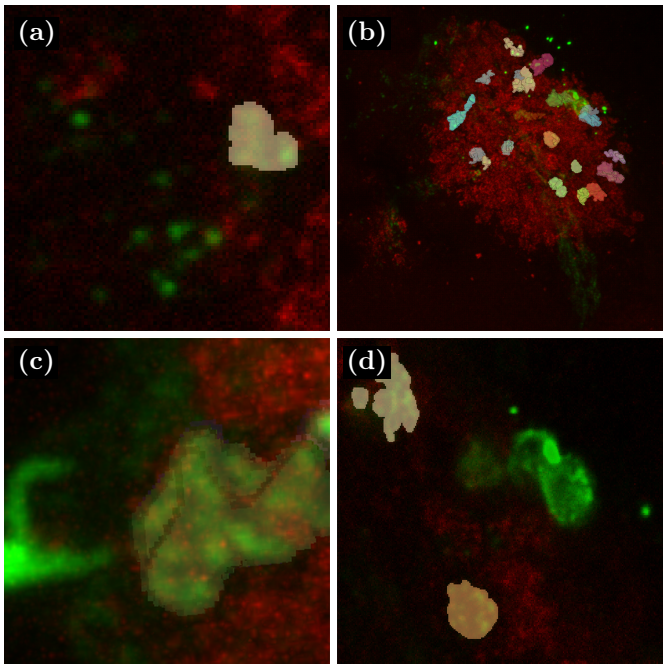
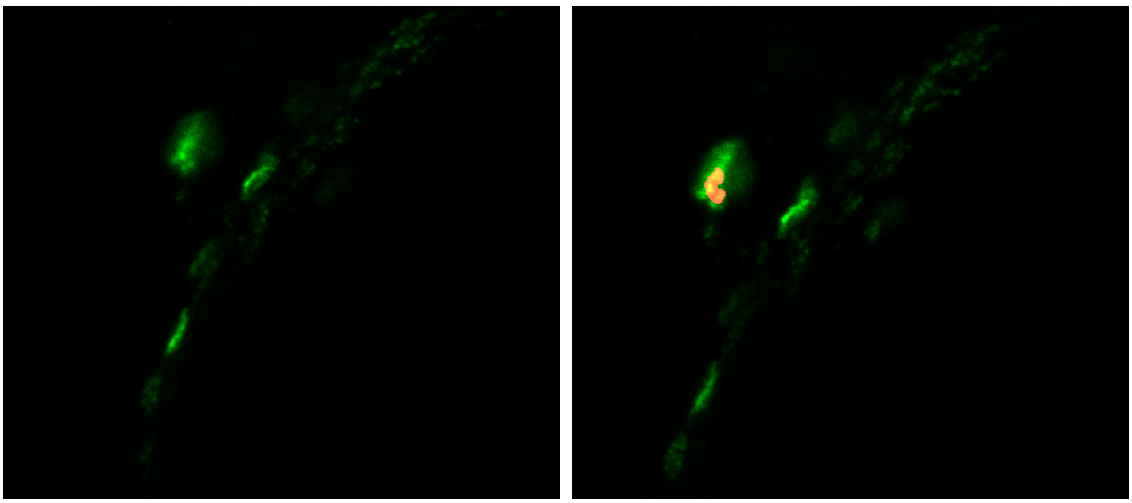


Figure 3.7: Calyx-region and marker-based exclusion examples from the training partition. The panel-specific annotation decisions are discussed in the text.

A fundamental constraint applied throughout annotation is that only structures located within the main calyx region of interest, as established by the spatial extent of the BRP-shortcherry signal in the red channel, are eligible for annotation as bouton instances. In Figure 3.7(a), green puncta clusters are morphologically similar to legitimate calycal boutons but are excluded because they lack corresponding BRP-shortcherry signal. Figure 3.7(b) shows the most challenging case: structures exhibit both BRP-shortcherry signal and green channel signal consistent with a microglomerular arrangement, but lie outside the main calyx boundary, so their exclusion requires spatial knowledge of where the calyx region begins and ends. Figure 3.7(c) shows a red confirmed calycal instance that is annotated while neighboring green-only structures remain excluded. Figure 3.7(d) shows bright green signal outside the calyx boundary excluded alongside valid annotated calycal

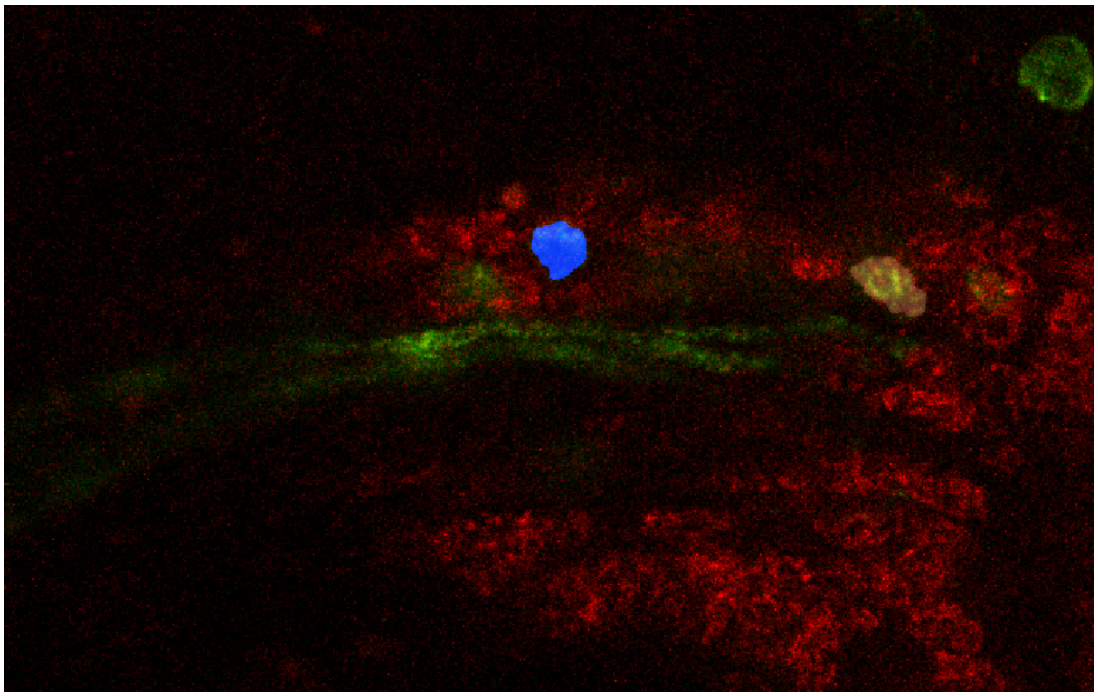
boutons. Together, these examples illustrate that morphological similarity to a bouton in the green channel is not sufficient for annotation eligibility without the corresponding presynaptic BRP-shortcherry marker, as described in Section 2.1. These examples are included in the training partition so that the model learns to apply both the spatial and signal criteria in combination rather than treating either green channel morphology or signal presence in isolation as sufficient evidence for a valid bouton prediction.

A common problem in *Drosophila* mushroom body calyx images is that BRP-shortcherry-positive signal from projection neuron axonal processes can superficially resemble the compact active zone puncta clusters of annotatable calycal microglomerular boutons, as described in Section 2.1. Figure 3.8 documents why axon like examples were retained in the training partition by design rather than treated only as qualitative edge cases. Because slices without any annotation mask were not used as supervised training examples, axon only slices would not by themselves teach the model that this fluorescence should be rejected. The informative training cases are therefore slices or crops in which valid bouton masks are present while nearby axon like signal remains unannotated; in these examples, the mask identifies the true target instance and simultaneously labels the surrounding axonal fluorescence as background. Careful inclusion of such masked images is necessary for the model to learn axon like structure rejection instead of treating every compact or bright BRP-shortcherry-positive region as a bouton. Figure 3.8 shows this distinction across two axial levels and one wider field of view example, making the extended axonal trajectory visible while preserving a supervised mask for the valid bouton signal. For visual clarity, all raw image panels in this figure are displayed with a linear twofold brightness increase.



(a) Axon like slice (b) Valid annotated slice

Figure 3.8: Axon like training examples at two axial levels, displayed with a linear twofold brightness increase. The valid slice contains one compact bouton mask, while visually similar axonal signal remains unannotated as non-target background.



(c) Additional wide axon like example

Figure 3.8: Additional wide field of view example showing an extended axon like trajectory near valid bouton signal. The panel is displayed with the same linear twofold brightness increase as Figure 3.8.

An initial annotation campaign was conducted on an earlier set of image volumes that were subsequently found to contain acquisition artifacts, including signal bloom and image degradation, that made reliable object boundary definition impossible. Those annotations were discarded in their entirety and a second annotation campaign was conducted on corrected image volumes, producing the final dataset described above.

3.1.4 Annotation Procedure

The annotation provenance differed between the training/validation partitions and the held out test partition. Training and validation labels were produced by the author using the Napari image viewer [65] in two-dimensional slice by slice mode. These labels were intended to serve as operational ground-truth targets for optimizing bouton detection under the thesis annotation rationale, but they were not created in isolation. Imaris annotations of the same image volumes were provided by the Tavosanis laboratory through Pallavi Syam and were used as rough expert guidance during annotation. These Imaris annotations originated from the laboratory's established semiautomatic surface mask workflow rather than from two-dimensional slice by slice labeling. Because this workflow depends on surface generation, threshold choices, and manual correction in three dimensions, the Imaris masks could contain errors and were therefore treated as localization cues and biological reference material rather than as definitive labels to copy. Initial author masks were constructed to approximate the expected ground truth, then discussed and validated with her for biological plausibility. After this validation, the masks were edited to incorporate the data-specific conventions described in this chapter, including slice based connected-component labels, size based exclusions, the handling of axon associated or otherwise

ambiguous bouton-like structures, and the false-positive-tolerant annotation strategy. The held out test ground truth was not defined by the author alone: because the final decision about whether ambiguous structures are true boutons requires biological expertise, she reviewed, corrected, and authorised the test labels, and these biologically reviewed labels were used as the evaluation ground truth. Masks generated by the author or model predictions for the test volumes served only as support material and were not treated as authoritative ground truth.

To reduce the manual effort required for creating author side annotations and candidate masks, a two stage workflow was adopted. In the first stage, one volume was fully annotated manually in Napari, using the available biological and Imaris based reference information only as guidance where applicable, to produce an initial thesis specific annotation. A rough segmentation model was then trained on this single annotated volume to generate initial mask predictions for the remaining volumes. In the second stage, these predictions were loaded directly into Napari as starting annotations for each subsequent volume, and the annotator manually corrected each slice by adding missed boutons, removing false positive detections, and adjusting boundaries where the predicted masks did not accurately reflect the target object extent defined by the thesis annotation rationale. The corrected training and validation masks were then aligned with the biological feedback described above and adjusted to the final data-specific annotation rules. The resulting labels were used directly for training and validation; for the test volumes, they served only as candidate material before biological review and final ground truth authorisation by the same expert.

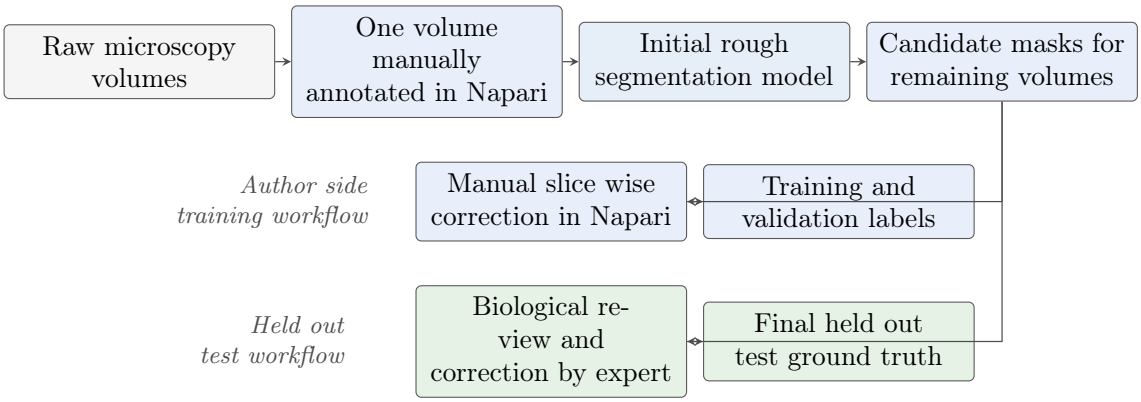


Figure 3.9: Schematic overview of the annotation workflow. One representative volume was first annotated manually in Napari and used to train an initial rough segmentation model. The resulting candidate masks were manually corrected slice by slice, checked against biological feedback and Imaris-based guidance, and then finalized as training and validation labels according to the thesis-specific annotation rules. For held out test volumes, author generated masks served only as candidate material before biological review and final ground truth authorisation.

Annotations were produced in two-dimensional mode by design rather than by labeling objects across three-dimensional volumes directly. Because MicroSAM processes images slice by slice, the model has no access to volumetric context during training and cannot resolve why two spatially disconnected regions within the same slice should share an identity. Instance labels were therefore assigned per connected component within each two-dimensional slice, so that disconnected signal regions within a single slice received separate instance identifiers even when they belonged to the same three-dimensional bouton. The

resulting two-dimensional target should therefore be understood as a bouton-structure detector rather than as the final biological instance definition: the model is trained to detect bouton-related connected regions in each slice, while final bouton identity is recovered only after stacking the predictions into a volume. During inference, these regions can likewise be detected as separate two-dimensional instances; if they are connected through foreground voxels in neighboring slices, the three-dimensional label stitching step described in Section 3.3.2 merges them automatically into one volumetric instance.

The same convention was enforced when the annotated volumes were exported for slice based model training. The thesis-side utility `images_2_slices.py` performed this offline conversion before training: it read each fixed-partition image and label volume, converted images to the canonical (Z, Y, X, C) layout, iterated over axial Z planes, discarded label-empty slices for the training export, converted each two-dimensional label slice to binary foreground, and relabeled connected components with compact sequential instance identifiers. This step removed gaps or duplicated label identifiers and ensured that disconnected foreground regions in the same slice were not treated as one slice-level training instance. It wrote the resulting slice-level image-label pairs to disk, but did not perform random patch sampling, augmentation, or 1024×1024 crop extraction; these operations belong to the model-specific data loaders described later.

This two-dimensional annotation convention also determined how axon associated ambiguity was handled, as illustrated by Figure 3.8. When the elongated axonal structure itself was visible in a slice, its fluorescence was left unannotated. In contrast, some axon associated puncta appeared in neighboring slices as isolated compact blobs: in the full three-dimensional stack they could be understood as being connected to an axonal process, but in the individual slice available to a two-dimensional model they resembled ordinary bouton clusters. Such locally bouton like borderline regions were annotated as bouton instances, because excluding them on the basis of three-dimensional evidence unavailable to the model would create visually similar two-dimensional training examples with contradictory labels.

In cases where a compact signal region was ambiguous and its inclusion as a bouton instance was uncertain, the annotator chose to include it rather than exclude it. This applied to axon associated puncta such as the red mask region in Figure 3.3, where the exclusion criterion depended on three-dimensional context unavailable to the slice based model, as well as to other expert flagged bouton like structures for which no reproducible two-dimensional exclusion rule could be identified even though the biological collaborator did not consider the structure a true bouton. The convention reflects an explicit preference for false positive over false negative errors during annotation, motivated by a practical asymmetry in the downstream correction cost: extra detections that should not have been included can be removed with minimal interaction in subsequent analysis, whereas missed boutons that were not annotated require the substantially more labour intensive process of identifying and adding them back. Borderline cases were therefore marked consistently to prevent the model from learning an inconsistent decision boundary between included and excluded objects of similar two-dimensional appearance.

3.2 Dataset Creation and Pre-processing

This section describes the pre-processing used to construct the stored input representations of the annotated volumes. These steps were applied to the raw volumetric acquisitions before model training and inference, and the resulting volumes were saved as separate image

files. The original, DoG sharpened, and RL deconvolved representations are combined in the dataset distribution reported in Table 3.1, while all subsequent model input normalization and post-processing steps remain identical across representations.

Only standard confocal LSM volumes were processed by the enhancement pipelines. Airyscan volumes were excluded from the DoG and RL pre-processing branches because the Airyscan reconstruction performed by the ZEN acquisition software already includes pixel reassignment and linear deconvolution, as described in Section 3.1.1. Applying additional deconvolution or sharpening to these volumes would therefore have introduced a second, modality specific enhancement step and would have made the Airyscan and standard confocal inputs less comparable. Airyscan volumes consequently contributed only their original reconstructed form to the dataset.

Common Background Correction. Both enhancement pipelines began with rolling ball background subtraction, following the background removal principle introduced in Section 2.2. The operation was applied independently to each two-dimensional axial slice and to each fluorescence channel using a ball radius of 50 pixels. This radius corresponds to $50 \times 0.0709 \approx 3.55$ µm in the lateral image plane. The choice was guided by the published bouton volume scale rather than by the volume of any single crop: at the common voxel size of $0.0709 \times 0.0709 \times 0.3 \approx 0.00151$ µm^3, the reported 1.3–39.9 µm^3 bouton range corresponds to approximately 860–26,500 voxels, while the 10–30 µm^3 range containing most boutons corresponds to approximately 6,600–19,900 voxels [64]. Since rolling ball correction is applied slice by slice, the relevant parameter is the lateral scale in an axial plane: the equivalent spherical radius of a 10–30 µm^3 bouton is approximately 1.3–1.9 µm, or 19–27 pixels, and even a 39.9 µm^3 bouton has an equivalent radius of approximately 30 pixels. A 50-pixel rolling ball radius is therefore larger than the expected bouton scale foreground radius while remaining small enough to follow broader slice level background variation. Background subtraction was performed on the raw unsigned 16-bit intensity values and produced a background corrected float32 volume. This shared initial step was used to reduce slowly varying illumination, out of focus haze, and broad background fluorescence before either deconvolution or sharpening was applied.

Richardson–Lucy Deconvolution Pipeline. The Richardson–Lucy (RL) pipeline was designed as one of the stored standard confocal input representations to test whether explicit deconvolution based enhancement improves downstream instance segmentation relative to the unenhanced and DoG sharpened representations. Its purpose was not to reconstruct a quantitatively exact optical volume, but to increase the contrast and separability of punctate BRP-shortcherry signal before model training and inference. The theoretical basis and limitations of Richardson–Lucy deconvolution are discussed in Section 2.2; the present section specifies the concrete implementation used to generate the RL dataset variant.

After rolling ball background subtraction, the fluorescence signal was normalized across the full three-dimensional stack using percentile clipping. Intensities below the 0.0001th percentile and above the 99.9999th percentile were clipped, and the remaining values were linearly rescaled to [0.0, 1.0] as float32 data. For the thesis experiments, the two fluorescence channels were handled as a combined-channel representation for the Richardson–Lucy step rather than being deconvolved independently with separate channel-specific point spread functions. Richardson–Lucy deconvolution was then applied to each two-dimensional axial

slice of this combined representation using `skimage.restoration.richardson_lucy` from the scikit-image library [66]. The implementation was called with `num_iter = 15` and `clip = True`, so the estimate was constrained to [0, 1] during the iterative update. This clipping was retained to avoid numerical divergence and excessive amplification of high intensity artifacts.

The point spread function was approximated by a single analytically derived two-dimensional Gaussian kernel with a 15×15-pixel support, based on the acquisition numerical aperture of 1.4 and the immersion oil refractive index of 1.518. Because this implementation used a combined-channel representation, it did not apply separate wavelength-specific PSFs to the BRP-shortcherry and Kenyon-cell channels. After deconvolution, residual low amplitude artifacts were suppressed by setting all deconvolved values below the second percentile of the deconvolved stack to zero. The cleaned deconvolved signal was then added to the normalized input representation,

$$I_{\text{final}} = I_{\text{normalized}} + I_{\text{deconvolved}},$$

and the resulting representation was min–max rescaled to [0.0, 1.0]. The final RL deconvolved volumes were stored as float32 TIFF files.

Difference-of-Gaussians Sharpening Pipeline. The Difference-of-Gaussians (DoG) pipeline was used as a computationally simpler alternative to deconvolution for enhancing punctate bouton related signal in standard confocal LSM volumes. As described conceptually in Section 2.2, DoG filtering acts as a band pass operation that emphasizes structures within a selected spatial scale range. In this work, the DoG pipeline was applied directly to the rolling ball corrected float32 data rather than to a percentile normalized input volume.

For each two-dimensional axial slice and each channel, two Gaussian smoothed images were computed with $\sigma_{\text{low}} = 1.5$ pixels and $\sigma_{\text{high}} = 3.0$ pixels. The broader smoothing was subtracted from the narrower smoothing to isolate local bright structure, and only positive responses were retained. The sharpened image was computed as

$$I_{\text{sharp}} = I + 1.5 \times \max\left(G_{\sigma_{\text{low}}} * I - G_{\sigma_{\text{high}}} * I,\, 0\right),$$

where the factor 1.5 controls the strength of the added band pass component. Negative DoG responses were discarded to avoid introducing dark ringing artifacts around bright puncta. After sharpening, the complete volume was min–max rescaled independently per channel to [0.0, 1.0] and saved as a float32 TIFF file.

Original Variant. The original representation used the unenhanced source volumes directly. No rolling ball background subtraction, Richardson–Lucy deconvolution, or Difference-of-Gaussians sharpening was applied during dataset creation for this representation. Original volumes were normalized only later during slice extraction, using the shared model input normalization procedure described in Section 3.3.1. This representation therefore served as the unenhanced reference input for comparisons with the DoG sharpened and RL deconvolved representations.

3.3 Image Pre-processing and Post-processing

This section describes only the processing applied at model input time and after model prediction. The dataset creation steps used to generate the original, DoG sharpened, and RL deconvolved representations are described separately in Section 3.2.

3.3.1 Slice Extraction and Normalization

Training and inference inputs were extracted as two-dimensional axial slices from the corresponding three-dimensional image volumes. At slice extraction time, each slice was normalized using percentile based clipping over the full source volume and channel. Intensities below the 0.00001th percentile and above the 99.99999th percentile were clipped, and the remaining range was linearly rescaled to [0.0, 1.0]. For MicroSAM training, these exported two-dimensional slices were stored on disk as `float32` arrays in this [0, 1] range, so slice creation was a one-time dataset preparation step rather than part of each training iteration. Conversion to [0, 255] was deferred to the model input pipeline: during training it occurred after patch sampling and intensity augmentation, and during inference it occurred after slice loading and before SAM encoder normalization. This percentile based normalization was applied consistently to original, DoG sharpened, and RL deconvolved inputs.

3.3.2 Post-processing

After per slice two-dimensional instance predictions were produced for each axial slice of a volume, three post-processing steps were applied in sequence to obtain the final three-dimensional instance segmentation: three-dimensional label stitching via connected component analysis, minimum-volume filtering, and axial span filtering.

Three-dimensional Label Stitching. Per slice instance label maps were stacked along the axial dimension into a three-dimensional array, and instance identifiers were reassigned globally using three-dimensional connected component analysis with full 26-connectivity, implemented via `scipy.ndimage.label` with a fully connected structuring element generated by `generate_binary_structure(3, 3)`. Under this approach, two foreground voxels in adjacent slices are merged into the same three-dimensional instance if and only if they are topologically adjacent in the 26-connected neighborhood, without requiring an explicit overlap fraction criterion. This topology based merging is consistent with the connector region annotation strategy described in Section 3.1.3, in which annotation masks for large or spatially distributed instances were intentionally extended to ensure spatial continuity across slices and thus support correct three-dimensional merging.

Size Filtering. After three-dimensional label assignment, detections were filtered by minimum physical size and axial extent to remove connected components unlikely to represent genuine calycal PN bouton instances. The lower size threshold was derived from the EM based morphological measurements reported by Yang *et al.* [64], in which 89 volumetrically reconstructed PN boutons in the MB calyx ranged from 1.3 μm^3 to 39.9 μm^3. These physical bounds were converted to voxel counts using the common voxel volume of

$0.0709 \times 0.0709 \times 0.3 \approx 0.001508$ µm^3, which applies equally to standard confocal volumes and downsampled Airyscan volumes, as established in Section 3.1.2.

The lower volume filter removed instances occupying fewer than 862 voxels, corresponding to $862 \times 0.001508 \approx 1.30$ µm^3. This filter is necessary because the models are trained and inferred on two-dimensional slices: the terminal part of a true bouton can occupy only a very small label area in an individual slice, so these small slice-level examples are retained during training to teach the model to detect bouton endings. However, the same supervision can also make the model sensitive to small noise-like or redundant structures. Filtering after three-dimensional stitching distinguishes such isolated small detections from small terminal slices that are connected to a larger volumetric bouton.

The reported upper bouton volume of approximately 39.9 µm^3, corresponding to about 26,524 voxels at the common voxel size, was not used as a hard removal threshold. This choice follows the false-positive-tolerant annotation strategy described above: a prediction may represent an otherwise largely correct bouton but include some attached extra foreground because ambiguous regions were preferentially included during training. Removing such an object entirely merely because its volume exceeds the published upper range would discard the valid part of the segmentation and impose a larger correction burden on the end user than retaining an oversized candidate for review or downstream refinement. Finally, instances spanning three or fewer axial slices were removed, so that retained objects had to be present in at least four consecutive optical sections. No additional intensity based mask shrinking was applied.

3.4 MicroSAM

MicroSAM [17], described in Section 2.7.3, was selected as the primary segmentation model for this thesis on the basis of its dedicated automatic instance segmentation mode and its domain specific fine tuning on light microscopy data. This section describes the selected primary model variant, the training data preparation procedure, the fine tuning configuration, the inference procedure used for evaluation, and the smaller encoder size comparison variant.

3.4.1 Model Variant

The primary MicroSAM configuration used in this thesis was the `vit_l_lm` model variant, hereafter referred to as MicroSAM Large. This variant uses the large Vision Transformer encoder of SAM [16] and was fine tuned by Archit et al. [17] on a curated dataset of over 17,000 light microscopy images, producing a generalist microscopy encoder that serves as the starting point for the domain specific fine tuning described below. Automatic Instance Segmentation (AIS) mode was used for both fine tuning and inference, as it trains an additional decoder to predict foreground probability, centroid distance, and boundary distance maps that are combined by a watershed step to produce instance labels without requiring manual prompts, and it is substantially faster than the grid point based Automatic Mask Generation mode of the original SAM while yielding better results on data similar to the fine tuning domain [17].

3.4.2 Training Data Preparation

Training data were prepared as two-dimensional axial slices extracted from the three-dimensional pre-processed volumes. The slice-level export and connected-component relabeling followed the annotation convention described in Section 3.1.4 and were implemented by `images_2_slices.py`. Consequently, the MicroSAM training pipeline consumed stored two-dimensional image-label pairs from the fixed partition directories rather than resampling volumes or redefining labels dynamically at each training run. Images were exported in a canonical (Z, Y, X, C) layout, lateral (XY) slices were used exclusively, and XZ/YZ reslicing was not performed. Only slices containing at least one foreground label voxel were retained for training, because label-empty slices provide no positive supervision for the AIS decoder. Patch extraction, foreground-based crop rejection, augmentation, and batching were deferred to the MicroSAM `SegmentationDataset`/data-loader pipeline during fine tuning.

Because SAM expects three channel RGB inputs and the imaging data consist of two fluorescence channels, a channel adaptation strategy was required to construct valid model inputs. Each two-dimensional slice was arranged as a three channel array in which channel 1 contains the BRP-shortcherry signal and channel 2 contains the Kenyon cell signal, while channel 3 was set to zero throughout. Archit *et al.* [17] evaluated two strategies for this adaptation: appending an empty channel and averaging the available channels into a single channel duplicated three times. The averaging strategy was found to perform better in their experiments; the empty channel strategy was adopted in the present work for implementation simplicity, with the understanding that this constitutes a minor limitation relative to the alternative. Each slice was normalized over the full three-dimensional volume using percentile based clipping at the 0.00001th and 99.99999th percentiles. This export-time normalization converted the input data from their original integer representation, such as `uint8` or `uint16`, to `float32`, clipped extreme intensities, and rescaled the retained intensity range to [0.0, 1.0]. The exported two-dimensional files remained on disk in this normalized `float32` representation. The [0, 255] scaling expected by the SAM image encoder was performed later inside the training or inference input pipeline, after the intensity augmentation step during training.

3.4.3 Fine tuning Configuration

Fine tuning was performed using the `train_sam` function from the MicroSAM library with the AIS decoder enabled. The optimizer was AdamW with an initial learning rate of 1×10^{-5} and a ReduceLROnPlateau scheduler. Training ran for a maximum of 200 epochs with early stopping applied at a patience of 40 epochs. The batch size was 1, and the nominal patch size of 1024×1024 pixels was passed to the MicroSAM data-loading configuration. Patch extraction, foreground-based rejection, raw image transformation, label transformation, and geometric augmentation were handled inside the `SegmentationDataset`/data-loader pipeline before `train_sam` consumed the resulting batches. Figure 3.10 summarizes this data flow.

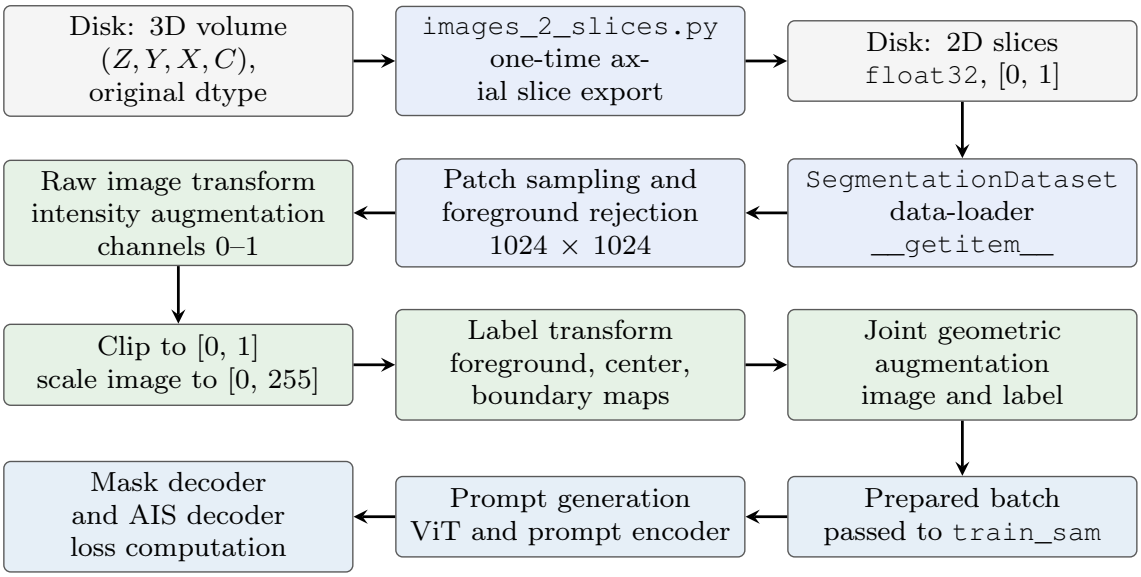


Figure 3.10: Data flow from disk to the MicroSAM training loss. Augmentation and label-target construction, including intensity augmentation, distance-transform label encoding, and joint geometric augmentation, happen inside the `SegmentationDataset`/data loader before `train_sam` is invoked. The `train_sam` routine consumes already prepared 1024×1024 batches and performs SAM-specific prompt generation, encoding, decoding, and loss computation.

Data augmentation was applied to training slices only; validation slices received no geometric or intensity augmentation. Intensity augmentations were applied first on the normalized [0, 1] image values and independently to channels 0 and 1 only, leaving the zero valued third channel unchanged. They comprised additive Gaussian noise with $\sigma = 0.05$ applied per channel with probability 0.5, and random brightness scaling by a factor sampled uniformly from [0.8, 1.2] per channel with probability 0.5. After intensity augmentation, all values were clipped to [0, 1] and the image was rescaled to [0, 255]. Geometric augmentations were then applied jointly to the image and label using a Kornia augmentation pipeline [67, 68]. They comprised random horizontal flip with probability 0.5, random vertical flip with probability 0.5, random rotation by multiples of 90° with probability 0.5 using nearest neighbor interpolation, and random elastic deformation with probability 0.3 using nearest neighbor interpolation. This order was used because the noise scale $\sigma = 0.05$ is calibrated to normalized [0, 1] intensities, whereas geometric transformations depend on spatial coordinates rather than on the absolute intensity range.

3.4.4 Inference

During inference, each axial slice of the pre-processed volume was processed independently using the AIS decoder with the final nominal 1024×1024 tiled setting. In this setting, each tile used an 896×896-pixel nonoverlapping core and a halo overlap of 64 pixels on each side, yielding a 1024×1024 context window per tiled prediction and preventing boundary artifacts at tile edges. No explicit confidence threshold was set; the default AIS parameters of the MicroSAM library were used. The resulting per slice instance label maps were assembled into a three-dimensional volume and subjected to the post-processing pipeline

described in Section 3.3.2, comprising three-dimensional label stitching via 26-connected component analysis, minimum-volume filtering, and axial span filtering.

3.4.5 MicroSAM Base Comparison Variant

In addition to the primary MicroSAM Large configuration described above, a MicroSAM Base model was trained for encoder size comparison. This comparison variant used the `vit_b_lm` light microscopy checkpoint, which uses the same MicroSAM AIS framework but replaces the large Vision Transformer encoder with the smaller base encoder. It was not treated as the main proposed MicroSAM configuration. Instead, it was fine tuned and evaluated to test whether the lower capacity encoder provides a more favourable tradeoff between recall, false positive behavior, and boundary accuracy on the bouton dataset. Unless stated otherwise in the results, MicroSAM Base used the same data preparation, geometric and intensity augmentation, fine tuning settings, AIS inference procedure, and post-processing pipeline described in Sections 3.4.2–3.4.4.

3.5 Baseline and Comparison Models

Three additional instance segmentation models were trained and evaluated alongside MicroSAM to provide a systematic comparison across architectures and inference strategies. CellposeSAM was evaluated with both a standard XY slice training set and an XYZ slice training set to assess whether orthogonal training views improved performance on the anisotropic bouton data. The comparison models were trained independently on the pre-processing representations generated by the pre-processing procedure described in Section 3.2, using the same training and held out partitions as MicroSAM. For models that produce semantic foreground predictions rather than instance level output, instance labels were recovered by applying the size filtering step described in Section 3.3.2 to connected components extracted from the foreground mask.

3.5.1 CellposeSAM

CellposeSAM, described in Section 2.7.3, was fine tuned using the default `cpsam` configuration, which combines the SAM Vision Transformer image encoder with the Cellpose flow based decoder [63]. Two CellposeSAM training modes were prepared. The standard mode used lateral XY slices only, while the XYZ mode extended the training set with XZ and YZ slices so that the two-dimensional Cellpose objective received examples from three orthogonal planes of the bouton volumes. Training inputs were prepared with the same percentile based normalization used for MicroSAM, with intensities clipped at the 0.00001th and 99.99999th percentiles and rescaled to [0, 1]. The BRP-shortcherry channel was used as the primary segmentation channel and the Kenyon cell channel as the auxiliary channel, following the two channel convention of the Cellpose framework.

For the XYZ training mode, the orthogonal XZ and YZ slices required explicit anisotropy handling because the resampled volumes have lateral spacings of $\Delta x = \Delta y = 0.0709$ µm but an axial spacing of $\Delta z = 0.3$ µm. Thus, one axial voxel corresponds to approximately $0.3/0.0709 \approx 4.23$ lateral voxels. If XZ or YZ slices were used on the raw voxel grid, the

axial dimension would be physically compressed relative to the lateral dimension, causing bouton morphology to appear flattened to the two-dimensional training objective. To preserve approximate physical proportions, the z-axis of each extracted XZ or YZ slice was therefore stretched by this anisotropy factor during resampling. Image slices were resampled with bilinear interpolation, whereas label slices were resampled with nearest neighbor interpolation to preserve integer instance identifiers. Orthogonal slices were extracted at a stride of 100 planes to include volumetric context without introducing every highly redundant XZ and YZ slice.

No additional custom image augmentation pipeline was applied for CellposeSAM, because the Cellpose training procedure already includes its own internal training transforms. For each training mode, the same trained weights were evaluated with slice wise 2D inference and volumetric 3D inference, and resulting instance predictions were subjected to the common three-dimensional stitching and size filtering pipeline described in Section 3.3.2. For volumetric inference, the same anisotropy factor was supplied to CellposeSAM so that orthogonal plane flow estimation was performed in the physical voxel geometry of the resampled dataset. The optimizer was AdamW with a learning rate of 1×10^{-5} and a batch size of 1. Training ran for a maximum of 300 epochs with early stopping at a patience of 40 epochs.

3.5.2 Swin UNETR

Swin UNETR, described in Section 2.7.2, was configured as a volumetric semantic segmentation model operating directly on three-dimensional image patches [62]. Unlike MicroSAM, which required adapting two fluorescence channels to a three channel RGB array, Swin UNETR accepts an arbitrary number of input channels natively; the model was therefore configured with two input channels corresponding directly to the BRP-shortcherry and Kenyon cell fluorescence signals, with no channel padding or averaging required. Training patches were sampled at a spatial resolution of $32 \times 192 \times 192$ voxels in the Z, Y, and X dimensions respectively, corresponding to a physical extent of approximately $9.6 \times 13.6 \times 13.6$ µm^3. This anisotropic patch geometry reflects the z-to-lateral anisotropy of the dataset: the 32-voxel axial extent covers approximately the same physical distance as the 192-voxel lateral extent, ensuring that the model receives volumetrically balanced spatial context. Four random patches were sampled per volume at each training step, with a 75 percent foreground and 25 percent background sampling ratio enforced via `RandCropByPosNegLabeld`. Sliding window inference used the same region of interest size of $32 \times 192 \times 192$ voxels with an overlap fraction of 0.5 and a sliding window batch size of 2, with Gaussian weighting applied at window boundaries to reduce stitching artifacts.

The training objective combined a Dice loss weighted by a factor of 2.0 and a cross entropy loss weighted by a factor of 0.5, with the cross entropy component further weighted by inverse class frequencies to address the imbalance between foreground and background present in the bouton segmentation data. The optimizer was AdamW with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-5}, paired with a cosine annealing learning rate schedule. Since Swin UNETR produces voxel wise semantic foreground predictions rather than instance level output, instance labels were recovered by applying three-dimensional connected component analysis and the size filtering steps described in Section 3.3.2 directly to the predicted foreground mask.

3.5.3 nnU-Net

nnU-Net, described in Section 2.5.1, was applied using its self configuring pipeline without modification to the default architectural or training parameters [52]. The `3d_fullres` configuration was selected, in which a single full resolution three-dimensional U-Net is trained on the complete volumetric data without a low resolution pre-processing stage. Training volumes were converted to the NIfTI format with voxel spacings of $0.3 \times 0.0709 \times 0.0709$ µm embedded in the image affine, enabling nnU-Net's dataset fingerprinting mechanism to automatically derive patch size, network depth, and normalization strategy appropriate for the anisotropic voxel geometry without manual intervention. Two channel inputs were provided, with the BRP-shortcherry channel stored as `_0000` and the Kenyon cell channel stored as `_0001`, and binary foreground labels were provided as the segmentation target.

nnU-Net's standard five fold cross validation was not used, as this would have produced a different training and validation data partition from those used by the other models in this comparison. Instead, all five fold entries in the dataset split configuration were overridden with the same training and validation assignment used throughout this thesis, ensuring that nnU-Net was trained and validated on identical data to MicroSAM, CellposeSAM, and Swin UNETR. A single model corresponding to fold 0 was trained and used for inference. Inference was performed using `nnUNetPredictor` with a tile step size of 0.5, Gaussian boundary weighting, and test time mirroring augmentation enabled, using the best validation checkpoint. As with Swin UNETR, nnU-Net produces semantic foreground predictions, and instance labels were recovered by applying three-dimensional connected component analysis and the size filtering steps described in Section 3.3.2 to the predicted foreground mask.

Chapter 4

Experiments and Results

This chapter presents the experiments conducted to evaluate whether domain-adapted deep learning segmentation can provide a reliable and scalable alternative to the existing semi-manual bouton quantification workflow, as formulated in Section 1.4. The final evaluation compares the trained deep-learning model families, additional CellposeSAM inference variants, and the Imaris reference baseline across the available pre-processing configurations. Each variant was evaluated on a held-out test set comprising two brain specimens acquired with different confocal microscopy systems, as described in Chapter 3. The chapter is structured as follows. Section 4.1 establishes the evaluation framework, including the rationale for metric selection and the fair comparison methodology developed for this study. Section 4.2 presents the ablation studies used to select the Brain2 inference pre-processing for all-pre-processing models and the CellposeSAM training/inference configuration, and summarizes the preliminary MicroSAM Large inference-setting checks. Section 4.3 then presents the cross-model and per-family comparison from the perspective of detection completeness, segmentation quality, and volume accuracy. Section 4.4 analyzes image-specific recall, matched-boundary quality, and false-positive behavior across the Airyscan and LSMPlus acquisitions. Section 4.5 synthesises the findings into a practical deployment recommendation.

4.1 Evaluation Framework

Reliable quantitative comparison of the trained model variants requires an evaluation framework that is both technically rigorous and aligned with the specific requirements of the bouton detection task. The standard metrics used in general instance segmentation benchmarks are not fully sufficient for this purpose for two reasons. First, the annotation strategy described in Section 3.1.3 introduces a known and intentional discrepancy between the training ground truth and the test ground truth: predictions extending into the dim inter-cluster regions of structurally complex boutons can be counted as false positives at evaluation even though they conform to the extended training annotation convention. This means that false positive counts may be inflated relative to what would be expected from a model trained on conservative annotations, and a metric that penalises false positives equally to false negatives can misrepresent the practical quality of those models. Second, the downstream use of the segmentation results in the Tavosanis laboratory context requires that the evaluation framework reflect the cost asymmetry between missed boutons

and spurious detections. A missed bouton constitutes a permanent counting error that cannot be recovered by downstream filtering, whereas a spurious detection can in principle be rejected by applying size, shape, or intensity filters to the prediction output. These considerations motivate the evaluation framework described in the following subsections.

The test set comprised two fluorescence microscopy volumes of the mushroom body calyx from different brain specimens and acquisition systems, as described in Section 3.1.2. The first test image, referred to throughout this chapter as Brain1, was acquired using Airyscan super-resolution confocal microscopy and contains 19 ground-truth bouton instances. The second test image, referred to as Brain2, was acquired using an LSMPlus confocal system and contains 14 ground-truth bouton instances. Brain2 was evaluated under three pre-processing conditions applied independently: the raw unprocessed volume, the Difference-of-Gaussians (DoG) sharpened version, and the PSF-deconvolved version. Consequently, models trained on the all-pre-processing dataset were evaluated on four images in total, corresponding to 61 ground-truth bouton instances.

Whenever Imaris is reported in this chapter, it denotes the existing semi-automatic Imaris-based workflow discussed in Section 2.4, not an additional learned model. The Imaris segmentations used for comparison were generated with the laboratory's default Imaris workflow and were provided by collaborating biologists in the Tavosanis laboratory, including segmentations prepared by Pallavi Syam. They therefore represent the expert-guided output that would have been used in the established bouton-quantification workflow before the automated models evaluated in this thesis. These Imaris results were also useful as expert reference material during dataset construction, but they are not treated here as an absolute ground truth. The evaluation ground truth follows the annotation methodology and object-definition rationale described in Section 3.1.4 and Section 3.1.3. Accordingly, the Imaris baseline is used to quantify how the proposed domain-adapted models compare with the previous expert-guided analysis workflow, rather than to define the target labels themselves.

4.1.1 Metric Selection

The evaluation reports the standard instance-detection metrics precision, recall, and F1, but the interpretation in the main model analysis prioritises recall, matched-instance mean IoU, and the recall-weighted quality score $\mathrm{PQ}_{\mathrm{rec}}$. Precision is defined as $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$, recall is defined as $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$, and F1 is the harmonic mean of the two. Before these counts were computed, predicted and ground-truth bouton regions were matched one-to-one using the Hungarian algorithm. The matching was based on the pairwise 3D intersection-over-union (IoU) between each predicted instance and each ground-truth instance, and the Hungarian algorithm selected the assignment with the highest overall overlap while ensuring that each prediction and each ground-truth bouton could be used at most once. A matched pair was accepted only if its IoU was at least 0.25. Accepted pairs were counted as true positives, predictions without an accepted match were counted as false positives, and ground-truth boutons without an accepted match were counted as false negatives. The 0.25 IoU threshold is therefore the evaluation criterion for matching predicted regions to ground truth, not an image-intensity or segmentation threshold.

The use of $\mathrm{IoU} \geq 0.25$ was introduced after an initial evaluation with the stricter conventional threshold $\mathrm{IoU} \geq 0.5$. Manual inspection showed that the stricter threshold conflated genuine missed boutons with boundary-imprecise but correctly localized detections. In the

initial Brain1 inspection under IoU ≥ 0.5, 14 false negatives were reported. Six of these cases had a plausible corresponding prediction with IoU values between 0.25 and 0.45, often because the bouton was detected only in a subset of slices, and one further case was a merge error in which one prediction spanned two ground-truth boutons. These cases are biologically different from a true miss, where no prediction is present near the bouton. Because bouton counting asks first whether the object was detected at all, the IoU ≥ 0.25 criterion is used for the detection-oriented true-positive assignment, while matched-instance mean IoU remains the metric that quantifies boundary quality after matching. The lower matching threshold therefore does not remove the penalty associated with borderline detections: predictions that are accepted only because their IoU lies between 0.25 and 0.5 enter the matched-instance mean IoU with these lower overlap values, so the gain in recall is counterbalanced by a lower boundary-quality term and by the corresponding recall-weighted quality scores. The evaluation therefore does not simply convert false negatives into unqualified successes, but instead treats correctly localized yet boundary-imprecise objects as detections while still penalising their spatial disagreement. This choice is interpreted together with mean IoU and false-positive counts, because a lower threshold could otherwise accept incorrect matches in crowded regions.

A threshold-sensitivity check on representative configurations from the final comparison table, Table 4.7, supports this interpretation. For example, in one MicroSAM Base configuration, moving from the stricter IoU ≥ 0.5 criterion to IoU ≥ 0.25 changed one borderline case: true positives increased from 32 to 33, false positives decreased from 7 to 6, false negatives decreased from 1 to 0, and recall increased from 0.974 to 1.000. At the same time, the matched-instance mean IoU decreased from 0.780 to 0.771 because the newly accepted object was not segmented with sufficiently accurate boundaries to pass the stricter threshold. A stronger threshold effect was seen for the Imaris baseline: true positives increased from 12 to 24 and false positives decreased from 17 to 5 under IoU ≥ 0.25, but the matched-instance mean IoU decreased from 0.615 to 0.505. These examples show that some detections counted as false positives at IoU ≥ 0.5 were not necessarily unrelated hallucinations; they were often predictions close enough to the ground truth to indicate bouton localization, but with imperfect or partial segmentation. The lower threshold therefore allows such cases to contribute to detection recall, while the reduced mean IoU still penalises their boundary disagreement.

Recall is used as the primary detection-completeness metric because missed boutons directly affect the biological readout of bouton number. False positives also matter because they would inflate bouton counts if accepted without correction; however, they are considered less severe in the intended semi-automatic workflow because they remain visible as candidate objects and can often be removed by size, shape, or intensity filtering, or by manual review. A false negative, in contrast, has no corresponding candidate mask in the output and is therefore missing entirely from downstream counting and volume analysis unless the raw image is re-examined. The mean IoU of matched instances is included as a measure of segmentation shape quality. In the context of this metric, which corresponds to the segmentation-quality term used in the Panoptic Quality framework [69], only true-positive matched pairs contribute, and the value reflects how closely the spatial extent of each predicted bouton boundary agrees with the corresponding ground-truth boundary. This metric is particularly relevant for the downstream application in the Tavosanis laboratory, where bouton volume measurements derived from voxel counts are used to characterize structural plasticity. A model that detects the correct instances but systematically over-segments or under-segments their boundaries would produce volume estimates that

are biased in ways not captured by detection counts alone.

The distinction between matched-instance IoU and false-positive counts is important for interpreting split detections. If a large bouton is predicted as two disconnected components, for example because an intermediate slice is not detected, the Hungarian matching procedure can assign only one of these predicted components to the ground-truth bouton. The remaining component is counted as a false positive even if it covers a biologically plausible part of the same bouton. This type of error is nevertheless penalised by the matched-instance mean IoU, because the accepted prediction covers only part of the ground-truth object and therefore receives a lower overlap score. The false-positive count is still reported separately because it describes the additional review or filtering burden and becomes more important in use cases where candidate detections cannot be manually checked.

The recall-weighted quality score $\mathrm{PQ}_{\mathrm{rec}} = \text{mean IoU} \times \text{recall}$ is inspired by the decomposition used in the Panoptic Quality framework, but the standard Panoptic Quality score is not used as a reported metric in the final comparison. Instead, $\mathrm{PQ}_{\mathrm{rec}}$ combines matched-instance boundary agreement with bouton recovery without directly including the precision term. This choice reflects the intended deployment scenario in which false-positive candidate regions can often be filtered or reviewed downstream, whereas missed boutons cannot be recovered after inference. Precision, F1 score, $F_{\beta=2}$ score, and false-positive counts are still reported separately where relevant to describe the cost of additional candidate detections. The mean relative volume error, defined as the mean absolute relative difference between predicted and ground-truth bouton volumes over all matched pairs, is included as a direct biological quality metric. This measure characterizes how accurately a model reproduces the physical volume of each detected bouton in cubic micrometers, and a value of zero indicates perfect volumetric agreement. The voxel volume used for volume computation was 0.001508 $\mu\mathrm{m}^3$, corresponding to an in-plane spacing of 0.0709 μm and a z-spacing of 0.3 μm, as established in Section 3.1.1.

Table 4.1: Evaluation metrics used for instance-level bouton segmentation.

Metric	**Formula**	**Purpose in this evaluation**
n_{gt}, n_{pred}	Counts of ground-truth and predicted instances	Measures the total number of annotated and predicted bouton instances.
precision	$\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$	Penalises spurious detections and is most relevant when false-positive candidates cannot be filtered or reviewed.
recall	$\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$	Measures detection completeness and penalises missed boutons.
F1 score	$2PR/(P+R)$	Provides a standard detection summary with equal weight on precision and recall.
mean IoU	$\frac{1}{\lvert\mathrm{TP}\rvert}\sum \mathrm{IoU}(g_i, p_i)$	Measures boundary agreement over Hungarian-matched true-positive bouton pairs.
recall-weighted quality score $\mathrm{PQ}_{\mathrm{rec}}$	mean IoU × recall	Score inspired by the Panoptic Quality decomposition and used to combine bouton recovery with matched-boundary agreement.
mean relative volume error	$\frac{1}{\lvert\mathrm{TP}\rvert}\sum \lvert v_{p_i} - v_{g_i}\rvert / v_{g_i}$	Measures biological volume accuracy over matched bouton pairs.
$F_{\beta=2}$ score	$5PR/(4P+R)$	Standard recall-priority detection metric that weights recall four times more strongly than precision.

The threshold-sensitivity check in Table 4.2 summarizes how this decision affected the all-pre-processing training variants on the common two-image evaluation set consisting of Brain1 and PSF-deconvolved Brain2. These values are not a per-pre-processing breakdown and do not describe the separately trained DoG-only or PSF-only models. The increase in recall should therefore not be read as an improvement in segmentation quality itself, but as evidence that part of the apparent false-negative burden at IoU ≥ 0.5 came from partial detections, split or merge behavior, and boundary disagreement rather than from complete localization failure.

Table 4.2: IoU-threshold sensitivity for all-pre-processing training variants evaluated on Brain1 together with PSF-deconvolved Brain2. Recall is shown after Hungarian instance matching at IoU ≥ 0.5 and IoU ≥ 0.25. Best values are shown in bold and second-best non-tied values are underlined. The table is not a per-pre-processing comparison and excludes the separately trained DoG-only and PSF-only models. Non-augmented MicroSAM Base variants are omitted to keep the comparison aligned with the final model-selection tables.

All-pre-processing model	Recall @ 0.5	Recall @ 0.25	Difference
MicroSAM-B (aug.)	<u>0.975</u>	**1.000**	<u>+0.025</u>
MicroSAM-L	**0.988**	**1.000**	+0.012
SwinUNETR	0.950	<u>0.974</u>	+0.024
nnU-Net	0.895	0.947	**+0.052**

4.1.2 Fair Comparison Framework

In this chapter, a pre-processing configuration denotes the evaluation label attached to a model trained on a particular selection of stored input representations, not a separate biological dataset. As described in Section 3.1.2, a volume is one annotated three-dimensional acquisition after modality alignment and partition assignment, whereas a version is one stored representation of that same volume after optional pre-processing. The Original configuration uses the unenhanced source volumes. The DoG configuration uses standard confocal LSM volumes sharpened with a Difference-of-Gaussians filter, while Airyscan volumes remain in their original reconstructed form. The PSF-deconvolved configuration uses the Richardson–Lucy deconvolved LSM volumes described in Section 3.2, again with Airyscan volumes retained in their original reconstructed form. The All-pre-processing configuration combines the original, DoG-sharpened, and PSF-deconvolved LSM versions within one training dataset. Models trained on Original, DoG, or PSF-deconvolved data are therefore referred to as single-pre-processing models, whereas models trained on the combined dataset are referred to as all-pre-processing models.

Table 4.3: Pre-processing configurations used for fair comparison.

Configuration	Airyscan training volumes	Standard confocal LSM training volumes
Original	Original reconstructed	Original unenhanced
DoG	Original reconstructed	DoG-sharpened
PSF-deconvolved	Original reconstructed	Richardson–Lucy deconvolved
All-pre-processing	Original reconstructed	Original, DoG-sharpened, and Richardson–Lucy deconvolved

The final evaluation compares model variants across these pre-processing configurations, as summarized in Table 4.3; the dataset distribution used for the main deployment-oriented model is reported in Table 3.1. A naive comparison that averages metrics across all available test images and pre-processing variants would be misleading for two compounding reasons.

First, the three Brain2 test images represent the same biological specimen acquired with the same instrument but presented to the models as raw, DoG-sharpened, or PSF-deconvolved inputs. A single-pre-processing model is evaluated on Brain1 and on the Brain2 representation matching its own training configuration. By contrast, an all-pre-processing model can be applied to all three Brain2 representations. The fair-comparison strategy therefore first selects the most suitable Brain2 inference pre-processing within each all-pre-processing model and then uses that selected Brain2 condition for later comparison with the single-pre-processing models. This avoids treating three differently processed versions of the same Brain2 specimen as three independent test cases when the goal is to compare model configurations.

Second, the two test brain specimens were acquired with different microscopy systems: Brain1 was acquired using Airyscan super-resolution confocal microscopy, while Brain2 was acquired using an LSMPlus confocal system. Several model families show substantially better performance on the LSMPlus acquisition than on the Airyscan acquisition, as shown in Section 4.4. Averaging over a test set that contains three Brain2 images and only one Brain1 image would cause the overall summary score to be dominated by Brain2 performance and would obscure acquisition-modality-specific failure modes.

To address these concerns, the main analysis separates three complementary views. Section 4.2 first selects the Brain2 inference pre-processing for models trained with all pre-processing representations, using recall and matched-instance mean IoU as within-model decision criteria. It then selects the CellposeSAM training and inference configuration using aggregate pre-processing-specific results. Section 4.3 reports the cross-model comparison, and Section 4.4 examines image-specific recall, matched-instance mean IoU, and false-positive behavior to interpret acquisition-dependent performance. The All-pre-processing configuration is therefore treated as a candidate deployment setting rather than as an automatically superior choice; the ablation checks which Brain2 pre-processing option should represent an all-pre-processing model in fair comparison and which option should be used when such a model is applied in practice.

4.2 Ablation Study

4.2.1 Inference Pre-processing Selection for All-Pre-processing Models

This ablation step evaluates only the all-pre-processing training mode. In this setting, the models were trained on the combined input-representation dataset described in Section 3.1.2. Following the fair-comparison rationale in Section 4.1.2, the ablation then evaluates each trained model on Brain1 and on the three independently stored Brain2 representations: raw LSMPlus, DoG-sharpened LSMPlus, and Richardson–Lucy deconvolved LSMPlus, referred to here as the PSF/deconvolved representation. The processing details are given in Section 3.2: the DoG image was generated by adding the positive DoG detail component back to the source image before normalization. The DoG and Richardson–Lucy

operations were not applied sequentially to the same input. Instead, they were applied independently to the same Brain2 image, yielding two distinct pre-processed representations of the same biological specimen.

The purpose of this ablation is therefore not yet to compare the all-pre-processing model against the single-pre-processing models, nor to rank different model families against one another. Rather, as established in Section 4.1.2, the ablation compares each model only against its own Brain2 pre-processing variants to determine which LSMPlus input representation is most suitable. This serves two purposes. First, it identifies the Brain2 modality that can represent the all-pre-processing model in a later fair comparison. This distinction is important because an all-pre-processing model can be evaluated on four images, whereas a single-pre-processing model is evaluated only on Brain1 and the Brain2 image corresponding to its own pre-processing condition. If the four-image all-pre-processing aggregate were compared directly against two-image single-pre-processing results, the comparison would mix different evaluation-set sizes and would overweight Brain2. After this modality-selection step, the downstream comparison can use Brain1 together with the selected Brain2 modality for the all-pre-processing model, making it comparable to models trained on a specific modality such as a DoG-trained MicroSAM Large model. Second, the same selection answers a practical deployment question: when an all-pre-processing model can be applied to raw, DoG-sharpened, or deconvolved LSMPlus data, the ablation indicates which pre-processing setting should be used for inference in a real-world workflow. All values in this ablation use the same Hungarian instance matching with an IoU threshold of 0.25 described in Section 4.1.1; this threshold determines which predicted bouton regions are accepted as matches to ground-truth boutons before true positives, false positives, and false negatives are counted.

For the modality-selection ablation in Table 4.4, recall and matched-instance mean IoU are used as the main decision criteria because the immediate goal is to identify the most suitable Brain2 inference modality within each model, not to make the final model-ranking decision. Recall prioritises true-positive recovery by measuring how many ground-truth boutons are successfully detected after Hungarian matching at IoU ≥ 0.25, following the task-specific cost asymmetry and metric rationale established in Section 4.1.1. Matched-instance mean IoU then checks whether the recovered boutons also have reasonable spatial agreement with the ground truth. False-positive-sensitive metrics are therefore not used to choose the representative modality in this ablation, because false positives can partly reflect the conservative test annotation and may be filtered or reviewed downstream. This does not mean that false positives are ignored in the final experiments: precision, F1 score, $F_{\beta=2}$ score, and false-positive counts are used in the later full comparison and deployment analysis. CellposeSAM is shown in a separate table because its all-pre-processing comparison also distinguishes training and inference variants, and the detailed CellposeSAM configuration table later uses the recall-weighted quality score $\mathrm{PQ}_{\mathrm{rec}}$ as its main selection metric.

Table 4.4: All-pre-processing ablation for MicroSAM, SwinUNETR, and nnU-Net using Hungarian matching at IoU ≥ 0.25. The total row sums ground-truth instances, true positives, and false negatives; mean IoU is reported as the aggregate matched-instance value from the evaluation summary. Best Total/avg. values are shown in bold and second-best non-tied Total/avg. values are underlined.

Model	Image	GT	TP	FN	Recall	mIoU
MicroSAM-B	Brain1	19	19	0	1.000	0.794
MicroSAM-B	Brain2 deconv.	14	14	0	1.000	0.747
MicroSAM-B	Brain2 DoG	14	14	0	1.000	0.742
MicroSAM-B	Brain2 raw	14	14	0	1.000	0.735
MicroSAM-B	Total/avg.	61	**61**	**0**	**1.000**	0.755
MicroSAM-L	Brain1	19	19	0	1.000	0.825
MicroSAM-L	Brain2 deconv.	14	14	0	1.000	0.749
MicroSAM-L	Brain2 DoG	14	14	0	1.000	0.732
MicroSAM-L	Brain2 raw	14	14	0	1.000	0.725
MicroSAM-L	Total/avg.	61	**61**	**0**	**1.000**	**0.758**
SwinUNETR	Brain1	19	17	2	0.895	0.732
SwinUNETR	Brain2 deconv.	14	14	0	1.000	0.732
SwinUNETR	Brain2 DoG	14	14	0	1.000	0.681
SwinUNETR	Brain2 raw	14	14	0	1.000	0.690
SwinUNETR	Total/avg.	61	59	2	0.974	0.709
nnU-Net	Brain1	19	15	4	0.789	0.655
nnU-Net	Brain2 deconv.	14	14	0	1.000	0.702
nnU-Net	Brain2 DoG	14	14	0	1.000	0.689
nnU-Net	Brain2 raw	14	14	0	1.000	0.673
nnU-Net	Total/avg.	61	57	4	0.947	0.680

Table 4.5: All-pre-processing ablation for CellposeSAM training and inference variants. Best Total/avg. values are shown in bold and second-best non-tied Total/avg. values are underlined.

Configuration	Image	GT	TP	FN	Recall	mIoU
CellposeSAM 3D	Brain1	19	4	15	0.211	0.450
CellposeSAM 3D	Brain2 deconv.	14	6	8	0.429	0.405
CellposeSAM 3D	Brain2 DoG	14	5	9	0.357	0.405
CellposeSAM 3D	Brain2 raw	14	0	14	0.000	0.000
CellposeSAM 3D	Total/avg.	61	15	46	0.249	0.315
CellposeSAM 2D	Brain1	19	12	7	0.632	0.655
CellposeSAM 2D	Brain2 deconv.	14	14	0	1.000	0.732
CellposeSAM 2D	Brain2 DoG	14	13	1	0.929	0.698
CellposeSAM 2D	Brain2 raw	14	0	14	0.000	0.000
CellposeSAM 2D	Total/avg.	61	39	22	0.640	0.521
CellposeSAM XYZ 3D	Brain1	19	15	4	0.789	0.705
CellposeSAM XYZ 3D	Brain2 deconv.	14	13	1	0.929	0.675
CellposeSAM XYZ 3D	Brain2 DoG	14	13	1	0.929	0.656
CellposeSAM XYZ 3D	Brain2 raw	14	0	14	0.000	0.000
CellposeSAM XYZ 3D	Total/avg.	61	41	20	0.662	0.509
CellposeSAM XYZ 2D	Brain1	19	17	2	0.895	0.791
CellposeSAM XYZ 2D	Brain2 deconv.	14	14	0	1.000	0.737
CellposeSAM XYZ 2D	Brain2 DoG	14	14	0	1.000	0.745
CellposeSAM XYZ 2D	Brain2 raw	14	0	14	0.000	0.000
CellposeSAM XYZ 2D	Total/avg.	61	**45**	**16**	**0.724**	**0.568**

The modality decision in Table 4.4 is made within each model and is based on the Brain2 rows only; the Brain1 and total rows provide context but are not used to choose the Brain2 inference pre-processing. For the MicroSAM variants, recall is 1.000 for all three

Brain2 pre-processing states, so the Brain2 decision is determined by matched-instance mean IoU. MicroSAM Base performs best on the deconvolved Brain2 input, with a mean IoU of 0.747 compared with 0.742 for DoG-sharpened input and 0.735 for raw input. Therefore, if the all-pre-processing MicroSAM Base model is used in a practical LSMPlus workflow, the deconvolved input is the preferred inference setting. The same within-model pattern is observed for MicroSAM Large, SwinUNETR, and nnU-Net: all of these models and variants recover all Brain2 boutons across the three pre-processing states, and the deconvolved Brain2 input gives the highest matched-instance mean IoU for each of these models.

Table 4.5 reports the corresponding all-pre-processing comparison for CellposeSAM. These rows use the same recall and matched-instance mean IoU criteria as Table 4.4, but compare the four CellposeSAM training and inference variants against one another. Detailed CellposeSAM metrics and pre-processing-specific behavior are analyzed later in Section 4.2.2.

The zero true-positive values on raw Brain2 do not mean that all CellposeSAM variants produced empty outputs. Rather, the raw/original LSMPlus representation is the outlier in Table 4.5: every CellposeSAM variant reaches zero accepted detections on this input, whereas the deconvolved and DoG-sharpened Brain2 variants recover most or all boutons for the stronger configurations. This behavior is consistent with the motivation for the pre-processing variants themselves. In the raw LSMPlus image, bouton boundaries are often difficult to decipher because weak edges blend into surrounding noise and neighboring fluorescence. DoG sharpening and Richardson–Lucy deconvolution were included to mitigate exactly this problem by enhancing local contrast or improving apparent boundary definition before inference.

Figures 4.1 and 4.2 visualize this failure mode as three-dimensional renderings of the Brain2 region, not as individual slice-by-slice two-dimensional views. They therefore provide a whole-region perspective on where the ground-truth boutons and CellposeSAM candidates lie within the raw volume. Two variants produced visible candidate masks, but none overlapped a ground-truth bouton sufficiently to satisfy the IoU ≥ 0.25 matching threshold; one visualized case produced no candidate predictions. Some visible candidates appear as small fragments along elongated bouton structures rather than as one continuous object. This can arise from the slice-wise inference and subsequent three-dimensional stitching pipeline: if an elongated bouton is missed in one or more intermediate slices, the remaining detected parts are no longer connected during stitching and can become several smaller components. The common post-processing described in Section 3.3.2 then removes components below the minimum-volume or axial-span thresholds. Thus, the poor raw-CellposeSAM result may reflect both weak raw-image boundary evidence and a reconstruction/post-processing failure mode in which fragmented parts of a true bouton are either counted as separate low-overlap candidates or removed before evaluation.

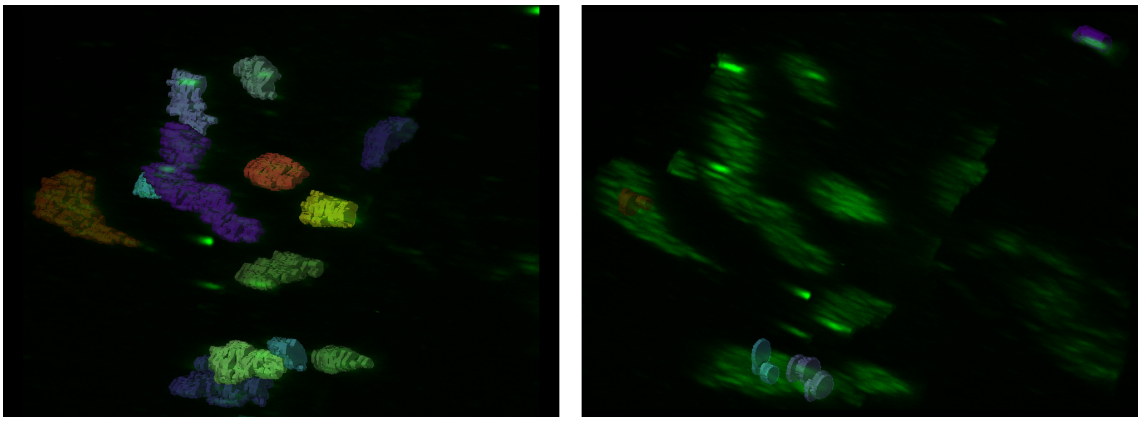


Figure 4.1: Whole-region 3D view of the CellposeSAM raw Brain2 failure mode for the XY-trained 2D inference variant. The prediction contains visible candidate masks, but none reaches the IoU ≥ 0.25 matching threshold against the ground truth.

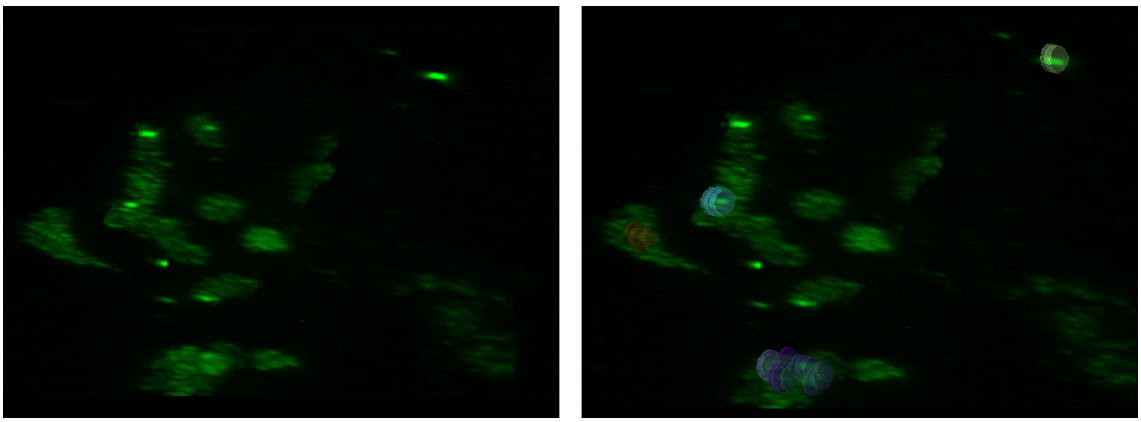


Figure 4.2: Whole-region 3D view of the CellposeSAM raw Brain2 failure modes for the XYZ-trained variants. The 3D inference variant produces no visible candidate predictions, whereas the 2D inference variant produces candidates that remain below the IoU ≥ 0.25 matching threshold.

Table 4.5 and Figures 4.1 and 4.2 together show that all CellposeSAM variants fail on raw Brain2 in the all-pre-processing setting, with zero recovered boutons. Among the enhanced Brain2 inputs, CellposeSAM XYZ 2D is the strongest CellposeSAM variant: it reaches complete recall on both deconvolved and DoG-sharpened Brain2, and the DoG-sharpened input gives the slightly higher matched-instance mean IoU. This configuration also gives the strongest Brain1 performance among the CellposeSAM variants, with a recall of 0.895 and a matched-instance mean IoU of 0.791. CellposeSAM XYZ 2D with DoG-sharpened Brain2 inference is therefore the best CellposeSAM image-detection configuration in this ablation, while the more detailed Cellpose-specific trade-offs are left to Section 4.2.2.

4.2.2 CellposeSAM Training and Inference Selection

Before comparing CellposeSAM with the other model families, the Cellpose-specific design choices were reduced to a single representative configuration. Two training modes and two inference modes were evaluated because the anisotropic voxel geometry creates a

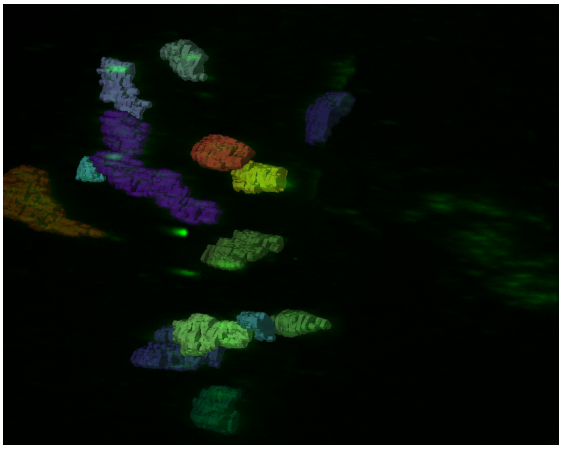
trade-off between preserving consistent lateral training context and exposing the model to through-plane bouton morphology. The standard CellposeSAM training mode used only lateral XY slices, thereby retaining the native square field of view and consistent patch geometry of the image planes. The XYZ training mode instead used a combined slice set from the XY, XZ, and YZ planes. As described in Section 3.5.1, the orthogonal XZ and YZ slices were resampled to account for the physical voxel anisotropy, but their axial extent remained much smaller than the lateral image extent. This mode therefore prioritised explicit orthogonal-plane morphological context over strict patch-size consistency. Independently of the training mode, inference was run either with Cellpose's orthogonal-plane `do_3D` mode or with a slice-wise 2D strategy followed by the common three-dimensional stitching and connected-component post-processing. This yielded four combinations for testing whether Cellpose's internal volumetric flow aggregation or a more stable slice-wise reconstruction was better suited to the anisotropic bouton volumes. In the tables, the prefix "XYZ" therefore denotes the three-axis training set, while the suffix "2D" or "3D" denotes the inference mode. For example, CellposeSAM XYZ 2D refers to a model trained on XY, XZ, and YZ slices and inferred slice-wise in 2D.

Table 4.6 compares the CellposeSAM configurations across the pre-processing settings available in the final evaluation summary. The values are not obtained by pooling the all-pre-processing, DoG, original, and PSF rows into a single aggregate. Instead, each row corresponds to one trained model and one evaluation setting. Every row is evaluated on two images: Brain1 and one Brain2 representation. The combined TP/FP/FN entry displays true positives, false positives, and false negatives over these two images; the continuous metrics are macro-averaged over the same pair. To keep the all-pre-processing rows comparable to the single-pre-processing rows, they use Brain1 and the best Brain2 representation selected for each CellposeSAM variant: deconvolved Brain2 for CellposeSAM 3D, CellposeSAM 2D, and CellposeSAM XYZ 3D, and DoG-sharpened Brain2 for CellposeSAM XYZ 2D. The single-pre-processing rows use Brain1 and the corresponding Brain2 representation only. The original/raw modality is reported only for the XY-trained CellposeSAM variants; because this modality consistently produced weaker CellposeSAM results, XYZ training was not repeated for the original configuration in order to limit training time and computational cost, and those rows are therefore not included in the ablation table. The main selection metric is PQ_{rec}, reported in the final column.

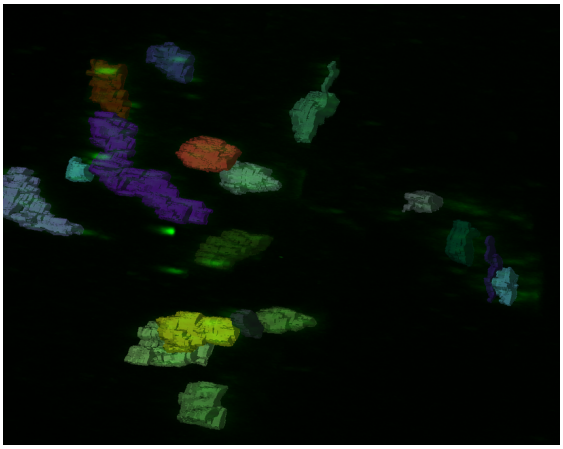
Table 4.6: CellposeSAM training and inference ablation across pre-processing settings. Best values are shown in bold and second-best non-tied values are underlined for the main comparison metrics.

Configuration	Training slices	Inf.	Preproc.	TP/FP/FN	Prec.	Recall	F1	mIoU	PQ_{rec}
CellposeSAM 3D	XY	3D	All (deconv.)	10/15/23	0.404	0.320	0.356	0.427	0.137
CellposeSAM 3D	XY	3D	DoG	14/22/19	0.389	0.425	0.404	0.432	0.184
CellposeSAM 3D	XY	3D	Original	9/2/24	0.409	0.237	0.300	0.212	0.050
CellposeSAM 3D	XY	3D	PSF	17/34/16	0.345	0.523	0.415	0.528	0.276
CellposeSAM 2D	XY	2D	All (deconv.)	26/10/7	0.738	0.816	0.774	0.693	0.565
CellposeSAM 2D	XY	2D	DoG	30/6/3	0.831	0.912	0.869	0.697	0.636
CellposeSAM 2D	XY	2D	Original	17/6/16	0.447	0.447	0.447	0.343	0.153
CellposeSAM 2D	XY	2D	PSF	31/5/2	0.861	0.947	0.897	0.747	0.707
CellposeSAM XYZ 3D	XY+XZ+YZ	3D	All (deconv.)	28/11/5	0.746	0.859	0.797	0.690	0.593
CellposeSAM XYZ 3D	XY+XZ+YZ	3D	DoG	28/12/5	0.733	0.859	0.789	0.650	0.558
CellposeSAM XYZ 3D	XY+XZ+YZ	3D	PSF	26/19/7	0.616	0.797	0.692	0.580	0.462
CellposeSAM XYZ 2D	XY+XZ+YZ	2D	All (DoG)	31/3/2	**0.910**	0.947	0.926	**0.768**	0.727
CellposeSAM XYZ 2D	XY+XZ+YZ	2D	DoG	**33/4/0**	0.890	**1.000**	<u>0.942</u>	<u>0.766</u>	**0.766**
CellposeSAM XYZ 2D	XY+XZ+YZ	2D	PSF	**33/4/0**	<u>0.898</u>	**1.000**	**0.946**	0.741	<u>0.741</u>

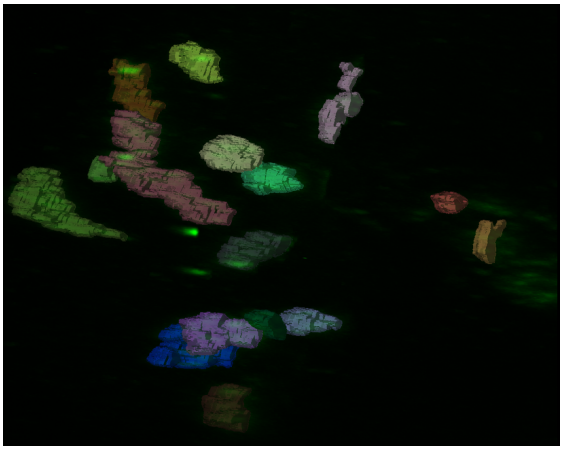
The expanded CellposeSAM ablation shows that inference mode and pre-processing both have a strong effect on performance. Using PQ_{rec} as the main decision metric, replacing Cellpose's 3D inference with slice-wise 2D inference improves the PSF result from 0.276 to 0.707 and also improves the selected all-pre-processing comparison from 0.137 to 0.565 for the standard XY training mode. The original/raw Brain2 condition remains difficult for both standard CellposeSAM variants, which explains the weaker original-pre-processing result despite reasonable performance on Brain1. The XYZ-trained 2D configuration is the strongest CellposeSAM variant overall: it achieves the highest PQ_{rec} in the table under DoG pre-processing, remains close under PSF pre-processing, and gives the best selected all-pre-processing CellposeSAM result. Therefore, CellposeSAM XYZ 2D is selected as the representative CellposeSAM configuration for later comparison with the other model families.



(a) Ground truth



(b) CellposeSAM 2D prediction



(c) CellposeSAM XYZ 2D prediction

Figure 4.3: Whole-region 3D comparison of Brain2 ground truth, standard CellposeSAM 2D prediction, and CellposeSAM XYZ 2D prediction. The three renderings are shown as separate panels to preserve the regional detail of each instance distribution while keeping the comparison within a single figure.

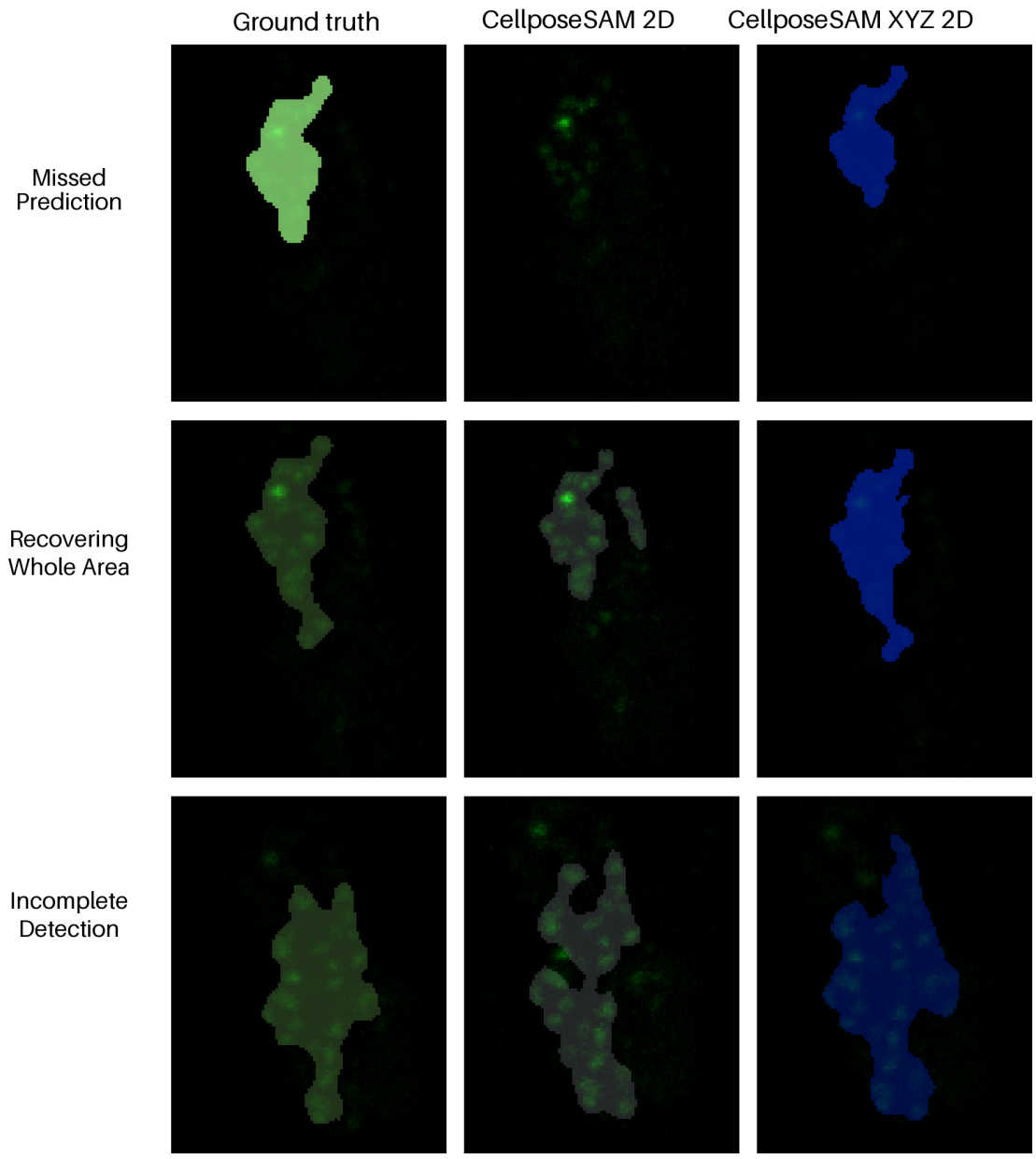


Figure 4.4: Representative two-dimensional slice examples comparing ground truth, standard CellposeSAM 2D prediction, and CellposeSAM XYZ 2D prediction. Rows correspond to the matched example names from the result folders and illustrate missed, incomplete, and spatially limited detections.

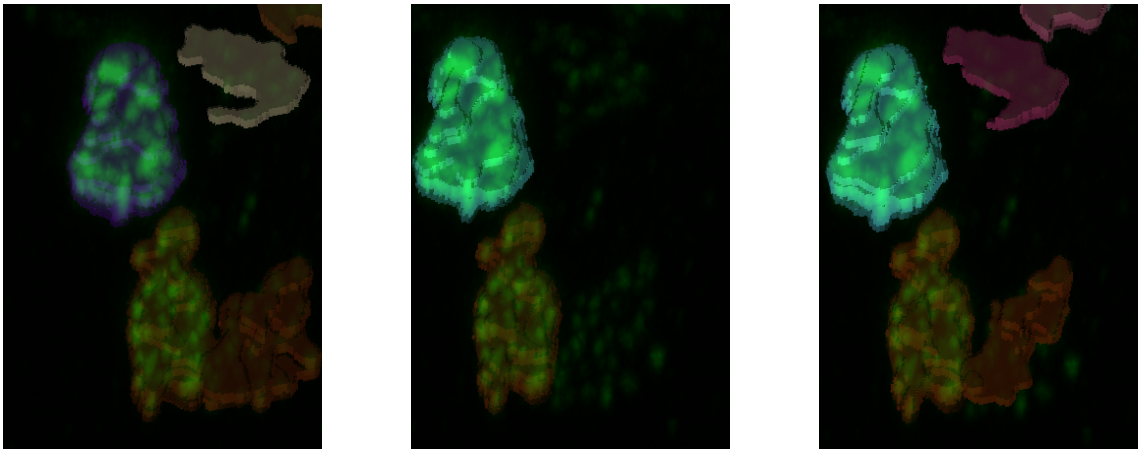
Figures 4.3 and 4.4 provide qualitative context for this selection by comparing the standard XY-trained 2D model with the XYZ-trained 2D model. The first figure uses whole-region 3D renderings to show the spatial distribution of predicted instances in the Brain2 volume, while the second figure shows matched two-dimensional slice examples from the same named cases. These figures are intended as visual support for the quantitative ablation rather than as a replacement for the IoU-based metrics in Table 4.6.

The separated 3D panels show that both prediction variants recover the ground-truth boutons visible in Figure 4.3(a) at the whole-region level. In Figure 4.3(b) and Figure 4.3(c),

the main ground-truth structures have corresponding predicted instances, but both predictions also introduce additional bouton candidates on the right side of the volume. A related effect is visible for the upper-right bouton that appears as a green instance in Figure 4.3(b) and as a pink instance in Figure 4.3(c): the predicted mask includes an extended section adjacent to the main bouton body. This can be interpreted as local over-segmentation relative to the conservative test annotation, and it should be read in the context of the annotation-boundary trade-off described in Section 3.1.3.

Figure 4.4 shows three slice-level examples of boutons whose fluorescence is distributed across spatially separated puncta rather than concentrated in one compact cluster. This is the difficult morphology discussed in Section 3.1.3, where weak intermediate signal can make the distinction between one complex bouton and several nearby regions ambiguous. The qualitative examples in Figure 4.4 show the practical value of the selected annotation convention without restating the full rationale. The standard XY-trained CellposeSAM 2D model is less reliable on these problematic boutons, producing a missed prediction or only partial recovery, whereas CellposeSAM XYZ 2D more consistently recovers the separated bouton structure as a coherent object. This supports the quantitative selection of CellposeSAM XYZ 2D in Table 4.6, because orthogonal XZ and YZ training slices improve robustness to spatially distributed bouton morphologies that are difficult to represent from lateral slices alone.

The same Brain1 case is further documented in the following two three-dimensional renderings, each shown as its own comparison figure. These views provide additional qualitative context for editing and interpretation: the one-sided rendering emphasizes local instance geometry from a restricted viewing angle, whereas the PSF-deconvolved rendering shows the same comparison under a pre-processing condition that makes distributed puncta and extended candidate regions easier to inspect. In this example, CellposeSAM XYZ 2D provides full coverage of the red bouton, while the standard CellposeSAM 2D variant does not capture the full instance. This is the same type of connected-but-weak structure discussed in Section 3.1.3 and illustrated in Figure 3.5; the standard 2D variant also misses the two boutons in the upper-right region, whereas the XYZ-trained model detects them.



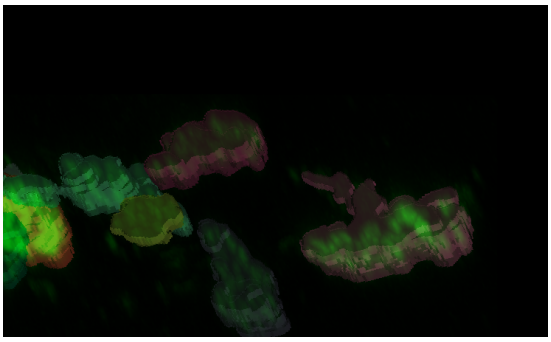
(a) Ground truth

(b) CellposeSAM 2D prediction

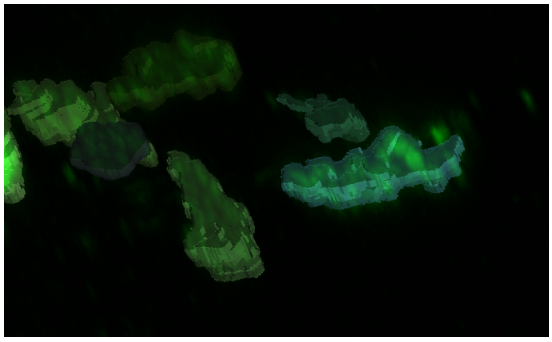
(c) CellposeSAM XYZ 2D prediction

Figure 4.5: Brain1 one-sided 3D rendering comparing the ground truth, standard CellposeSAM 2D prediction, and CellposeSAM XYZ 2D prediction. The restricted viewing angle highlights local instance geometry and the spatial relation between recovered boutons and additional candidate regions.

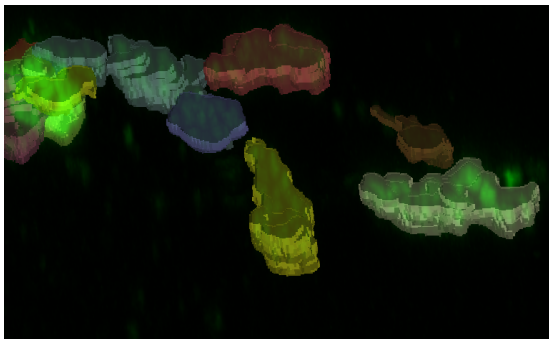
Figure 4.5 can be used to discuss how the two CellposeSAM variants behave when only one side of the bouton structure is visually prominent. The comparison is intended to support a qualitative description of whether the predictions preserve the dominant ground-truth structures and how much additional candidate volume is introduced around them.



(a) Ground truth



(b) CellposeSAM 2D prediction



(c) CellposeSAM XYZ 2D prediction

Figure 4.6: Brain1 PSF-deconvolved 3D rendering comparing the ground truth, standard CellposeSAM 2D prediction, and CellposeSAM XYZ 2D prediction. The panels are cropped to a comparable aspect ratio; the ground-truth panel is cropped to remove a small white rendering artifact in the upper-right corner. This view provides complementary context for the effect of pre-processing on visible bouton extent and candidate-instance recovery.

Figure 4.6 provides a complementary view of the same Brain1 region under PSF-deconvolved input. In this three-dimensional section, most boutons are covered over the expected spatial extent, but both CellposeSAM variants show difficulty at the same location. For the rightmost bouton, the prediction detects a separate section that appears to belong to the same bouton and should therefore be connected to the main instance, but remains disconnected. A closer slice-level inspection indicates that the connector region between these parts has very weak and ambiguous fluorescence signal, which likely contributed to this split prediction. The corresponding two-dimensional slice crops are shown in Figure 4.7 to support further analysis of this failure case.

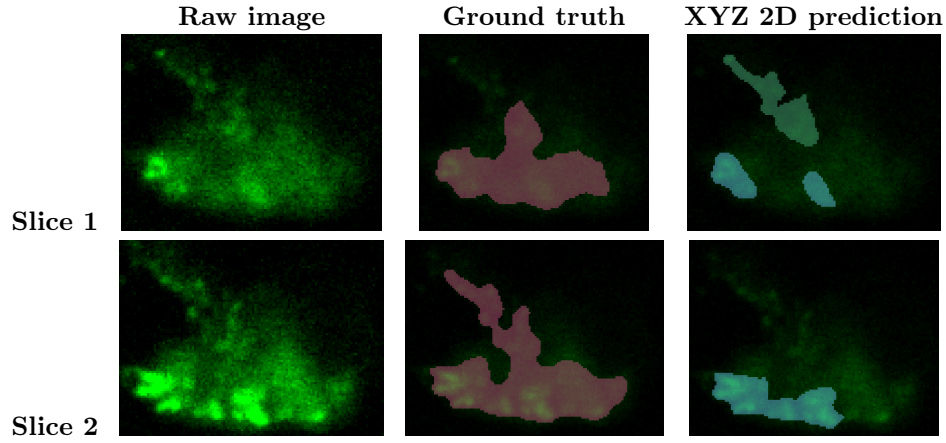


Figure 4.7: Two-dimensional slice crops of the ambiguous Brain1 bouton highlighted in the PSF-deconvolved 3D view. The two rows show neighboring slices from the same region. In both slices, weak fluorescence connects brighter bouton subregions. The ground truth annotates this structure as one connected bouton instance, whereas the CellposeSAM XYZ 2D prediction leaves parts of the signal separated into disconnected candidate regions.

In Figure 4.7, the raw-image panels are brightness-enhanced for visual clarity, but the connector between the brighter signal regions remains weak and ambiguous. The separated region recovered by CellposeSAM XYZ 2D is therefore not necessarily an implausible prediction: depending on the annotator, display settings, and biological interpretation, the same dim structure could reasonably be annotated either as one connected bouton or as partially separated candidate regions. This ambiguity is especially visible because the lower part of the fluorescence signal is brighter than the upper connector, creating the visual impression of two blobs separated by a low-intensity gap. The output may also reflect competing training cues: the dataset contains examples in which nearby boutons should remain separate, as shown in Figure 3.1, and examples in which low-intensity subclusters should be connected into one instance, as shown in Figure 3.5. Only the XYZ 2D result is shown here because the standard CellposeSAM 2D prediction produced a similar qualitative split, and the XYZ panel is sufficient to illustrate that this case lies close to an annotation-boundary decision rather than representing an unequivocal model hallucination.

4.2.3 Preliminary MicroSAM Large Inference-Setting Check

Before the final MicroSAM Large evaluation pipeline was fixed, several Automatic Instance Segmentation (AIS) inference settings were inspected on a small sample subset of the dataset. The AIS mode and its use in the MicroSAM pipeline are described in Section 3.4.1. These exploratory checks included smaller 512×512 inference windows, the nominal 1024×1024 setting used in the final pipeline, and variants with and without tiled inference. Because these trials were performed during the initial phase of the thesis and were not processed through the final Hungarian-matching evaluation scripts, complete quantitative outputs are not available for those settings. The final setting retained for all reported MicroSAM Large results was the tiled nominal 1024×1024 configuration described in Section 3.4.4. In side-by-side visual inspection, this configuration produced

clearly better instance continuity and boundary quality than the smaller or non-tiled alternatives. A caveat is that the MicroSAM models were also fine-tuned with 1024×1024 training patches, which is the default patch size used by the MicroSAM workflow. The visual superiority of the final inference setting should therefore be interpreted as support for the selected training-and-inference configuration rather than as a fully quantified inference-only ablation.

4.3 Per-Model Family Analysis

Table 4.7 compares the selected model variants across all available pre-processing configurations. Following the modality-selection decision in Section 4.2.1, the all-pre-processing rows do not average over all three Brain2 representations. Instead, they use Brain1 together with the selected Brain2 representation: DoG-sharpened Brain2 for CellposeSAM XYZ 2D and PSF-deconvolved Brain2 for MicroSAM, SwinUNETR, and nnU-Net. This makes the all-pre-processing rows comparable to the single-pre-processing rows, which also contain Brain1 and one Brain2 representation. Consequently, every row in the table represents the same two-image comparison set and the image-count column is omitted. To keep the comparison aligned with the preceding ablation decisions, the CellposeSAM family is represented only by CellposeSAM XYZ 2D, and the non-augmented MicroSAM Base variants are excluded. The Imaris reference is labeled Default because it was generated with the laboratory's default Imaris workflow rather than with a PSF-deconvolved model-input configuration.

The interpretation of Table 4.7 follows the metric hierarchy established in Section 4.1.1. Recall measures whether boutons are recovered, matched-instance mean IoU measures the spatial agreement of the recovered boutons, and $\mathrm{PQ}_{\mathrm{rec}}$ combines these two quantities without directly penalising additional candidate objects. The TP, FP, and FN counts are therefore used as contextual information. False negatives indicate missed boutons and are central for the biological counting task. False positives indicate additional candidates that may need filtering or manual review and are especially relevant in use cases where the output is used without downstream correction.

The table supports a clearer interpretation when the models are analyzed together rather than as isolated model-family summaries. The first observation is that the strongest recall-oriented methods are the MicroSAM variants and CellposeSAM XYZ 2D under fixed enhanced pre-processing. MicroSAM Large and MicroSAM Base recover all 33 boutons in their selected All–PSF rows, giving recall values of 1.000 and $\mathrm{PQ}_{\mathrm{rec}}$ values of 0.787 and 0.771 respectively. CellposeSAM XYZ 2D also reaches complete recall in the DoG and PSF single-pre-processing rows, with $\mathrm{PQ}_{\mathrm{rec}}$ values of 0.766 and 0.741. These numbers show that the highest bouton recovery is not unique to one architecture, but depends on matching the model to a suitable pre-processing regime.

The second observation is that the ranking changes when matched-boundary quality is considered together with recall. MicroSAM Large gives the highest selected-row mean IoU at 0.787 and therefore the highest selected-row $\mathrm{PQ}_{\mathrm{rec}}$, while MicroSAM Base is close behind with a mean IoU and $\mathrm{PQ}_{\mathrm{rec}}$ of 0.771 in the selected All–PSF row. CellposeSAM XYZ 2D is strongest among the CellposeSAM variants and remains competitive under DoG pre-processing, where it reaches recall 1.000 and mean IoU 0.766. SwinUNETR and nnU-Net are weaker by this recall-weighted quality criterion: their selected-row $\mathrm{PQ}_{\mathrm{rec}}$ values are

Table 4.7: Overall comparison of selected model variants on the common two-image evaluation set. All–DoG and All–PSF use Brain1 plus the selected Brain2 representation, and Imaris denotes the laboratory default workflow; counts are summed and continuous metrics are image-averaged. Best values are shown in bold and second-best non-tied values are underlined for the main comparison metrics.

Model	Config.	GT	Pred.	TP	FP	FN	Prec.	Rec.	F1	$F_{\beta=2}$	mIoU	PQ_{rec}	Vol. err.
CellposeSAM XYZ 2D	All–DoG	33	34	31	3	2	0.910	0.947	0.926	0.938	0.768	0.726	0.120
	DoG	33	37	33	4	0	0.890	**1.000**	0.942	0.976	0.766	0.766	0.120
	PSF	33	37	33	4	0	0.898	**1.000**	0.946	**0.978**	0.741	0.741	0.142
Imaris	Default	33	29	24	5	9	0.844	0.735	0.773	0.747	0.505	0.371	0.399
MicroSAM-B (aug.)	All–PSF	33	39	33	6	0	0.851	**1.000**	0.919	0.966	0.771	0.771	0.131
	Original	33	40	33	7	0	0.821	**1.000**	0.899	0.956	0.780	0.780	0.130
	DoG	33	43	31	12	2	0.720	0.947	0.817	0.890	0.772	0.729	0.158
	PSF	33	43	32	11	1	0.741	0.974	0.840	0.915	0.770	0.750	0.128
MicroSAM-L	All–PSF	33	49	33	16	0	0.670	**1.000**	0.801	0.909	**0.787**	**0.787**	**0.117**
	Original	33	41	33	8	0	0.800	**1.000**	0.888	0.951	0.784	0.784	0.118
	DoG	33	35	31	4	2	0.889	0.947	0.910	0.930	0.776	0.733	0.130
	PSF	33	40	33	7	0	0.825	**1.000**	0.899	0.955	0.763	0.763	0.157
SwinUNETR	All–PSF	33	32	31	**1**	2	**0.967**	0.947	**0.955**	0.950	0.732	0.693	0.160
	Original	33	39	32	7	1	0.824	0.974	0.885	0.934	0.707	0.688	0.191
	DoG	33	34	30	4	3	0.882	0.921	0.896	0.910	0.717	0.661	0.179
	PSF	33	40	32	8	1	0.807	0.974	0.874	0.928	0.706	0.688	0.212
nnU-Net	All–PSF	33	31	29	2	4	0.941	0.895	0.917	0.903	0.678	0.609	0.165
	Original	33	31	28	3	5	0.912	0.868	0.889	0.876	0.689	0.602	0.174
	DoG	33	32	26	6	7	0.820	0.816	0.816	0.816	0.659	0.540	0.196
	PSF	33	31	27	4	6	0.882	0.842	0.861	0.849	0.693	0.584	0.149

0.693 and 0.609 respectively, mainly because of lower matched-instance mean IoU and, for nnU-Net, additional missed boutons. Compared directly with SwinUNETR, CellposeSAM XYZ 2D is therefore a useful alternative when higher boundary agreement is preferred over the lowest possible false-positive count: its selected All–DoG row has higher mean IoU and PQ_{rec} than SwinUNETR's selected All–PSF row, 0.768 versus 0.732 and 0.726 versus 0.693 respectively, while producing more false positives, 3 rather than 1.

The third observation concerns false positives and false negatives, which provide context not captured directly by PQ_{rec}. MicroSAM Large gives the best recall-weighted quality but also produces the largest number of additional candidates in the selected All–PSF row, with 16 false positives. MicroSAM Base reduces this burden to 6 false positives while retaining complete recall and only a small reduction in mean IoU. SwinUNETR produces the cleanest selected all-pre-processing output, with only 1 false positive, but this comes with 2 false negatives and a lower mean IoU. nnU-Net is similarly conservative, with 2 false positives but 4 false negatives. False-positive counts remain practically important because every additional candidate either inflates the bouton count if accepted automatically or creates extra filtering and manual-review work before the predictions can be used for counting and volume analysis.

Taken together, Table 4.7 identifies three practically distinct operating regimes. MicroSAM Large is the quality-oriented setting when recall and matched-boundary agreement are prioritised and additional candidate cleanup is acceptable. MicroSAM Base is the most balanced substitute because it preserves complete recall, maintains high mean IoU, reduces false positives, and requires less computation than the large encoder. SwinUNETR is

the conservative low-false-positive option for workflows in which additional candidates are more costly than occasional missed boutons. CellposeSAM XYZ 2D is a strong fixed-pre-processing candidate and a practical alternative to SwinUNETR when the use case can tolerate a few more false positives in exchange for better matched-boundary quality, particularly for DoG or PSF inputs. It is not the most robust general deployment model because of its raw Brain2 failure. The image-specific analysis below examines these trade-offs at the level of individual acquisitions.

4.4 Image-Specific Analysis

This section delves into the intricacies of the results and provides the reader with additional context and analysis for the findings displayed in the previous subsection.

Table 4.8 reports the image-specific values for the highest-PQ_{rec} configuration of each model block in Table 4.7. For CellposeSAM XYZ 2D this is the DoG row, for MicroSAM Base it is the Original row, and for MicroSAM Large, SwinUNETR, and nnU-Net it is the selected All–PSF row. The Imaris Default row is included as the expert-guided baseline. The Total/avg. rows reproduce the corresponding two-image totals from Table 4.7; counts are summed over Brain1 and Brain2, while recall, matched-instance mean IoU, and PQ_{rec} follow the same averaged two-image comparison.

Table 4.8: Image-specific metrics for the best PQ_{rec} configuration of each model block in Table 4.7. Best Total/avg. values are shown in bold and second-best non-tied Total/avg. values are underlined.

Model	Mode	Image	GT	Pred.	TP	FP	FN	Recall	mIoU	PQ_{rec}
CellposeSAM XYZ 2D	DoG	Brain1	19	21	19	2	0	1.000	0.781	0.781
CellposeSAM XYZ 2D	DoG	Brain2 DoG	14	16	14	2	0	1.000	0.751	0.751
CellposeSAM XYZ 2D	DoG	Total/avg.	33	37	**33**	4	**0**	**1.000**	0.766	0.766
Imaris	Default	Brain1	19	13	13	0	6	0.684	0.509	0.348
Imaris	Default	Brain2	14	16	11	5	3	0.786	0.501	0.394
Imaris	Default	Total/avg.	33	29	24	5	9	0.735	0.505	0.371
MicroSAM-B (aug.)	Original	Brain1	19	21	19	2	0	1.000	0.817	0.817
MicroSAM-B (aug.)	Original	Brain2 raw	14	19	14	5	0	1.000	0.744	0.744
MicroSAM-B (aug.)	Original	Total/avg.	33	40	**33**	7	**0**	**1.000**	<u>0.780</u>	<u>0.780</u>
MicroSAM-L	All–PSF	Brain1	19	26	19	7	0	1.000	0.825	0.825
MicroSAM-L	All–PSF	Brain2 deconv.	14	23	14	9	0	1.000	0.749	0.749
MicroSAM-L	All–PSF	Total/avg.	33	49	**33**	16	**0**	**1.000**	**0.787**	**0.787**
SwinUNETR	All–PSF	Brain1	19	17	17	0	2	0.895	0.732	0.655
SwinUNETR	All–PSF	Brain2 deconv.	14	15	14	1	0	1.000	0.732	0.732
SwinUNETR	All–PSF	Total/avg.	33	32	31	**1**	2	0.947	0.732	0.693
nnU-Net	All–PSF	Brain1	19	17	15	2	4	0.789	0.655	0.517
nnU-Net	All–PSF	Brain2 deconv.	14	14	14	0	0	1.000	0.702	0.702
nnU-Net	All–PSF	Total/avg.	33	31	29	<u>2</u>	4	0.895	0.678	0.609

The best-configuration comparison separates two effects that are partially merged in the aggregate table: performance on the Airyscan Brain1 image and performance on the selected LSMPlus Brain2 representation. On Brain1, MicroSAM Large and MicroSAM Base recover all 19 boutons and give the highest matched-instance mean IoU values, 0.825 and 0.817 respectively. CellposeSAM XYZ 2D also recovers all Brain1 boutons in its best DoG

configuration and remains close in boundary quality with mean IoU 0.781, while producing fewer false positives than MicroSAM Large. SwinUNETR is the cleanest Brain1 output with zero false positives, but it misses two boutons; nnU-Net and Imaris miss four and six Brain1 boutons respectively.

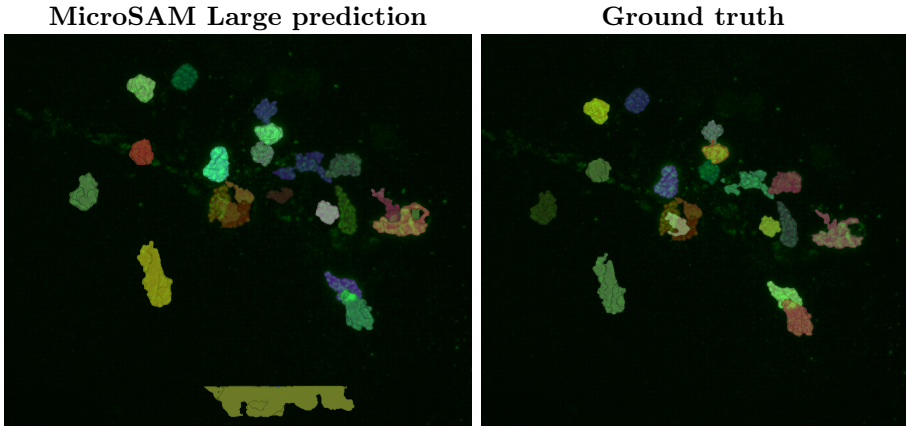
On Brain2, the best deep-learning modes all recover the 14 boutons, whereas Imaris recovers 11. The main distinction among the deep-learning models is therefore boundary agreement and false-positive burden. CellposeSAM XYZ 2D, MicroSAM Large, and MicroSAM Base have very similar Brain2 mean IoU values in their best modes, 0.751, 0.749, and 0.744 respectively. SwinUNETR and nnU-Net are cleaner on Brain2, with one and zero false positives respectively, but their matched-instance mean IoU values are lower than the CellposeSAM and MicroSAM values.

The Total/avg. rows show why the best-mode ranking in Table 4.7 is close among the leading methods. MicroSAM Large gives the highest total PQ_{rec} at 0.787, followed by MicroSAM Base Original at 0.780 and CellposeSAM XYZ 2D DoG at 0.766. The practical difference between these rows is therefore not only the score but also the false-positive burden: MicroSAM Large produces 16 false positives, MicroSAM Base produces 7, and CellposeSAM XYZ 2D produces 4. SwinUNETR remains the strongest low-false-positive alternative, with only one false positive across the two-image comparison, but its lower PQ_{rec} reflects the two missed Brain1 boutons and lower matched-boundary quality.

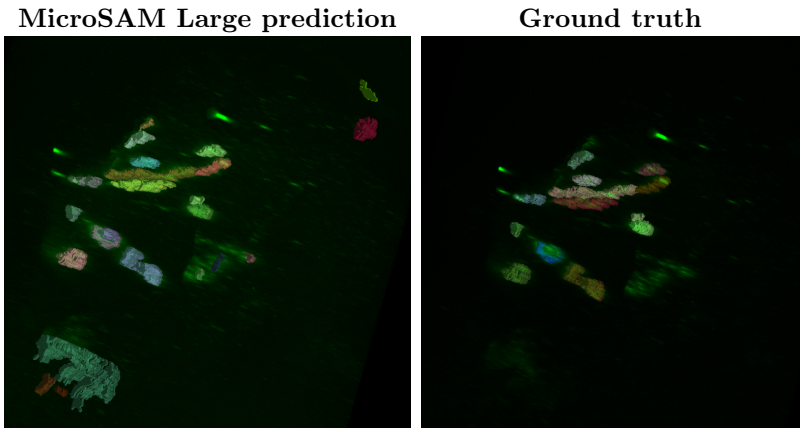
4.4.1 Qualitative Image-Level Error Analysis

This section examines the qualitative details behind the quantitative results and provides additional context for the comparisons displayed in the previous subsection. The qualitative examples therefore focus on the model outputs for which additional inspection is most informative for the final deployment recommendation. A separate CellposeSAM-focused qualitative discussion is not repeated here, because the relevant CellposeSAM behavior, including the comparison between standard 2D and XYZ-informed 2D inference, was already analyzed in the pre-processing and model-ablation study in Section 4.2. SwinUNETR is also not examined in a separate qualitative figure block: although it is useful as a low-false-positive alternative, it produced false negatives in the selected fair-comparison row and reached a lower PQ_{rec} than the MicroSAM variants and CellposeSAM XYZ 2D, so it is less informative for the recall- and boundary-quality-oriented recommendation developed below.

Before moving forward with the per-image analysis, one common peculiarity was noted across the evaluated model families: the Figure 4.7 example. In the manual ground-truth annotation, the weakly connected fluorescence signal is treated as one bouton instance. In the deep-learning model outputs, however, the same structure is consistently interpreted as two bouton-like regions rather than one connected object. Among the inspected Brain1 errors, this is the only bouton for which the evaluated deep-learning model families agree on the same alternative interpretation. This agreement across architectures suggests that the case is not simply a model-specific hallucination, but reflects a genuine annotation-boundary ambiguity caused by the weak connector signal between brighter puncta. It also explains why false positives and matched-instance mean IoU must be interpreted together: under the Hungarian matching protocol, only one predicted component can be matched to the manually annotated bouton, while any second component is counted as a false positive even if it represents a biologically plausible part of the same fluorescence structure.



(a) Brain1



(b) Brain2

Figure 4.8: MicroSAM Large three-dimensional examples: Brain1 and Brain2 predictions compared with the corresponding ground truth.

Figure 4.8(a) shows the Brain1 three-dimensional prediction and Figure 4.8(b) shows the corresponding Brain2 example. Two observations are especially noticeable across the two subfigures. First, all regions marked in the ground truth are also found in the MicroSAM Large predictions, which is consistent with the complete recall reported for the selected All–PSF comparison. Second, the prediction panels also contain additional detected regions in areas where no corresponding fluorescence signal is visible in the ground-truth panels. These signal-poor candidate regions illustrate the false-positive burden that accompanies this high-recall setting. Together, Figure 4.8(a) and Figure 4.8(b) support the interpretation of MicroSAM Large as a recall- and boundary-quality-oriented deployment choice rather than a conservative low-false-positive setting.

The next slice-level comparison, Figure 4.9, focuses on two representative two-dimensional cases that explain where the same high-recall behavior is beneficial and where it can create connector-like overprediction.

In the Brain1 slice in Figure 4.9, MicroSAM Large shows its strength relative to the incomplete-coverage example discussed for Figure 4.4: the predicted region overlaps the annotated bouton well and would therefore be expected to give a good matched-instance IoU. The small extension on the prediction remains an annotation-boundary ambiguity

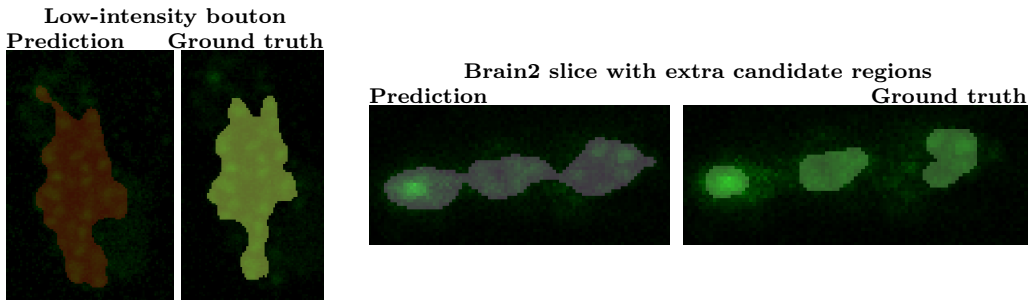


Figure 4.9: Two-dimensional MicroSAM Large slice examples showing recovered ground-truth bouton regions together with additional candidate detections.

rather than an unequivocal error, because a slightly more inclusive annotation convention could also treat this weak neighboring signal as part of the same bouton. The Brain2 slice shows the opposite side of the same problem: MicroSAM Large connects neighboring blobs, whereas the evaluation ground truth keeps them as three separate bouton regions. This is another example of the annotation-boundary trade-off discussed in Section 3.1.3 and already illustrated for CellposeSAM in Figure 4.4. In this MicroSAM Large case, the same tendency supports recovery of weakly connected bouton structures but can also connect nearby regions that the evaluation ground truth keeps separate.

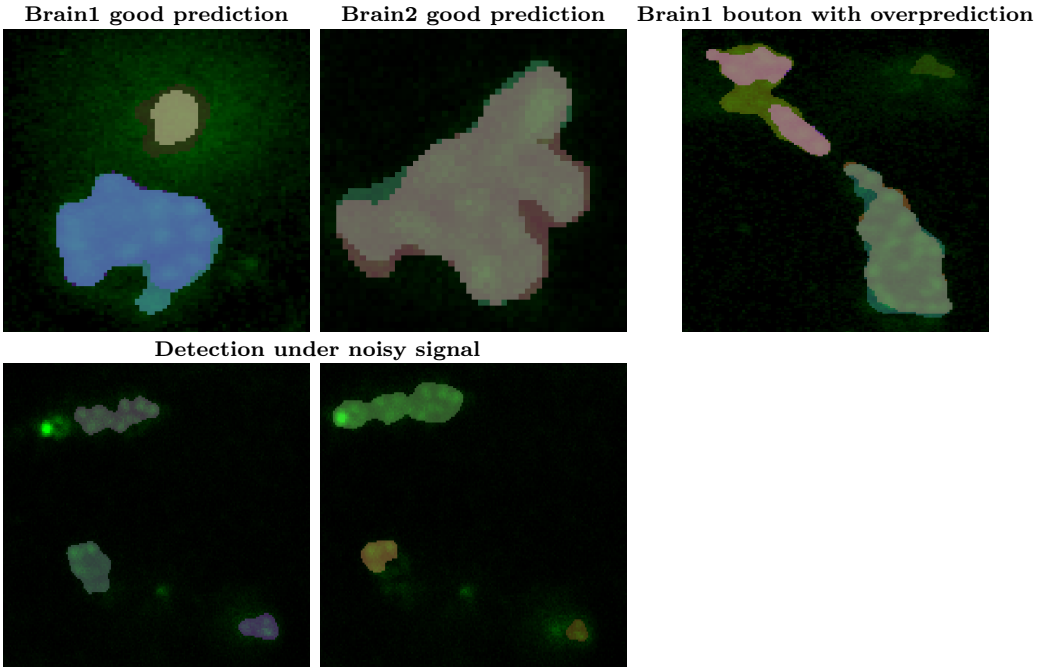
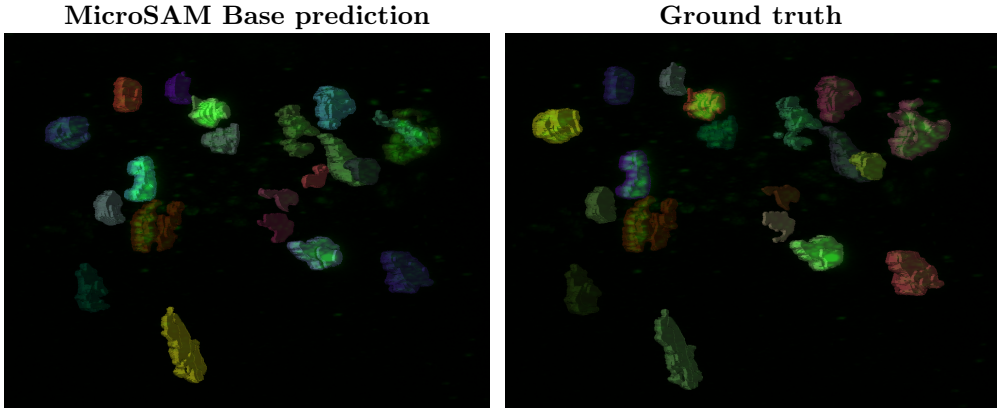


Figure 4.10: Additional two-dimensional MicroSAM Large examples showing good predictions, bouton overprediction, and detection under noisy signal.

Figure 4.10 collects the remaining MicroSAM Large slice examples. The first two good-prediction panels illustrate the segmentation capability of this model: the predicted masks follow the main annotated bouton regions closely and therefore visually support the strong matched-instance quality reported for MicroSAM Large. The Brain1 bouton with over-prediction example shows a related strength on a more complex local morphology. In the upper bouton of this panel, the smaller region corresponds to the evaluation ground truth,

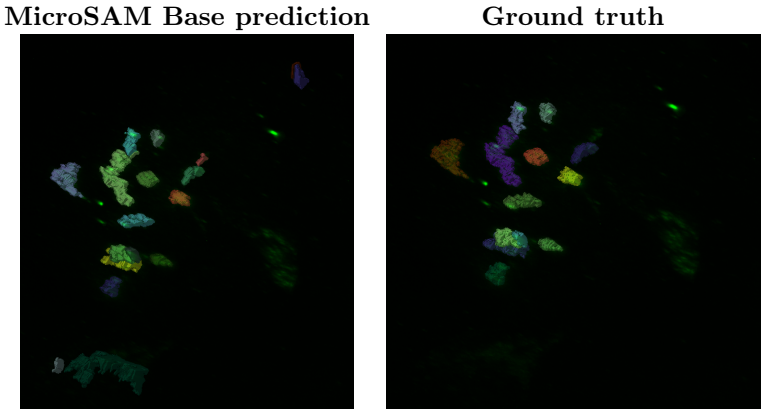
whereas the larger surrounding region corresponds to the MicroSAM Large prediction. The model captures the nearby bouton well, but it also adds an extended region around the complex structure, which is consistent with the same connector-like behavior discussed for difficult boutons in Figure 4.9. In the detection-under-noisy-signal comparison, the model shows a more general overprediction tendency by adding candidate regions beyond the ground-truth mask. One likely contributor is the bright spot visible in the image: because it is brighter than much of the surrounding bouton signal, MicroSAM Large may interpret it as a noisy bouton-like signal rather than background. Together, these examples show both sides of the MicroSAM Large output: strong local segmentation of true bouton regions, but also a tendency to extend masks or add candidates when bright or spatially complex signal is present.

Because MicroSAM Base is the closest practical alternative to MicroSAM Large in the quantitative comparison, its qualitative behavior is shown directly after the MicroSAM Large examples. Figure 4.11 is split across two placements: Figure 4.11 shows Brain1, and Figure 4.11 shows Brain2. The panels support the interpretation from Table 4.7: MicroSAM Base preserves complete bouton recovery in the selected All–PSF row, so all annotated boutons are detected in both Brain1 and Brain2. At the same time, it is less prone than MicroSAM Large to signal-poor detections in Brain1: the Brain1 prediction follows the ground-truth distribution without the visually obvious extra candidates in regions where no corresponding bouton signal is apparent. This advantage is not absolute across acquisitions. In Brain2, MicroSAM Base still introduces additional candidate regions that are weakly supported or not clearly supported by visible fluorescence signal, although the burden is lower than for MicroSAM Large. Its selected-row mean IoU of 0.771 remains close to the MicroSAM Large value of 0.787, while the false-positive count is reduced from 16 to 6. Qualitatively, this makes MicroSAM Base a less aggressive but still high-recall variant rather than a clearly inferior segmentation setting.



(a) Brain1

Figure 4.11: MicroSAM Base three-dimensional prediction examples compared with the corresponding ground truth.



(b) Brain2

Figure 4.11: MicroSAM Base three-dimensional prediction examples compared with the corresponding ground truth.

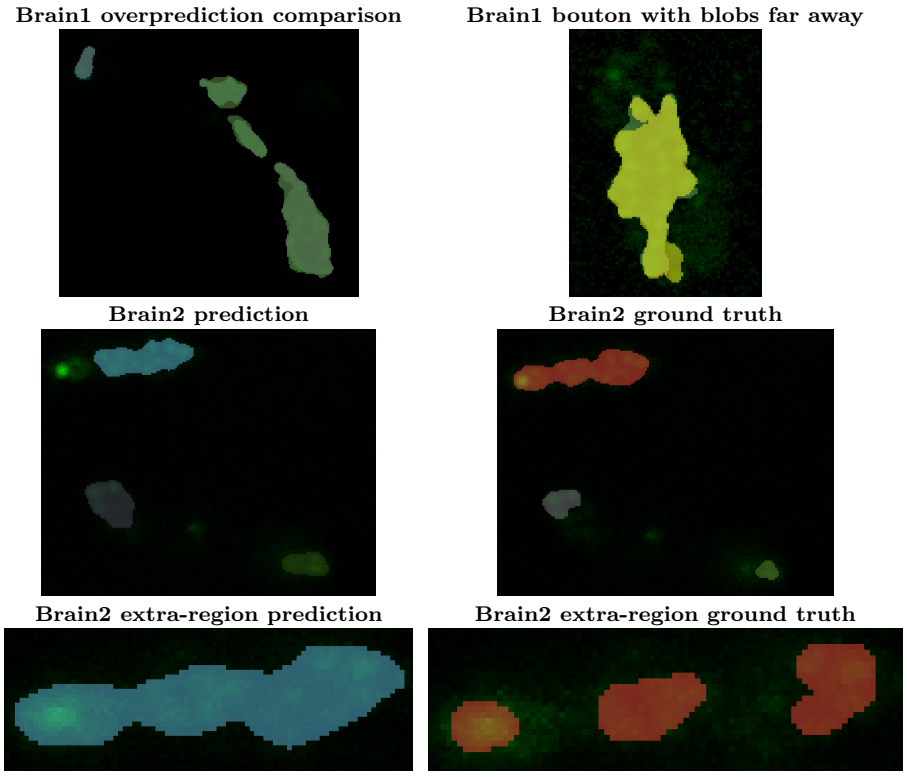
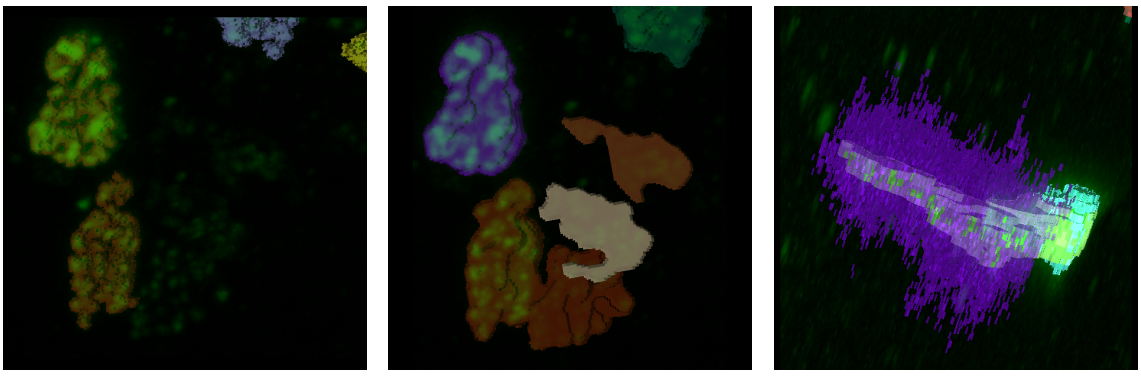


Figure 4.12: Two-dimensional MicroSAM Base examples showing Brain1 comparison cases and Brain2 prediction–ground-truth slice comparisons.

Figure 4.12 shows slice-level examples that mirror the MicroSAM Large analysis. The first Brain1 comparison should be read together with the Brain1 bouton with overprediction panel in Figure 4.10. In that MicroSAM Large example, the smaller region on the upper bouton is the ground-truth annotation, whereas the larger surrounding region is the prediction. MicroSAM Base performs better in this local case because it keeps the bouton region separated and does not produce the extra connector-like region that MicroSAM Large adds under the complex-bouton interpretation. Depending on the downstream analysis require-

ment, this can be considered a preferable detection, especially when conservative boundary placement and a lower manual-cleanup burden are more important than maximally inclusive masks. The Brain1 bouton with blobs far away comparison is the standard slice used here to compare the two MicroSAM variants and should be read against the low-intensity far-blob example in Figure 4.9. The ground-truth bouton is also visible in the MicroSAM Base image; because the prediction and ground truth overlap, the remaining excess region corresponds to prediction-only extent, which shows that the Base variant handles this case better than the Large variant while still recovering the relevant bouton. The two Brain2 image pairs correspond to the detection-under-noisy-signal comparison in Figure 4.10 and the bottom Brain2 row in Figure 4.9. For these Brain2 cases, the suboptimal predictions are similar for both MicroSAM variants: both can add extra regions or connect neighboring signal beyond the evaluation ground truth, so the Base model's qualitative advantage is clearer on Brain1 than on these Brain2 local examples.

Although Imaris represents an expert-guided semi-automatic workflow, it still produces substantial false negatives in the best-configuration comparison: six missed boutons on Brain1, three on Brain2, and nine missed boutons in total. Its matched-instance mean IoU is also moderate, with a Total/avg. value of 0.505 in Table 4.8. This does not mean that the Imaris result is biologically invalid; rather, it shows that the established semi-manual workflow and the evaluation ground truth do not always encode identical object-boundary decisions. Figures 4.13, 4.14, and 4.15 provide qualitative examples for interpreting whether missed detections and lower IoU arise from intensity-biased surface selection, volume-boundary disagreement, merged detections, pixelated slice-level masks, or disagreement in ambiguous regions such as the bouton shown in Figure 4.7.

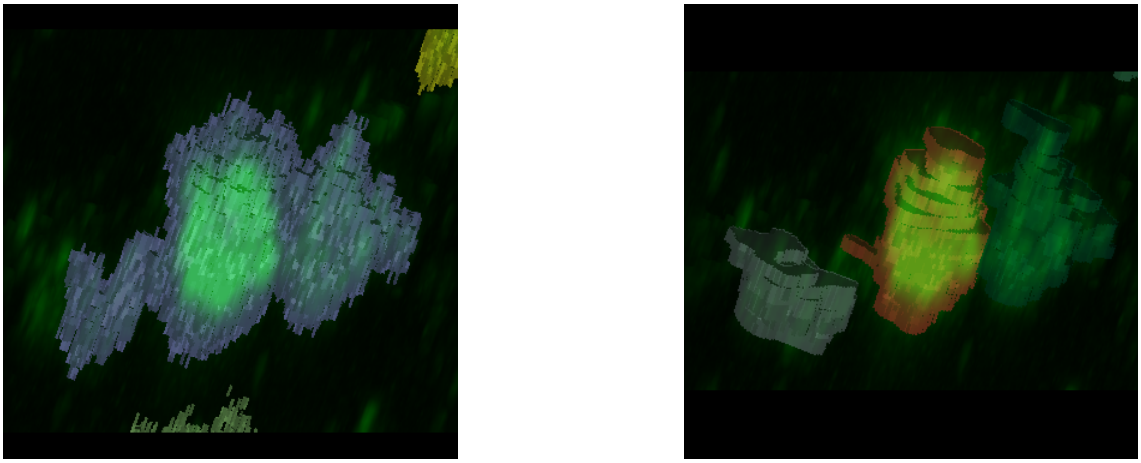


(a) Imaris output **(b)** Ground truth **(c)** Imaris surface vs. ground truth

Figure 4.13: Qualitative Imaris examples showing brightness-biased surface selection and volume mismatch. In the Imaris-output panels, the purple surface denotes the Imaris detection. In the volume-mismatch panel, the solid structure contained inside the larger purple Imaris surface is the evaluation ground-truth bouton.

Figure 4.13 shows that the brightness-imbalance failure mode is not unique to a trained deep-learning model. A surface- and intensity-driven Imaris workflow can likewise focus on the dominant bright signal and miss the same two upper boutons that were difficult for the standard XY-trained CellposeSAM 2D prediction in Figure 4.5. In Figure 4.13(c), the purple region denotes the Imaris-detected surface, whereas the smaller solid structure inside it corresponds to the annotated ground-truth bouton. Qualitatively, this can look like

a detection because the Imaris surface contains bouton signal. For the fixed quantitative protocol, however, it must be treated consistently with all other model outputs: a prediction is accepted only if its IoU with a ground-truth bouton reaches 0.25 after one-to-one Hungarian matching. The oversized Imaris surface fails this criterion because its spatial extent is several times larger than the actual bouton region and appears noisy, scattered, and spiky rather than compact and continuous. Therefore, this case contributes both a false-positive prediction, because the large Imaris surface is not accepted as a valid match, and a false-negative ground-truth bouton, because no prediction overlaps the annotated bouton sufficiently to pass the matching threshold. This strict interpretation avoids giving credit to arbitrarily large detections that include some bouton signal only because they cover a broad region, and keeps the Imaris baseline comparable to the learned models.



(a) Imaris prediction for close boutons

(b) Ground truth

Figure 4.14: Imaris prediction for close boutons compared with the ground truth.

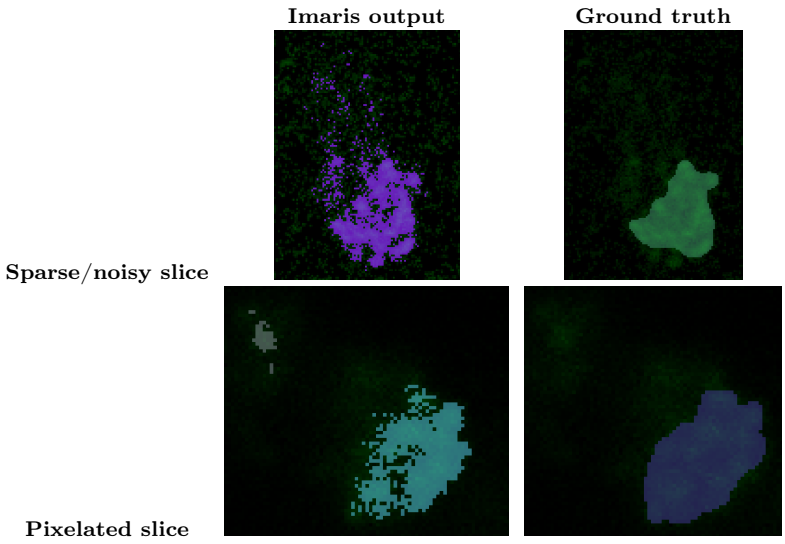


Figure 4.15: Two-dimensional slice examples from the Imaris baseline showing sparse/noisy output and pixelated boundary disagreement relative to the evaluation ground truth.

Figure 4.14 illustrates the close-bouton ambiguity introduced in Section 3.1.3. Here, the evaluation ground truth resolves the region as three separate bouton instances, whereas Imaris groups them into one surface. Since the Hungarian matching protocol allows each prediction to match at most one ground-truth object, this region can contribute only one true positive; the remaining two ground-truth boutons are counted as false negatives.

Figure 4.15 complements the three-dimensional renderings by showing that the same limitations are visible in individual image planes. The sparse/noisy example is a two-dimensional slice from the same bouton shown in the volume-mismatch example in Figure 4.13(c). In this slice, the Imaris output contains scattered purple pixels around the bouton region rather than a compact continuous mask. Such sparse and irregular pixel patterns do not resemble a plausible biological bouton object and therefore help explain why the corresponding three-dimensional surface becomes spatially noisy and suboptimal for volume-based analysis. A related pixelated behavior is visible in the second row of Figure 4.15: the output boundary is blocky and fragmented, with an additional small component, whereas the ground-truth object remains a continuous bouton region. These slice-level artifacts are consistent with the pixel- and intensity-driven nature of the Imaris segmentation workflow, where local pixel selections can accumulate into surfaces that broadly overlap fluorescence signal but still produce low IoU, extra false positives, or missed ground-truth boutons under the fixed matching protocol. Together with the close-bouton merging in Figure 4.14, these examples show that the Imaris baseline is useful mainly as a reference for the previous expert-guided workflow, not as evidence that it is sufficient for accurate quantitative analysis. For the fixed evaluation protocol used here, they support the conclusion that a learned segmentation protocol is better suited for consistent bouton counting and boundary-quality assessment.

4.5 Final Model Selection and Deployment Recommendation

For a deployment scenario that requires a single model and a single standardized preprocessing pipeline for new acquisitions, the selection must balance detection completeness, matched-boundary quality, false-positive burden, and volume accuracy. Under the recall-weighted interpretation used in this chapter, the decisive criteria are recall, matched-instance mean IoU, and PQ_{rec}, with false-positive and false-negative counts used to describe the practical correction burden.

Under this metric hierarchy, MicroSAM Large is the strongest quality-oriented deployment choice. In the all-pre-processing ablation it recovered all 61 ground-truth boutons and achieved the highest aggregate matched-instance mean IoU of 0.758. In the fair two-image All–PSF comparison it also achieved the highest PQ_{rec} at 0.787, with complete recall and the highest matched-instance mean IoU in Table 4.7. This makes MicroSAM Large the preferred model when complete bouton recovery and spatial agreement are more important than the number of additional candidate detections.

The main drawback of MicroSAM Large is its false-positive burden. In the all-pre-processing aggregate it produced 30 false positives, and in the fair All–PSF row it produced 16 false positives. These additional candidates may partly include split components of correctly localized large boutons, but they still require downstream filtering or manual review. MicroSAM Base is therefore the strongest practical alternative to MicroSAM Large when compute, GPU memory, or false-positive cleanup effort are limiting factors. It achieved

almost comparable segmentation performance while requiring the smaller encoder: it also recovered all 61 boutons in the all-pre-processing ablation, achieved an aggregate mean IoU of 0.755 compared with 0.758 for MicroSAM Large, and reached a fair-comparison PQ_{rec} of 0.771 compared with 0.787 for MicroSAM Large. At the same time, it reduced the false-positive count to 12 in the full all-pre-processing aggregate and 6 in the selected All–PSF row.

SwinUNETR should only be considered when false-positive suppression is the dominant practical constraint. It produced the cleanest selected all-pre-processing output, with only one false positive in the All–PSF row, and also performed robustly across the Brain2 pre-processing variants. However, it missed two boutons in the fair comparison and two boutons in the full all-pre-processing aggregate, and its PQ_{rec} remained below the MicroSAM variants because of lower matched-instance mean IoU. Therefore, unless false positives are the overriding concern and missed boutons are acceptable, SwinUNETR should not be selected as the main deployment model.

For LSMPlus-only applications, nnU-Net is also a strong candidate because it achieved perfect detection on the deconvolved and DoG-sharpened Brain2 variants and high performance on raw Brain2. Its weaker Airyscan Brain1 performance makes it less suitable as the main mixed-acquisition recommendation. CellposeSAM XYZ 2D performs very well under fixed DoG or PSF pre-processing, including complete recall in the corresponding single-pre-processing rows, but its raw Brain2 failure makes it unsuitable as the general all-pre-processing deployment model.

In summary, the recommended final configuration for recall- and boundary-quality-oriented mixed-acquisition deployment is MicroSAM Large trained on all pre-processing variants and applied with the selected deconvolved inference input. MicroSAM Base is the recommended substitute for environments and devices without sufficient compute or GPU memory for the large encoder, because it provides nearly comparable recall-weighted segmentation quality at lower computational cost while also reducing the false-positive cleanup burden.

Chapter 5

Conclusion and Outlook

5.1 Conclusion

This thesis investigated whether domain-adapted deep-learning segmentation can provide a reliable and scalable alternative to the existing semi-manual Imaris-based workflow for quantifying projection-neuron boutons in the mushroom body calyx of *Drosophila melanogaster*. The task is challenging because BRP-shortcherry-labeled boutons do not appear as smooth compact objects, but as irregular three-dimensional structures composed of multiple fluorescent active-zone puncta. In addition, the available data combine Airyscan and LSMPlus confocal acquisitions with different lateral image dimensions (1834×1834 pixels for Airyscan and 1101×1101 pixels for LSMPlus), contain anisotropic image stacks, and include background signal from neighboring axonal and neuropil structures. These acquisition-specific dimensions add further complexity to pre-processing, batching, and dimensionality handling. These properties make the problem substantially more difficult than applying a generic object detector to isolated bright blobs.

The first contribution of this work is the construction of a thesis-specific annotated dataset for bouton segmentation. The dataset combines manually corrected annotations, explicit object-definition rules, and multiple pre-processing representations for the LSMPlus data. The annotation rationale was important for obtaining labels that are usable for model training, but it also exposed a central ambiguity of the biological structure: spatially separated puncta can either represent one complex bouton or several nearby boutons depending on the intensity of the connecting signal and the chosen annotation convention. This ambiguity is not a minor technical detail, because it directly affects false-positive counts, matched-instance IoU, and the interpretation of split or connected predictions.

The second contribution is the systematic benchmarking of several model families and inference configurations on this domain-specific segmentation task. The evaluation showed that the choice of metric strongly affects the practical interpretation of model quality. Precision-sensitive scores are useful when false-positive candidates cannot be reviewed, but the biological readout of bouton number makes recall and matched-instance boundary quality especially important. For this reason, the analysis prioritised recall, matched-instance mean IoU, and PQ_{rec}, while still reporting false-positive counts and volume error as practical correction and quantification measures. Direct volume agreement was not used as a primary model-selection criterion, because a wrong segmentation can still produce a similar total volume if it misses bouton voxels in one region but includes a comparable

amount of non-bouton space elsewhere. Since bouton volume is ultimately calculated from the voxel volume multiplied by the segmented voxel support, spatially correct segmentation is the prerequisite for reliable volume estimation. Segmentation-quality metrics therefore provide the more meaningful basis for assessing volume-related reliability in this task than total volume alone.

Under this recall- and boundary-quality-oriented interpretation, MicroSAM Large trained on all pre-processing variants and applied with the selected deconvolved inference input is the strongest final deployment choice. It recovered all ground-truth boutons in the relevant all-pre-processing comparisons and achieved the highest recall-weighted segmentation quality among the evaluated mixed-acquisition settings. MicroSAM Base is the closest practical alternative. Its performance is nearly comparable to MicroSAM Large while using the smaller encoder, and it substantially reduces the false-positive burden. It is therefore the preferred substitute when computational resources, GPU memory, or manual cleanup effort are limiting factors. SwinUNETR should only be considered when suppressing false positives is the dominant practical constraint, because its lower false-positive count is obtained at the cost of missed boutons and lower PQ_{rec}. CellposeSAM XYZ 2D showed that incorporating axial context improves CellposeSAM performance, especially under suitable pre-processing, but the raw Brain2 failure makes it less suitable as the general all-pre-processing deployment model. The Imaris comparison further showed that the established semi-manual workflow remains useful as expert reference material, but is not sufficient as a scalable and consistent quantitative segmentation solution under the fixed evaluation protocol used here.

Taken together, the results support the central hypothesis of this thesis: domain-adapted deep-learning segmentation can provide a more scalable and quantitatively consistent basis for bouton detection and volume estimation than the previous semi-manual workflow. The recommended deployment model is not a fully automatic black-box replacement for expert judgment, but a practical segmentation component that can recover difficult bouton structures with high recall and reduce the amount of manual annotation and correction needed for future biological analyses.

5.2 Limitations

The main limitation of this study is the size and composition of the dataset. Although the data include both Airyscan and LSMPlus acquisitions, the number of biological volumes is small, and the final test set contains only two brain specimens. The evaluation was therefore designed to be as informative as possible for the available data, but it cannot fully characterize variability across laboratories, staining batches, microscope settings, genotypes, developmental stages, or larger biological cohorts. The selected test images cover important signal and morphology differences, yet additional independent data would be required before treating the reported numbers as definitive performance estimates for routine high-throughput deployment. In particular, some blob-like or borderline bouton structures may not have been present, or may not have been represented sufficiently often, in the test data. As a result, certain morphology-specific failure modes may not have been tested accurately, and the reported performance should be interpreted as evidence for the available test distribution rather than as exhaustive coverage of all possible bouton appearances.

A related limitation is the uneven training distribution between the acquisition-specific models and the model trained on all pre-processing variants. In the all-pre-processing setting, the same LSMPlus image content was effectively presented multiple times through different pre-processing representations, whereas the Airyscan data did not receive the same degree of representation. Consequently, this model was exposed to more LSMPlus-derived samples per epoch than the models trained on a single pre-processing condition, which could in principle bias the learned representation toward LSMPlus image characteristics. The empirical results did not indicate a clear disadvantage on the Airyscan test image, but this observation is based on a very small test set with only one image for each relevant acquisition/pre-processing condition. The limited number of test images therefore amplifies the uncertainty of this comparison: with a larger and more diverse test cohort, additional failure points or acquisition-specific biases might have become visible.

A second limitation is the remaining dependence on two-dimensional slice-wise consistency when segmenting and correcting intrinsically three-dimensional boutons. Even if a model identifies most of a bouton correctly, missing the connecting signal in only one or a few adjacent slices can break a single volumetric bouton into separate components. Under the object-level evaluation protocol, this can have a disproportionate effect: one fragment may be counted as an additional false-positive object, while the fragment that is matched to the ground truth can receive a weak IoU because it represents only part of the true bouton volume. Conversely, post-processing rules that remove components below a minimum acceptable bouton volume can suppress small fragments and reduce false positives, but they can also remove a real low-volume bouton or a weakly segmented part of a true bouton. These slice-wise discontinuities therefore create a practical trade-off between avoiding split-object false positives and preserving small or weak boutons. For an end user, such cases still require visual proofreading and may create extra annotation work, because a removed or split bouton has to be manually restored, merged, or corrected before reliable biological quantification.

A third limitation is that the segmentation pipeline still requires post-processing and, depending on the application, manual review. MicroSAM Large provides the best recall-oriented segmentation quality, but it also produces additional candidate regions. Some of these may be split components of true boutons or plausible weak-signal extensions, whereas others are likely false detections in signal-poor regions. The current evaluation reports this burden explicitly, but the thesis does not yet provide a fully optimized downstream filtering strategy based on intensity, shape, size, or anatomical context. As a result, the final recommendation should be understood as a strong segmentation starting point rather than a completely finished biological quantification pipeline.

Finally, the evaluation focuses on instance-level segmentation quality and derived volume agreement, but does not yet validate the complete biological endpoint in a large experimental study. The most important downstream question for the Tavosanis laboratory is whether the automated workflow reproduces biologically meaningful differences in bouton number and bouton volume across experimental conditions. The present work establishes the technical basis for that workflow, but the final biological validation must be performed on larger experimental cohorts.

As a final note for interpreting the reported results, the annotation provenance should also be kept in mind. The training and validation annotations were generated by the author according to the explicit thesis annotation protocol described in Section 3.1.3. They therefore provide consistent operational training targets, but they should not be interpreted as an

independent biological consensus. The held-out test ground truth was deliberately treated differently: it was reviewed together with collaborating biologists from the Tavosanis laboratory and used as the biologically grounded reference for final evaluation. This reduces the risk that the reported test performance merely reflects agreement with the author's annotation logic, and it aligns the evaluation more closely with the object definitions and practical correction needs of the intended end users.

This point is best understood as an interpretive caveat rather than as a major limitation of the study, because it only affects a small source of uncertainty around rare ambiguous structures. Complex boutons can contain spatially separated puncta, bright signal on one side, and low-intensity signal on the other side, making it difficult to determine the true bouton extent with complete certainty. In borderline cases, as illustrated in Figure 4.7, a reasonable expert decision can depend on the visibility of weak connector signal, which may be affected by display calibration, contrast settings, monitor quality, and expert judgment about where a bouton should start or end. These uncertainties should not be understood as errors by individual annotators and do not undermine the value of the test set. Rather, they define a narrow margin of label uncertainty around ambiguous structures. Consequently, the final metrics should be interpreted as estimates against the best available biologically informed reference, not as measurements against an entirely unambiguous ground truth. Future work should quantify inter-annotator variability and use consensus labels or uncertainty labels for ambiguous bouton instances.

5.3 Outlook

The most direct next step is to expand the annotated test dataset. Additional Airyscan and LSMPlus stacks should be collected and annotated across more brain preparations, imaging sessions, and signal conditions. This would make it possible to separate model selection from final testing more robustly, quantify confidence intervals for the main metrics, and evaluate whether the recommended MicroSAM configuration generalizes beyond the small held-out set used in this thesis. A multi-annotator test subset would be especially valuable for estimating the uncertainty of complex bouton definitions and for identifying cases where the ground truth should encode ambiguity rather than a single hard label.

A second important direction is the development of an automated candidate-cleanup stage. Since MicroSAM Large and MicroSAM Base recover all boutons in the most relevant comparisons but differ in their false-positive burden, the next practical improvement should focus on testing whether additional candidates can be filtered without sacrificing recall. This post-processing stage should be developed empirically on expanded validation data before being fixed as part of the final quantification workflow, because the most reliable filtering criteria cannot be determined from the current small test set alone.

The segmentation workflow could also benefit from stronger three-dimensional consistency. Several models and annotation steps operate on two-dimensional slices, whereas the biological objects are volumetric structures with anisotropic resolution. Future work could therefore develop a CellposeSAM XYZ-inspired `do_3d` strategy for MicroSAM Large or MicroSAM Base, in which inference is performed along three orthogonal axes and the resulting predictions are combined into a volumetric instance segmentation. This type of multi-axis inference could help reduce dependence on a single slice orientation, especially for boutons that contain separated puncta connected by weak signal, because these are the

cases where two-dimensional appearance and three-dimensional biological interpretation can diverge.

A more fundamental future direction would be to replace slice-wise prediction with direct volumetric segmentation. Fully three-dimensional SAM-style prediction was not pursued in this thesis because it was not computationally practical within the available scope and hardware, especially for large anisotropic microscopy volumes. However, if future 3D foundation models, voxel-based architectures, or other 3D-context segmentation methods become sufficiently mature and computationally feasible, they could be fine-tuned directly on bouton volumes. Such models would be a promising way to reduce slice-by-slice discontinuities, because they could learn weak inter-slice connector signal and object continuity as part of the prediction task rather than relying on post hoc merging of two-dimensional masks.

Future versions of the workflow should also test native-resolution processing for Airyscan acquisitions. In the present thesis, Airyscan volumes were downsampled to the standard confocal lateral sampling so that all model inputs shared a common voxel geometry. This made mixed-modality training and fair comparison more straightforward, but it also removed some of the lateral detail available in the original Airyscan data. A resolution-aware pipeline could preserve native Airyscan sampling during model inference, or use modality-specific processing before converting final bouton counts and volumes back into common physical units.

Future work should also revisit the PSF-deconvolution pre-processing itself. In this thesis, the Richardson–Lucy branch used a combined-channel representation rather than deconvolving the BRP-shortcherry and Kenyon-cell channels separately. A channel-specific or single-channel deconvolution variant, especially one optimized for the BRP-shortcherry bouton signal or using wavelength-specific PSFs for each fluorescence channel, could preserve marker-specific structure more faithfully and may yield a modest improvement in segmentation performance. This should be tested as a controlled pre-processing ablation on expanded validation data before being adopted as the default workflow.

To support practical deployment, a prototype software workflow was developed for applying the trained models to new microscopy data, as summarized in Appendix A. The prototype is intended as an initial end-user route from input image volumes to reviewable segmentations and quantitative outputs for the collaborating laboratory. It supports model selection, pre-processing, prediction, napari-based visual proofreading, measurement of bouton volume and surface area, and manual deletion of false-positive candidates. Future development should turn this prototype into a stable and documented laboratory tool with standardized pre-processing, model inference, candidate filtering, visual proofreading, and export of bouton counts and volumes in a format compatible with downstream statistical analysis. Because several trained models are available, the interface should also expose a simple model selector: MicroSAM Large could remain the default recall- and quality-oriented option, MicroSAM Base could be selected when runtime or GPU memory is limited, and more conservative models could be tested when suppressing false-positive candidates is the main practical concern. An active-learning strategy would be useful in this setting: uncertain or manually corrected predictions could be fed back into the annotation pool, allowing the model to improve as new biological data are processed.

Beyond the immediate bouton-quantification task, the same framework could be adapted to related synaptic-marker or microglomeruli analysis problems. Testing transfer to other projection-neuron populations, additional fluorescence markers, or different microscopy

systems would clarify whether the learned representation captures general bouton morphology or mainly the specific imaging domain used here. In this broader view, the present thesis provides a foundation for replacing isolated manual segmentation steps with reproducible, reviewable, and domain-adapted deep-learning tools for quantitative neuroanatomy.

Bibliography

[1] R. Lamprecht and J. LeDoux, "Structural plasticity and memory," *Nature Reviews Neuroscience*, vol. 5, no. 1, pp. 45–54, 2004. DOI: 10.1038/nrn1301

[2] R. L. Davis, "Learning and memory using *drosophila melanogaster*: A focus on advances made in the fifth decade of research," *Genetics*, vol. 224, no. 4, iyad085, 2023. DOI: 10.1093/genetics/iyad085

[3] M. Heisenberg, "Mushroom body memoir: From maps to models," *Nature Reviews Neuroscience*, vol. 4, no. 4, pp. 266–275, 2003. DOI: 10.1038/nrn1074

[4] S. Lin, "The making of the *drosophila* mushroom body," *Frontiers in Physiology*, vol. 14, p. 1 091 248, 2023. DOI: 10.3389/fphys.2023.1091248

[5] L. McGurk, A. Berson, and N. M. Bonini, "*Drosophila* as an *in vivo* model for human neurodegenerative disease," *Genetics*, vol. 201, no. 2, pp. 377–402, 2015. DOI: 10.1534/genetics.115.179457

[6] S. Lenz, P. Karsten, J. B. Schulz, and A. Voigt, "*Drosophila* as a screening tool to study human neurodegenerative diseases," *Journal of Neurochemistry*, vol. 127, no. 4, pp. 453–460, 2013. DOI: 10.1111/jnc.12436

[7] L. Baltruschat et al., "Circuit reorganization in the *drosophila* mushroom body calyx accompanies memory consolidation," *Cell Reports*, vol. 34, no. 11, p. 108 871, 2021. DOI: 10.1016/j.celrep.2021.108871

[8] Oxford Instruments. "How to extract a 3d object in imaris using the contour surface," Accessed: May 11, 2026. [Online]. Available: https://imaris.oxinst.com/learning/view/article/contour-surface

[9] Oxford Instruments. "Masking properties of imaris," Accessed: May 11, 2026. [Online]. Available: https://imaris.oxinst.com/learning/view/article/masking-properties-of-imaris

[10] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979. DOI: 10.1109/TSMC.1979.4310076

[11] J. Serra, *Image Analysis and Mathematical Morphology*. London: Academic Press, 1982, ISBN: 0126372403.

[12] L. Vincent and P. Soille, "Watersheds in digital spaces: An efficient algorithm based on immersion simulations," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 13, no. 6, pp. 583–598, 1991. DOI: 10.1109/34.87344

[13] M. C. Kremer et al., "Structural long-term changes at mushroom body input synapses," *Current Biology*, vol. 20, no. 21, pp. 1938–1944, 2010. DOI: 10.1016/j.cub.2010.09.060

[14] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, 2015, pp. 234–241. DOI: 10.1007/978-3-319-24574-4_28

[15] Ö. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, "3D U-Net: Learning dense volumetric segmentation from sparse annotation," in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2016*, 2016, pp. 424–432. DOI: 10.1007/978-3-319-46723-8_49

[16] A. Kirillov et al., "Segment anything," *arXiv preprint arXiv:2304.02643*, 2023. DOI: 10.48550/arXiv.2304.02643

[17] A. Archit et al., "Segment anything for microscopy," *Nature Methods*, vol. 22, pp. 579–591, 2025. DOI: 10.1038/s41592-024-02580-4

[18] K. G. Hales, C. A. Korey, A. M. Larracuente, and D. M. Roberts, "Genetics on the fly: A primer on the *drosophila* model system," *Genetics*, vol. 201, no. 3, pp. 815–842, 2015. DOI: 10.1534/genetics.115.183392

[19] L. T. Reiter, L. Potocki, S. Chien, M. Gribskov, and E. Bier, "A systematic analysis of human disease-associated gene sequences in *drosophila melanogaster*," *Genome Research*, vol. 11, no. 6, pp. 1114–1125, 2001. DOI: 10.1101/gr.169101

[20] U. B. Pandey and C. D. Nichols, "Human disease models in *drosophila melanogaster* and the role of the fly in therapeutic drug discovery," *Pharmacological Reviews*, vol. 63, no. 2, pp. 411–436, 2011. DOI: 10.1124/pr.110.003293

[21] A. H. Brand and N. Perrimon, "Targeted gene expression as a means of altering cell fates and generating dominant phenotypes," *Development*, vol. 118, no. 2, pp. 401–415, 1993. DOI: 10.1242/dev.118.2.401

[22] N. Y. Masse, G. C. Turner, and G. S. X. E. Jefferis, "Olfactory information processing in *drosophila*," *Current Biology*, vol. 19, no. 16, R700–R713, 2009. DOI: 10.1016/j.cub.2009.06.026

[23] N. J. Butcher, A. B. Friedrich, Z. Lu, H. Tanimoto, and I. A. Meinertzhagen, "Different classes of input and output neurons reveal new features in microglomeruli of the adult *drosophila* mushroom body calyx," *Journal of Comparative Neurology*, vol. 520, no. 10, pp. 2185–2201, 2012. DOI: 10.1002/cne.23037

[24] V. M. Puñal, M. Ahmed, E. M. Thornton-Kolbe, and E. J. Clowney, "Untangling the wires: Development of sparse, distributed connectivity in the mushroom body calyx," *Cell and Tissue Research*, vol. 383, no. 1, pp. 91–112, 2021. DOI: 10.1007/s00441-020-03386-4

[25] K. Yasuyama, I. A. Meinertzhagen, and F.-W. Schürmann, "Synaptic organization of the mushroom body calyx in *drosophila melanogaster*," *Journal of Comparative Neurology*, vol. 445, no. 3, pp. 211–226, 2002. DOI: 10.1002/cne.10155

[26] A. D. Elliott, "Confocal microscopy: Principles and modern practices," *Current Protocols in Cytometry*, vol. 92, no. 1, e68, 2020. DOI: 10.1002/cpcy.68

[27] J. Jonkman, C. M. Brown, G. D. Wright, K. I. Anderson, and A. J. North, "Tutorial: Guidance for quantitative confocal microscopy," *Nature Protocols*, vol. 15, pp. 1585–1611, 2020. DOI: 10.1038/s41596-020-0313-9

[28] M. Minsky, "Memoir on inventing the confocal scanning microscope," *Scanning*, vol. 10, no. 4, pp. 128–138, 1988. DOI: 10.1002/sca.4950100403

[29] K. Weisshart, "The basic principle of airyscanning," Carl Zeiss Microscopy GmbH, Technology Note, 2014.

[30] J. Huff, "The Airyscan detector from ZEISS: Confocal imaging with improved signal-to-noise ratio and super-resolution," *Nature Methods*, vol. 12, pp. i–ii, 2015. DOI: 10.1038/nmeth.f.388

[31] K. Korobchevskaya, B. C. Lagerholm, H. Colin-York, and M. Fritzsche, "Exploring the potential of Airyscan microscopy for live cell imaging," *Photonics*, vol. 4, no. 3, p. 41, 2017. DOI: 10.3390/photonics4030041

[32] J. C. Waters, "Accuracy and precision in quantitative fluorescence microscopy," *Journal of Cell Biology*, vol. 185, no. 7, pp. 1135–1148, 2009. DOI: 10.1083/jcb.200903097

[33] T. Peng et al., "A BaSiC tool for background and shading correction of optical microscopy images," *Nature Communications*, vol. 8, p. 14 836, 2017. DOI: 10.1038/ncomms14836

[34] K. Smith et al., "CIDRE: An illumination-correction method for optical microscopy," *Nature Methods*, vol. 12, pp. 404–406, 2015. DOI: 10.1038/nmeth.3323

[35] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed. Pearson, 2018.

[36] G. Litjens et al., "A survey on deep learning in medical image analysis," *Medical Image Analysis*, vol. 42, pp. 60–88, 2017. DOI: 10.1016/j.media.2017.07.005

[37] S. R. Sternberg, "Biomedical image processing," *Computer*, vol. 16, no. 1, pp. 22–34, 1983. DOI: 10.1109/MC.1983.1654163

[38] D. Marr and E. Hildreth, "Theory of edge detection," *Proceedings of the Royal Society of London. Series B. Biological Sciences*, vol. 207, no. 1167, pp. 187–217, 1980. DOI: 10.1098/rspb.1980.0020

[39] T. Lindeberg, *Scale-Space Theory in Computer Vision* (The Springer International Series in Engineering and Computer Science). Boston, MA: Kluwer Academic Publishers, 1994, vol. 256. DOI: 10.1007/978-1-4757-6465-9

[40] V. Kulikov et al., "DoGNet: A deep architecture for synapse detection in multiplexed fluorescence images," *PLOS Computational Biology*, vol. 15, no. 5, e1007012, 2019. DOI: 10.1371/journal.pcbi.1007012

[41] W. H. Richardson, "Bayesian-based iterative method of image restoration," *Journal of the Optical Society of America*, vol. 62, no. 1, pp. 55–59, 1972. DOI: 10.1364/JOSA.62.000055

[42] L. B. Lucy, "An iterative technique for the rectification of observed distributions," *The Astronomical Journal*, vol. 79, pp. 745–754, 1974. DOI: 10.1086/111605

[43] T. M. Lehmann, C. Gönner, and K. Spitzer, "Survey: Interpolation methods in medical image processing," *IEEE Transactions on Medical Imaging*, vol. 18, no. 11, pp. 1049–1075, 1999. DOI: 10.1109/42.816070

[44] J. Tilton, "Hybrid image segmentation for earth remote sensing data analysis," in *IGARSS '96. 1996 International Geoscience and Remote Sensing Symposium*, vol. 1, 1996, 703–705 vol.1. DOI: 10.1109/IGARSS.1996.516448

[45] E. Meijering, "Cell segmentation: 50 years down the road [life sciences]," *IEEE Signal Processing Magazine*, vol. 29, no. 5, pp. 140–145, 2012. DOI: 10.1109/MSP.2012.2204190

[46] M. Cybernetics. "Semantic vs. instance segmentation in microscopy: A complete guide | ai essentials," Accessed: Mar. 1, 2026. [Online]. Available: https://mediacy.com/blog/ai-essentials-semantic-vs-instance-segmentation-microscopy/

[47] O. Russakovsky et al., "ImageNet large scale visual recognition challenge," *International Journal of Computer Vision*, vol. 115, no. 3, pp. 211–252, 2015. DOI: 10.1007/s11263-015-0816-y

[48] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Computer Vision – ECCV 2014*, Springer, 2014, pp. 740–755. DOI: 10.1007/978-3-319-10602-1_48

[49] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015. DOI: 10.1038/nature14539

[50] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778. DOI: 10.1109/CVPR.2016.90

[51] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 4, pp. 640–651, 2017. DOI: 10.1109/TPAMI.2016.2572683

[52] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods*, vol. 18, pp. 203–211, 2021. DOI: 10.1038/s41592-020-01008-z

[53] U. Schmidt, M. Weigert, C. Broaddus, and G. Myers, "Cell detection with star-convex polygons," in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2018*, 2018, pp. 265–273. DOI: 10.1007/978-3-030-00934-2_30

[54] C. Stringer, T. Wang, M. Michaelos, and M. Pachitariu, "Cellpose: A generalist algorithm for cellular segmentation," *Nature Methods*, vol. 18, pp. 100–106, 2021. DOI: 10.1038/s41592-020-01018-x

[55] M. Weigert, U. Schmidt, R. Haase, K. Sugawara, and G. Myers, "Star-convex polyhedra for 3D object detection and segmentation in microscopy," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2020, pp. 3666–3673. DOI: 10.1109/WACV45572.2020.9093435

[56] M. Pachitariu and C. Stringer, "Cellpose 2.0: How to train your own model," *Nature Methods*, vol. 19, no. 12, pp. 1634–1641, 2022. DOI: 10.1038/s41592-022-01663-4

[57] C. Stringer and M. Pachitariu, "Cellpose3: One-click image restoration for improved cellular segmentation," *Nature Methods*, vol. 22, pp. 592–599, 2025. DOI: 10.1038/s41592-025-02595-5

[58] R. Bommasani, D. A. Hudson, E. Adeli, et al., "On the opportunities and risks of foundation models," *arXiv preprint arXiv:2108.07258*, 2021. DOI: 10.48550/arXiv.2108.07258

[59] A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html

[60] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations*, 2021. [Online]. Available: https://openreview.net/forum?id=YicbFdNTTy

[61] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, Oct. 2021, pp. 10 012–10 022.

[62] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. R. Roth, and D. Xu, "Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images," *arXiv preprint arXiv:2201.01266*, 2022. DOI: 10.48550/arXiv.2201.01266

[63] M. Pachitariu, M. Rariden, and C. Stringer, "Cellpose-SAM: Superhuman generalization for cellular segmentation," *bioRxiv*, 2025, Preprint. DOI: 10.1101/2025.04.28.651001

[64] K. Yang et al., "Classifying *drosophila* olfactory projection neuron boutons by quantitative analysis of electron microscopic reconstruction," *iScience*, vol. 25, no. 5, p. 104 180, 2022. DOI: 10.1016/j.isci.2022.104180

[65] napari contributors, *Napari: A multi-dimensional image viewer for Python*, 2019. DOI: 10.5281/zenodo.3555620 [Online]. Available: https://napari.org

[66] S. van der Walt et al., "Scikit-image: Image processing in Python," *PeerJ*, vol. 2, e453, 2014. DOI: 10.7717/peerj.453

[67] E. Riba, D. Mishkin, D. Ponsa, E. Rublee, and G. Bradski, "Kornia: An open source differentiable computer vision library for PyTorch," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2020, pp. 3674–3683. DOI: 10.1109/WACV45572.2020.9093363 [Online]. Available: https://openaccess.thecvf.com/content_WACV_2020/html/Riba_Kornia_an_Open_Source_Differentiable_Computer_Vision_Library_for_PyTorch_WACV_2020_paper.html

[68] Kornia contributors, *Kornia: Geometric computer vision library for spatial AI*, GitHub repository, 2026. Accessed: Jun. 23, 2026. [Online]. Available: https://github.com/kornia/kornia

[69] A. Kirillov, K. He, R. Girshick, C. Rother, and P. Dollár, "Panoptic segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 9404–9413. DOI: 10.1109/CVPR.2019.00963

Appendix A

Prototype Software Workflow

A prototype software workflow was developed to make the trained bouton-segmentation models usable on new microscopy data outside the evaluation scripts. The software is intended as a practical bridge between the trained models and the semi-automatic review workflow used in the collaborating laboratory. The source code is available at https://github.com/aditya0701/Fluorescent-Microscopy-Image-Segmentation-and-Quantification, and the selected trained model weights are available at https://huggingface.co/aditya0701/Drosophilla_melanogaster_Bouton_3d_segmentation. The model-weight repository contains the selected best-performing weights rather than every exploratory checkpoint produced during development.

The workflow allows a user to select an input image volume, choose one of the available trained models, and enter the voxel dimensions required for physical volume and surface-area measurements. After the user starts prediction, the corresponding pre-processing is applied automatically, the selected model is run, and the resulting label image is displayed together with the microscopy image in a napari-based viewer [65]. For computational efficiency, inference can be performed on downscaled data where required; the predicted labels are then mapped back to the original image size for visualization and quantitative review.

The review interface displays the image and predicted bouton masks at the original image scale. Users can inspect individual bouton candidates directly in the viewer and hover over objects to see their measured volume and surface area in a side panel. A linked table lists all detected boutons together with their volume and surface-area measurements. Selecting an entry in the table highlights the corresponding bouton in the viewer, and selecting a bouton in the viewer highlights the corresponding table entry. This bidirectional interaction is intended to make manual proofreading faster and less error-prone than inspecting masks only as static label images.

The software also supports manual deletion of candidate objects. If a user identifies a clear false-positive bouton, the corresponding object can be selected and removed from the prediction output. This functionality directly addresses the practical cleanup burden observed for high-recall models such as MicroSAM Large: additional candidate regions remain visible for review instead of being hidden in a fully automatic count, and obvious false positives can be removed before final bouton counts and volume measurements are exported.